Science AND Technology DPTE Notes

Diploma in Primary teacher Education (St. John's Teacher Training College, Kilimambogo)

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Diploma in primary education

Science and technology

UNIT 3 (47 HRS)

PREPARED AND COMPILED BY ODERO SAM MURANGA TTC 2021

THIS NOTES SHOULD ONLY ACT AS YOUR GUIDLINE ,USE DPTE CURRICULUM DESING

Living things

Classification

Introduction

- Classification is putting organisms into groups.
- Classification is based on the study of external characteristics of organisms.
- It involves detailed observation of structure and functions of organisms.
- Organisms with similar characteristics are put in one group.
- Differences in structure are used to distinguish one group from another.
- The magnifying lens is an instrument that assists in the observation of fine structure e.g. hairs by enlarging them.

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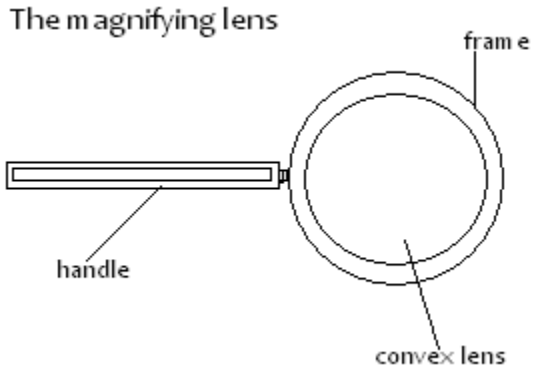

Using a Magnifying Lens

- A specimen is placed on the bench or held by hand,
- Then the magnifying lens is moved towards the eye until the object is dearly focused and an enlarged image is seen.
- The magnification can be worked out as follows:

$$\text{Magnification} = \frac{\text{length of the drawing}}{\text{length of the specimen}}$$

- ***Note: magnification has no units.***

Nececity/need for Classification

- To be able to identify organisms into their taxonomic groups.
- To enable easier and systematic study of organisms.
- To show evolutionary relationships in organisms.

Major Units of Classification (Taxonomic Groups)

- Taxonomy is the study of the characteristics of organisms for the purpose of classifying them.
- The groups are Taxa (singular Taxon).

The taxonomic groups include:

- ***Species:*** This is the smallest unit of classification. Organisms of the same species resemble each other. The number of chromosomes in their cells is the same. Members of a species interbreed to produce fertile offspring.
- ***Genus (plural genera):*** A genus is made up of a number of species that share several characteristics. Members of a genus cannot interbreed and if they do, the offspring are infertile.
- ***Family:*** A family is made up of a number of genera that share several characteristics.
- ***Order***: A number of families with common characteristics make an order.
- **Class**: Orders that share a number of characteristics make up a class.
- ***Phylum/Division:*** A number of classes with similar characteristics make up a phylum (plural phyla) in animals. In plants this is called a division.
- ***Kingdom:*** This is made up of several phyla (in animals) or divisions (in plants). It is the largest taxonomic unit in classification.

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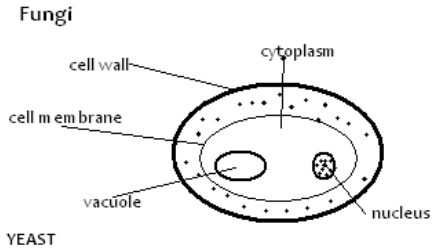
Kingdoms

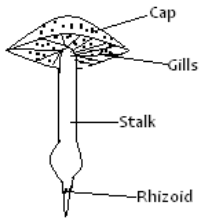
Living organisms are classified into five kingdoms;

- Monera,
- Protoctista,
- Fungi,
- Plantae
- Animalia.

Kingdom Fungi

- Some are unicellular while others are multicellular.
- They have no chlorophyll.
- Most are saprophytic e.g. yeasts, moulds and mushrooms.
- A few are parasitic e.g. ***Puccinia graminae.***





mushroom (Agaricus spp.)

Kingdom Monera (Prokaryota)

- These are very small unicellular organisms.
- They lack a nuclear membrane
- do not have any bound membrane organelles.
- Hence the name Prokaryota.
- They are mainly bacteria, e.g. ***Vibrio cholerae.***

Kingdom Protoctista

- They are unicellular organisms.
- Their nucleus and organelles are surrounded by membranes (eukaryotic).
- They include algae, slime moulds - fungi-like and protozoa

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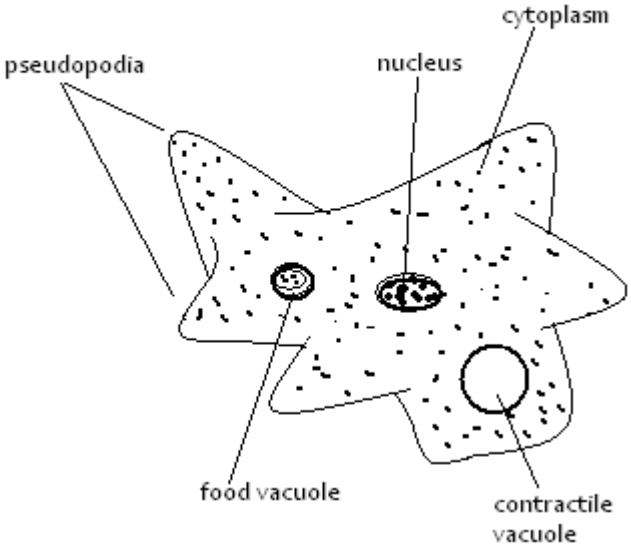
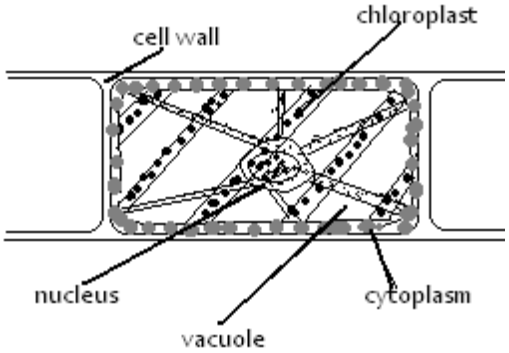
Amoeba-a protozoan(Protoctist)



Spyrogyra-filamentous green algae



Kingdom Plantae

- They are all multicellular.
- They contain chlorophyll and are all autotrophic.
- They include; Bryophyta (mossplant), Pteridophyta (ferns) and Spermatophyta (seed bearing plants).

Kingdom Animalia

- These are all multicellular and heterotrophic.
- Examples are annelida (earthworms), mollusca (snails),athropoda, chordata .
- Example of Arthropods are ticks, butterflies.
- Members of Chordata are fish, frogs and humans.

External Features of Organisms

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In plants we should look for:-

- Spore capsule and rhizoids in moss plants.
- Sori and fronds in ferns.
- Stem, leaves, roots, flowers, fruits and seeds in plants.

In animals, some important features to look for are:

- Segmentation, presence of limbs and, number of body parts, presence and number of antennae. These are found in phylum arthropoda:
- Visceral clefts, notochord, nerve tube, fur or hair, scales, fins, mammary glands, feathers and wings.
- These are found in chordata.

Binomial Nomenclature

- Organisms are known by their local names.
- Scientists use scientific names to be able to communicate easily among themselves.
- This method of naming uses two names, and is called Binomial nomenclature.
- The first name is the name of the genus: (generic name) which starts with a capital letter.
- The second name is the name of the species (specific name) which starts with a small letter.
- The two names are underlined or written in italics.
- Man belongs to the genus *Homo*, and the species, *sapiens.*
- The scientific name of man is therefore *Homo* *sapiens.*
- Maize belongs to the genus *Zea*, and the species *mays*.
- The scientific name of maize is *Zea* *mays.*

Practical Activities

- Use of Collecting Nets, Cutting Instruments and Hand Lens.
- Forceps are used to collect crawling and slow moving animals.
- Sweep nets are used to catch flying insects.
- Cutting instrument like scapel is used to cut specimen e.g. making sections.
- Hand lens is used to magnify small plants and animals.
- Drawing of the magnified organism are made and the linear magnification of each calculated.

Collection and Detailed Observation of Small Plants and Animals

e.g. moss, ferns, bean.

Look for the following:

- Moss plants: Rhizoids and spore capsules.
- Fern plants: Rhizomes with adventitious roots; large leaves (fronds) with Sori (clusters of sporangia).
- Seed plants: Tree/shrub (woody) or non-woody (herbs) e.g. bean.
- Root system - fibrous, adventitious and tap root.
- Stem - position and length of interrnodes.
- Type of leaves - simple or compound; arranged as alternate, opposite or whorled.

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- Flower - colour, number of parts, size and relative position of each:
- Fruits - freshy or dry; edible or not edible.
- Seeds - monocotyledonous or dicotyledonous.

Small animals e.g. earthworms, tick, grasshopper, butterfly, beetles.

Observe these animals to see:

- Number of legs.
- Presence or absence of wings.
- Number of antennae.
- Body covering.
- Body parts.

General Principles of Classification

- Classification is the science that puts organisms into distinct groups to make their study easy and systematic.
- Modern scientific classification is based on structure and functions.
- Organisms with similar anatomical and morphological characteristics are placed in one group while those with different structures are grouped separately.
- Modern studies in genetics and cell biochemistry are used to give additional help in classifying organisms.
- There are seven major taxonomic groups.
- The kingdom is the largest group.
- Others are phylum (division for plants) class, order, family, genus and species, the smallest.

Binomial Nomenclature

- Living organisms are named using Latin or Latinised names.
- Every organism has two names.
- This double naming is called ***binomial nomenclature.***
- This system of naming was devised by Carolus Linnaeus in the 18th Century.
- The first name is the generic name - the name of the genus.
- The second name is the name of the species.
- The generic name starts with a capital letter while that of the species starts with a small letter.
- The names are written in italics or are underlined in manuscripts.

Examples:

Bean =*Phaseolus vulgaris.*

- *Phaseolus* is the generic name,
- *vulgaris* is specific name.

Dog =*Canis familiaris.*

- *Canis* is the generic *name*
- *,familiaris* the specific name.

General Characteristics of Kingdoms

Organisms are classified into five kingdoms.

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- ***Monera,***
- ***Protoctista,***
- ***Fungi,***
- ***Plantae***
- ***Animalia.***

Viruses do not fit neatly into any of the above kingdoms.

- They are simple and not cellular.
- They are metabolically inactive outside the host cell.
- Most of them can be crystallised like chemical molecules.
- Therefore they do not exhibit the characteristics of living organisms.

Characteristic	*Monera*	*Protoctista*	*Fungi*	*Plantae*	*Animalia*
Cell type	Prokaryotic	Eucaryotic	Eucaryotic	Eucaryotic	Eucaryotic
U nicellularl Multicellular	Unicellular	Unicellular and multicellular	Unicellular and multicellular	Multicellular	Multicellular
Mode of Feeding	Autotrophic or heterotrophic by absorption	Autotrophic or heterotrophic by absorption or phagocytosis	Heterotrophis by absorption	Autotrophis	Heterotrophic by ingestion
Reproduction	Asexual by binary fission	Asexual binary fission, fragmentation, Sporulation	Asexual fission Fragmentation, sporulation	Asexual by sporulation and fragmentatio Sexual	Sexual

Examples of Organisms in Each Kingdom and Their Economic Importance

Kingdom Monera

General Characteristics

- Unicellular and microscopic
- Some single cells ,others colonial
- Nuclear material not enclosed within nuclear membrane-prokaryotic
- Have cell wall but not of cellulose.
- Have few organelles which are not membrane bound
- Mitochondria absent
- Mostly heterotrophic, feeding saprotrophically or parasitically,some are autotrophic.
- Reproduction mostly asexual through binary fission
- Most of them are anaerobes but others are aerobes
- Most move by flagella

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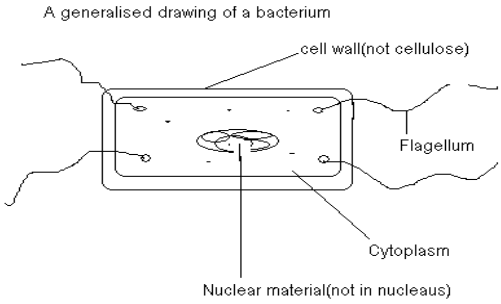

- Examples include ***Escherichia coli, Vibrio cholerae*** and ***Clostridium tetani.***
- Spherical known as Cocci.

- Rod shaped - e.g. *Clostridium tetani*
- Spiral shaped e.g. sprilla
- Coma shaped- Vibrios -e.g., *Vibrio cholerae.*

Economic importance of bacteria Benefits to man include:

- They are used in food processing e.g., *Lactobacillus* used in processing of cheese, yoghurt.
- Involved in synthesis of vitamin Band K, in humans and breakdown of cellulose in herbivores.

Genetic Engineering

- Bacteria are easily cultured and are being used for making antibiotics, aminoacids and enzymes e.g. amylase, and invertase e.g., ***Escherichia coli.***

Nutrient cycling:

- Saprophytes
- They are involved in decomposition of dead organic matter.
- They are useful in the nitrogen cycle.
- Nitrogen fixing and nitrifying bacteria.
- They increase soil fertility.
- Modem sewage works use bacteria in treatment of sewage.
- Cleaning oil spills in oceans and lakes.

Harmful Effects

- Bacteria cause disease:
- To humans (e.g. Cholera).
- To animals (e.g. Anthrax).
- Bacteria cause food spoilage.
- Others cause food poisoning e.g. *Salmonella.*
- Denitrifying bacteria reduce soil fertility e.g., *Pseudomonas denitrificans.*

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Kingdom Protoctista

Examples include ;

- Algae such as spirogyra, Chlamydomonas, euglena, Sargassum
- And protozoa such as amoeba, paramecium and Trypanosoma.

General Characteristics

- They are said to be eukaryotic since their nucleus is bound by a membrane
- Most are mobile, and use flagella, cilia and pseudopodia.
- Some are sessile.
- They reproduce mainly asexually, by binary fission, fragmentation and sporulation.
- Some reproduce sexually by conjugation.
- Some are heterotrophic e.g. paramecium.
- Others are autotrophic e.g. spirogyra.

Economic importance of protoctista

- Algae are the primary producers in aquatic food chains.
- They release a lot of oxygen to the atmosphere.
- Some cause human diseases like malaria and amoebic dysentry ,sleeping sickness
- Some are source of food for humans e.g. sargassum is a source of iodine
- Skeletons of diatoms used in paint making.

Spirogyra: They have spiral chloroplast.

- They are green, thread-like filaments

Chlamydomonas:

- This is a unicellular green algae and has a cup shaped chloroplast.
- They move towards light using the flagella
- Cilia assist the organism to move.
- The shape is due to the presence of a thin flexible pellicle.

Kingdom Fungi

- Multicellular fungi are made of thread-like structures called hyphae (singular hyphae) that form a mycelium.
- .e.g.Saccharomyces cereviseae(bread yeast).
- Others include Penicillium, Rhizopus, and edible mushroom

Economic Importance of Fungi

Beneficial Effects

- Some fungi are used as food e.g. mushrooms.
- Some are decomposers which enhance decay to improve soil fertility - recycling of nutrients e.g., toadstools.
- Some are useful in brewing and bread making e.g., yeast. Yeast is used as food - a rich source of Vitamin B.
- Some are useful in production of antibiotics e.g., *Penicillium griseofulvin.*
- Used in sewage treatment e.g., *Fusarium spp.*

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Harmful Effects

- Some cause food poisoning by producing toxic compounds e.g. *Aspergillus flavus* which produces aflatoxins.
- Some cause food spoilage, fabric and wood spoilage through decomposition.
- Some cause diseases to humans e.g., athlete's foot and ringworms.
- Others cause diseases to plants e.g., potato blight (Irish potatoes) rust in tomatoes and smuts in cereals.

Kingdom Plantae

General Characteristics

- They are multicellular and eukaryotic.
- They are photosynthetic and have a pigment chlorophyll.
- Their cells have cellulose cell walls.
- They reproduce sexually, others asexually.
- Kingdom Plantae has three major divisions:
  - Bryophyta,
  - Pteridophyta
  - Spermatophyta.

Division Bryophyta

These include mosses and liverworts.

- Plant body is not differentiated into root, stem and leaves.
- They have simple structures which resemble leaves and stems.
- They have rhizoids for absorbing water and anchoring the plant to substratum.
- Life cycle consists of two morphologically different plants, the gametophyte and sporophyte.
- The two alternate.
- They show alternation of generations.
- The gamete producing gametophyte is the persistent plant.
- The sporophyte is attached to the gametophyte and is nutritionally dependent on it.
- They lack vascular system.
- Sexual reproduction is dependent on water.

Division Pteridophyta:

These include ferns and horsetails.

General Characteristics

- They have root and shoot system.
- Leaves are compound known as fronds, they have a vascular system.
- They show alternation of generations whereby the spore bearing sporophyte is the main plant.
- Spores are borne in clusters on the underside of leaves making sari.
- The gametophyte is an independent minute structure called prothallus which is short lived.
- Sexual reproduction is dependent on water.

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Division Spermatophyta

- These are the seed bearing plants.

General Characteristics

- Plant body is differentiated into root, stem and leaves.
- Vascular tissue consists of xylem and phloem.
- Sexual reproduction is independent of water.
- Male gametophyte (pollen grain) germinates and grows to reach female gametophyte.
- They are divided into two sub-divisions:
  - Gymnosperms
  - Angiosperms.

Gymnosperms

- These are cone-bearing plants.
- Naked seeds.
- They are trees and shrubs.
- Xylem consists of tracheids only.
- Examples; pine, cypress and spruce.
- They show xerophytic characteristics like having needle-like leaves.

Angiosperms

- Seeds are enclosed within a fruit.
- They comprise trees, shrubs and herbs.
- Xylem consists of vessels of tracheids.
- These are the most advanced plants.
- Angiosperms has two classes;
  - Monocotyledonae
  - Dicotyledonae.

Comparison of Dicotyledonae and Monocotyledonae

Dicotyledonae	*Monocotyledonae*
• Embryo has two cotyledons.	• Embryo has one cotyledon.
• Leaves are broad and have network of veins.	• Leaves are long with parallel veins (have leaf sheath)
• T.S. of root has no pith.	• T.S. of root has pith.
• Have tap root system.	• Have fibrous root system.
• Cross section of stem reveals vascular bundles arranged in a ring.	• Cross section of stem reveals vascular bundles scattered all over.
• Vascular cambium present and have secondary growth.	• Vascular cambium absent and do not have secondary growth.
• Flower parts in four, five or multiples of these.	• Flower parts in three or multiples of three.
Examples: herbs e.g. tomatoe; shrubs e.g. tea,	Examples: grass, wheat, sugar-cane.

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hibiscus, lantana.	

Economic Importance of Spermatophyta

- They are a source of food for humans and other animals.
- Source of fue1- wood fuel and charcoal.
- Source of timber for building and for paper.
- Ornamental plants.
- Useful in textile industry.

Kingdom Animalia

- Most animals move from place to place in search of food.

Major phyla are:

- Platyhelminthes (Tapeworm).
- Nematoda (Ascaris).
- Annelida (Earthworm).
- Mollusca (Snails).
- Arthropoda
- chordata

Phylum Arthropoda

Distinguishing Characteristics

- They have jointed appendages, which are specialised for various functions.
- Their body is covered by a hardened exoskeleton made of chitin.
- It is shed at intervals to allow for growth.
- They have jointed body parts.
- Most are divided into head, thorax and abdomen.
- Some have two body parts,

General Characteristics

- Body is segmented.
- They have bilateral symmetry.
- Gaseous exchange is through tracheal system, book lungs or gills which opens to the outside through spiracles.
- Aquatic forms use gills.
- Reproduction is mainly sexual.
- They have an open circulatory system.

Phylum Arthropoda divided into five classes;

- Crustacea,
- Arachnida,
- Chilopoda,
- Diplopoda

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- Insecta

This division is based on:

- The number of limbs.
- Presence and number of antennae.
- Number of body parts.

Class Crustacea

- Most of them are aquatic, a few are terrestrial found in moist places e.g., woodlouse.

Distinguishing Characteristics

- Two body parts head and thorax are fused to form cephalothorax and an abdomen .
- They have two pairs of antennae; one is small and branched, the other is long.
- They have five or more parts of limbs.
- Some of these are modified for other functions e.g., locomotion, feeding and defence.
- Exoskeleton hardened with deposits of calcium carbonate i.e. carapace.

Other Characteristics

- Mouthparts include a pair of mandibles and two pairs of maxillae.
- Gaseous exchange is through gills.
- They have a pair of compound eyes.
- Most crustaceans are free-living but a few are parasitic e.g., barnacles.
- Examples are cray-fish and crab.

Class Arachnida

- Members are carnivorous and paralyse prey using poison produced from poison claws.

Distinguishing Characteristics

- The body has two parts: cephalothorax and abdomen.
- Cephalothorax is head fused to thorax.
- A pair of chelicerae, on ventral side of cephalothorax.
- They have four pairs of walking legs.
- They have no antennae.
- Instead they have a pair of short pedipalps which are sensitive to touch.
- Most arachnids use book lungs for gaseous exchange.
- Other characteristics include simple eyes.
- Examples include garden spider, ticks, scorpions.

Class Chilopoda

e.g. Centipede

Distinguishing Characteristics

- The body has 2 body parts, a head and trunk.
- The body is elongate, and has 15 or more segments.
- Has a pair of legs on each segment.
- The body is dorso-ventrally flattened.

Other characteristics include:

- Head has a pair of antennae.
- Gaseous exchange through tracheal system.

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- Are carnivorous.

Class Diplopoda e.g. Millipede

Distinguishing Characteristics

- Has two parts: head, short thorax and a trunk .
- Body elongate with 9-100 segments.
- Has two pairs of legs on each segment.
- They have a cylindrical body.
- Gaseous exchange is by tracheal system.

Other characteristics:

- Head has a pair of antennae.
- Are herbivorous.

Class Insecta

Distinguishing Characteristics

- Body is divided into three body parts head, thorax and abdomen.
- They have three pairs of legs ..
- Most insects have a pair or two of wings.

Other characteristics include:

- A pair of antennae.
- They breathe through spiracles, and gaseous exchange is through tracheal system.

The class is divided into several orders based on:

- Mouth parts- - type e.g. biting or piercing.
- Position of mouthparts - ventral or anterior.
- Wings - presence or absence; number of wing types, structure, texture.
- Size of legs.

Order Orthoptera

- Have biting and chewing mouthparts.
- Hind legs longer than other legs e.g. fore wings, leathery and longer than hind legs .
- e.g. locusts and grasshoppers .
- Swarming - locusts are a menace to farmers and the environment as they destroy crops and vegetation.

Order Diptera –

- True flies e.g. houseflies, and mosquitoes have sucking and piercing mouthparts, 1 pair of wings.
- The second pair is vestigial- acts as balancer.
- Mouthparts are ventral.
- These are disease vectors e.g., female anopheles mosquito transmits malaria.

Order Lepidoptera –

- Butterflies and moths have sucking mouthparts,

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- Two pairs of wings covered by scales.
- This group is important to farmers in pollination.

Order Hymenoptera –

- Bees ,wasps, ants.
- They have sucking mouthparts, two pairs of wings which are membranous.
- Some are non-winged e.g. some ants.
- Bees are important in pollination i.e. in production of honey.

Order Isoptera - Termites

- They have biting mouthparts which are anterior.
- Most are wingless,
- Those with wings they are membranous and of the same size.
- They are important in nutrient cycling as they feed on cellulose.

Order Coleoptera - Beetles

- Have biting mouthparts,
- Two pairs of wings,
- Fore wing hardened enclosing membranous wings.
- Destruction of stored grains and legumes (pulses)

Phylum Chordata

- This name is derived from the term notochord.
- This is a long flexible rod-like structure.
- The more familiar chordates are known as vertebrates.
- In vertebrates the notochord exists only in embryonic stages of development which in later stages is replaced by a vertebral column.

Main Characteristics of Vertebrates

- Members of the phylum have a notochord in early stages of development.
- They have visceral clefts - which are slits perforating the body wall at the pharynx.
- In fish these slits become gills while in higher chordates these slits are only present in embryo.
- They have a dorsal, hollow nerve cord.
- It develops into a brain at the anterior and spinal cord at the posterior end.
- The spinal cord is enclosed within the vertebral column.
- They have segmented muscle blocks known as myotomes on either side of the body.
- They possess a post-anal tail although rudimentary in some.
- They have a closed circulatory system.
- The heart is ventrally located.
- They possess an internal skeleton.

The main classes of phylum chordata are;

- Pisces,
- Amphibia,
- Reptilia,
- Aves

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- Mammalia.

Class Pisces

- These are the fishes.
- Some fish have a skeleton made of cartilage e.g. the shark.
- Others like Tilapia have a bony skeleton.

Distinguishing Characteristics

- They are aquatic.
- Movement is by means of fins.
- They have a streamlined body.
- They have a lateral line for sensitivity.
- Their heart has two chambers, the auricle and ventricle - simple circulatory system.

Other Characteristics

- Their body temperature changes according to the temperature of the environment.
- They are ectothermic (poikilothermic).
- Body covered with scales.
- They have gills for gaseous exchange.
- Exhibit external fertilisation.

Class Amphibia

- Larval forms are aquatic while adults are terrestrial.
- Adults return to water for breeding e.g. frogs, toads, newts, salamanders.

Distinguishing Characteristics

- Skin is soft and without scales.
- They have four well developed limbs.
- The hind limbs are longer and more muscular than forelimbs.
- The limb can be used for walking, jumping and swimming
- Gaseous exchange is through the skin, gills and lungs.
- Middle ear is present.

Other Characteristics

- They have a three-chambered heart with two atria and one ventricle.
- Fertilisation is external.
- They are ectothermic (poikilotherms).

Class Reptilia

- Examples are snakes, crocodiles, lizards, chameleons, tortoises and turtles.

Distinguishing Characteristics

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- The skin is dry and is covered by horny scales.
- Fertilisation is internal.
- Some species eggs contain a lot of yolk and have either leathery or calcareous shells.
- They have a double circulatory system.
- The heart has three chambers - two atria and a partly divided ventricle.
- However crocodiles have a four chamber heart.

Other Characteristics

- They are ectothermic (poikilothermic).
- Have 2 pairs of limbs.
- They use lungs for gaseous exchange.

Class Aves

- These are birds.
- They are terrestrial and arboreal and others are aquatic
- e.g. flamingo, goose, ostrich, penguin, hawk, dove.

Distinguishing Characteristics

- Body is covered by feathers and legs with horny scales.
- They have two pairs of limbs.
- Fore limbs modified to form wings for flight.
- Hind limbs are for walking or swimming.
- The mouth is a protruding beak.
- They have hollow bones.
- They have double circulation with a four-chambered heart (2 atria, 2 ventricles).
- They have lungs for gaseous exchange.
- Lungs are connected to air sacs in bones.
- Fertilisation is internal.
- They lay eggs with calcareous brittle shell.
- They have constant body temperatures hence are homoiotherms (endothermic).

Class Mammalia

- They are arboreal e.g. tree-squirrels,
- Others terrestrial e.g. humans
- Others are aquatic e.g. dolphins and whales.

Distinguishing Characteristics

- They have mammary glands hence name of the class.
- Body is covered with fur or hair.
- Their teeth are differentiated into four types (heterodont dentition).
- They have external ear-pinna.
- Most have sweat glands.
- They have a diaphragm that separates the body cavity into thoracic and abdominal.

Other Characteristics

- Internal fertilisation - most give birth.
- They have a double circulatory system with a four-chambered heart.
- They are endothermic (homoiotherms) .

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Eg Duck-billed Platypus (egg-laying mammal)

Eg.Kangaroo (pouched mammal)

- The young are born immature and are nourished in a pouch with milk from mammary glands.

Placental Mammals

- They give birth to fully developed young ones which are fed on milk from mammary glands.
- Some are aquatic. e.g. dolphins, whale,
- Others are flying e.g, bat;
- Most are terrestrial e.g. rabbits, elephants, buffalo, giraffe, antelope, cow, human being.

Placental mammals are divided into various orders:

- Rodentia: e.g. rats, mice - have one pair 9f upper incisors.
- Insectivora: e.g. mole-they are like rodents:
- Carnivora: e.g. dog; lion - flesh eaters, they have long pointed canines.
- Cetacea: e.g. whales and dolphins Aquatic mammals. Forelimbs are flippers.
- Chiroptera: e.g. bats - Forelimbs form wings.
- Artiodactyla: e.g. antelopes, cattle - they are even toed with split hooves.
- Perissodactyla: e.g. horse, donkey - they are odd toed with hooves.
- Proboscidea: e.g. elephant - upper lip and nose elongated to form trunk.
- Lagomorpha: e.g. rabbit, hare - mammals with upper and lower incisors. Have larger hind legs than forelegs.
- Primata: e.g. gorilla, orang utang, chimpanzee, monkeys - some are arboreal, with hand and foot for grasping.
- Human - *Homo sapiens* - upright gait, *opposable* thumb hence use of tools.

Construction and Use of Dichotomous Keys

- Biological keys are sets of statements that act as clues leading to the identification of an organism.
- By following the keys we can be able to place an organism in its group.
- The most common key is the dichotomous key.
- This is a biological tool for identification of unknown organisms.
- The word dichotomous means branching into two.
- A single characteristic is considered at a time.
- Two contrasting statements are put forward to describe the characteristics in such a way as to separate the organisms.
- This continues until all the organisms have been identified.

Rules Used to Construct a Dichotomous Key

- Use morphological characteristics as far as possible e.g. type of leaf - simple or compound.
- Select a single characteristic at a time and identify it by number. 1. Type of leaf. .
- Use identical forms of words for two contrasting statements e.g.:
  a) Flowers scented.

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b) Flowers not scented.

- Start with a major characteristic that divide the organisms into two large groups then proceed to lesser variations that would separate the organisms further into smaller groups.
- Use positive statements especially the first one.
- Avoid generalizations e.g. short plants. Be specific in your description e.g.:
  a) plants above 1m tall.
  b) plants below 1m tall.

Some Common Features Used for Identification

In Plants

Leaves

1. ***Type of leaf*** **Leaf**
   (a) Compound leaves. (b) Type of venation.

-
- Simple leaf
- Trifoliate
- Pinnate
- Type ofleaf margin.
- Type ofleaf arrangement on stem.
- The colour of leaf.
- The texture ofleaf; whether hairy or smooth.
- Shape of the leaf e.g. palmate.

Stem

- Type of stem - woody or herbaceous.
- Shape of stem - cylindrical or rectangular.
- Texture of stem smooth or spiny.

Infloresence

- Are flowers terminal or lateral
- For each flower:
- Is the flower regular or irregular?
- Number of floral parts for each whorl.
- Are floral parts free or fused?

Roots

- Type of root system- Taproot or fibrous?
- Function of the root.

In Animals

Features used to identify animals:

- Type of mouthparts.
- Type of skeleton.
- Presence or absence of antennae.
- Body segmentation.
- Body covering: scales, fur, hair or feathers.
- Number of body parts.
- Locomotory structures: legs, wings and fins.
- Presence or absence of vertebral column.

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- Presence and type of eves.

Practical Activities

To examine Bryophyta

- A mature moss plant is obtained.
- The specimen is observed using a hand -lens.
- A labelled drawing showing structures is made: rhizoids, set a capsule, gametophyte, sporophyte ..

To examine Pteridophyta

- A mature fern plant is obtained.
- **It** is observed using a hand lens.
- Sori can be seen on the lower side of fronds.
- A labelled drawing showing: frond, pinna, sorus, rhizome and adventitious roots.

To examine Spermatophyta

A mature twig of either cypress or pinus with cones is obtained.

- Observation of Male and female is made using a hand-lens.
- The naked seeds are noted.
- The leaves show xerophytic characteristics e.g. they are rolled, or needle-like.

A mature bean plant with pods is obtained,

- Observation of the leaves, stem and roots is made.
- **Leaves are compound, broad arid** have network of veins.
- The Ieaf-has a leaf stalk.
- They have a tap root system.
- Floral parts are in five e.g. 5 petals.
- A bean seed has two cotyledons.

A mature maize plant is obtained.

- Observation of the leaves, stems and roots is made.
- Leaves are simple, narrow and long with parallel veins ..
- The petiole is modified to form a leaf sheath.
- They have a-fibrous root system.
- Floral parts are in threes.
- A maize gram has one cotyledon,

Examination of Arthropoda

- Specimens of crayfish, millipede, centipede grasshopper and spider are obtained.
- Where specimens are not available photographs are used.
- External features of the specimens are observed.

The differences in the following are noted:

- Body parts.
- Antennae.
- Other appendages.
- Eyes.

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Examination of Chordata

- The following specimens are obtained:
- Tilapia, frog, Lizard, bird and rabbit.
- Using observable features each specimen is placed into its class.

Features used include:

- Body covering.
- Limbs.
- Type of teeth.

END

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Gaseous exchange in both

GASEOUS EXCHANGE IN PLANTS AND ANIMALS

Necessity for Gaseous Exchange in Living Organisms

- Living organisms require energy to perform cellular activities.
- The energy comes from breakdown of food in respiration.
- Carbon (IV) oxide is a by product of respiration and its accumulation in cells is harmful which has to be removed.
- Most organisms use oxygen for respiration which is obtained from the environment.
- Photosynthetic cells of green plants use carbon (Iv) oxide as a raw material for photosynthesis and produce oxygen as a byproduct.
- The movement of these gases between the cells of organisms and the environment comprises gaseous exchange.
- The process of moving oxygen into the body and carbon (Iv) oxide out of the body is called breathing or ventilation.
- Gaseous exchange involves the passage of oxygen and carbon (IV) oxide through a respiratory surface.
- Diffusion is the main process involved in gaseous exchange.

Gaseous Exchange in Plants

- Oxygen is required by plants for the production of energy for cellular activities.
- Carbon (IV) oxide is required as a raw material for the synthesis of complex organic substances.
- Oxygen and carbon (IV) oxide are obtained from the atmosphere in the case of terrestrial plants and from the surrounding water in the case of aquatic plants.
- Gaseous exchange takes place mainly through the stomata.

Structure of Guard Cells

- The stoma (stomata - plural) is surrounded by a pair of guard cells.
- The structure of the guard cells is such that changes in turgor inside the cell cause changes in their shape.
- They are joined at the ends and the cell walls facing the pore (inner walls) are thicker and less elastic than the cell walls farther from the pore (outer wall).
- Guard cells control the opening and closing of stomata.

Mechanism of Opening and Closing of Stomata

- In general stomata open during daytime (in light) and close during the night (darkness).

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- Stomata open when osmotic pressure in guard cells becomes higher than that in surrounding cells due to increase in solute concentration inside guard cells. Water is then drawn into guard cells by osmosis.
- Guard cells become turgid and extend.
- The thinner outer walls extend more than the thicker walls.
- This causes a bulge and stoma opens.
- Stomata close when the solute concentration inside guard cells become lower than that of surrounding epidermal cells.
- The water moves out by osmosis, and the guard cells shrink i.e. lose their turgidity and stoma closes.

Proposed causes of turgor changes in guard cells.

Accumulation of sugar.

- Guard cells have chloroplasts while other epidermal cells do not.
- Photosynthesis takes place during daytime and sugar produced raises the solute concentration of guard cells.
- Water is drawn into guard cells by osmosis from surrounding cells.
- Guard cells become turgid and stoma opens.
- At night no photosynthesis occurs hence no sugar is produced.
- The solute concentration of guard cells falls and water moves out of the guard cells by osmosis.
- Guard cells lose turgidity and the stoma closes.

pH changes in guard cells occur due to photosynthesis.

- In day time carbon (IV) oxide is used for photosynthesis. This reduces acidity while the oxygen produced increases alkalinity.
- Alkaline pH favours conversion of starch to sugar.
- Solute concentration increases inside guard cells, water is drawn into the cells by osmosis. Guard cells become turgid and the stoma opens.
- At night when no photosynthesis, Respiration produces carbon (IV) oxide which raises acidity .This favours conversion of sugar to starch. low sugar concentration lead to loss of turgidity in guard cells and stoma closes.

Explanation is based on accumulation of potassium ions

- In day time (light) adenosine triphosphate (ATP) is produced which causes potassium ions to move into guard cells by active transport.
- These ions cause an increase in solute concentration in guard cells that has been shown to cause movement of water into guard cells by osmosis.
- Guard cells become turgid and the stoma opens.
- At night potassium and chloride ions move out of the guard cells by diffusion and level of organic acid also decreases.
- This causes a drop in solute concentration that leads to movement of water out of guard cells by osmosis.
- Guard cells lose turgor and the stoma closes.

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Process of Gaseous Exchange in Root Stem and Leaves of Aquatic and Terrestrial Plants

Gaseous Exchange in leaves of Terrestrial Plants

- Gaseous exchange takes place by diffusion.
- The structure of the leaf is adapted for gaseous exchange by having intercellular spaces that are filled.
- These are many and large in the spongy mesophyll.
- When stomata are open,carbon(IV)oxide from the atmosphere diffuses into the substomatal air chambers.
- From here, it moves into the intercellular space in the spongy mesophyll layer.
- The CO_2 goes into solution when it comes into contact with the cell surface and diffuses into the cytoplasm.
- A concentration gradient is maintained between the cytoplasm of the cells and the intercellular spaces.
- CO_2 therefore continues to diffuse into the cells.
- The oxygen produced during photosynthesis moves out of the cells and into the intercellular spaces.
- From here it moves to the substomatal air chambers and eventually diffuses out of the leaf through the stomata.
- At night oxygen enters the cells while CO_2 moves out.

Gaseous exchange in the leaves of aquatic(floating)plants

- Aquatic plants such as water lily have stomata only on the upper leaf surface.
- The intercellular spaces in the leaf mesophyll are large.
- Gaseous exchange occurs by diffusion just as in terrestrial plants.

Observation of internal structure of leaves of aquatic plants

- Transverse section of leaves of an aquatic plant such as Nymphaea differs from that of terrestrial plant.

The following are some of the features that can be observed in the leave of an aquatic plant;

- Absence of cuticle
- Palisade mesophyll cells are very close to each other ie.compact.
- Air spaces (aerenchyma) in spongy mesophyll are very large.
- Sclereids (stone cells) are scattered in leaf surface and project into air spaces.
- They strengthen the leaf making it firm and assist it to float.

Gaseous Exchange Through Stems

Terrestrial Plants

- Stems of woody plants have narrow openings or slits at intervals called **lenticels.**
- They are surrounded by loosely arranged cells where the bark is broken.
- They have many large air intercellular spaces through which gaseous exchange occurs.
- Oxygen enters the cells by diffusion while carbon (IV) oxide leaves.
- Unlike the rest of the bark, lenticels are permeable to gases and water.

Aquatic Plant Stems

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- The water lily, *Salvia* and *Wolfia* whose stems remain in water are permeable to air and water.
- Oxygen dissolved in the water diffuses through the stem into the cells and carbon (IV) oxide diffuses out into the water.

Gaseous Exchange in Roots

Terrestrial Plants

- Gaseous exchange occurs in the root hair of young terrestrial plants.
- Oxygen in the air spaces in the soil dissolves in the film of moisture surrounding soil particles and diffuses into the root hair along a concentration gradient.
- It diffuses from root hair cells into the cortex where it is used for respiration.
- Carbon (IV) oxide diffuses in the opposite direction.
- In older roots of woody plants, gaseous exchange takes place through lenticels.

Aquatic Plants

- Roots of aquatic plants e.g. water lily are permeable to water and gases.
- Oxygen from the water diffuses into roots along a concentration gradient.
- Carbon (IV) oxide diffuses out of the roots and into the water.
- The roots have many small lateral branches to increase the surface area for gaseous exchange.
- They have air spaces that help the plants to float.
- Mangroove plants grow in permanently waterlogged soils, muddy beaches and at estuaries.
- They have roots that project above the ground level.
- These are known as breathing roots or pneumatophores.
- These have pores through which gaseous exchange takes place e.g. in *Avicenia* the tips of the roots have pores.
- Others have respiratory roots with large air spaces.

Gaseous Exchange in Animals

- All animals take in oxygen for oxidation of organic compounds to provide energy for cellular activities.
- The carbon (IV) oxide produced as a by-product is harmful to cells and has to be constantly removed from the body.
- Most animals have structures that are adapted for taking in oxygen and for removal of carbon (IV) oxide from the body.
- These are called "respiratory organs".
- The process of taking in oxygen into the body and carbon (IV) oxide out of the body is called breathing or ventilation.
- Gaseous exchange involves passage of oxygen and carbon (IV) oxide through a respiratory surface by diffusion.

Types and Characteristics of Respiratory surfaces

Different animals have different respiratory surfaces.

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- The type depends mainly on the habitat of the animal, size, shape and whether body form is complex or simple.
- ***Cell Membrane***: In unicellular organisms the cell membrane serves as a respiratory surface.
- ***Gills:*** Some aquatic animals have gills which may be external as in the tadpole or internal as in bony fish e.g. tilapia.
- They are adapted for gaseous exchange in water.
- ***Skin:*** Animals such as earthworm and tapeworm use the skin or body surface for gaseous exchange.
- The skin of the frog is adapted for gaseous exchange both in water and on land.
- The frog also uses ***epithelium lining of the mouth or buccal cavity*** for gaseous exchange.
- ***Lungs:*** Mammals, birds and reptiles have lungs which are adapted for gaseous exchange.

Characteristics of Respiratory Surfaces

- They are permeable to allow entry of gases.
- They have a large surface area in order to increase diffusion.
- They are usually thin in order to reduce the distance of diffusion.
- They are moist to allow gases to dissolve.
- They are well-supplied with blood to transport gases and maintain a concentration gradient.

Gaseous Exchange in Amoeba

- Gaseous exchange occurs across the cell membrane by diffusion.
- Oxygen diffuses in and carbon (IV) oxide diffuses out.
- Oxygen is used in the cell for respiration making its concentration lower than that in the surrounding water.
- Hence oxygen continually enters the cell along a concentration gradient.
- Carbon (IV) oxide concentration inside the cell is higher than that in the surrounding water thus it continually diffuses out of the cell along a concentration gradient.

Gaseous Exchange in Insects

- Gaseous exchange in insects e.g., grasshopper takes place across a system of tubes penetrating into the body known as the tracheal system.
- The main trachea communicate with atmosphere through tiny pores called spiracles.
- Spiracles are located at the sides of body segments;

- Two pairs on the thoracic segments and eight pairs on the sides of abdominal segments.
- Each spiracle lies in a cavity from which the trachea arises.
- Spiracles are guarded with valves that close and thus prevent excessive loss of water vapour.
- A filtering apparatus i.e. hairs also traps dust and parasites which would clog the trachea if they gained entry.
- The valves are operated by action of paired muscles.

Mechanism of Gaseous Exchange in Insects

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- The main tracheae in the locust are located laterally along the length of the body on each side and they are interconnected across.
- Each main trachea divides to form smaller tracheae, each of which branches into tiny tubes called tracheoles.
- Each tracheole branches further to form a network that penetrates the tissues. Some tracheoles penetrate into cells in active tissue such as flight muscles.
- These are referred to as intracellular tracheoles.
- Tracheoles in between the cells are known as intercellular tracheoles.
- The main tracheae are strengthened with rings of cuticle.
- This helps them to remain open during expiration when air pressure is low.

Adaptation of Insect Tracheoles for Gaseous Exchange

- The fine tracheoles are very thin about one micron in diameter in order to permeate tissue.
- They are made up of a single epithelial layer and have no spiral thickening to allow diffusion of gases.
- Terminal ends of the fine tracheoles are filled with a fluid in which gases dissolve to allow diffusion of oxygen into the cells.
- Amount of fluid at the ends of fine tracheoles varies according to activity i.e. oxygen demand of the insect.
- During flight, some of the fluid is withdrawn from the tracheoles such that oxygen reaches muscle cells faster and the rate of respiration is increased.
- In some insects, tracheoles widen at certain places to form air sacs.
- These are inflated or deflated to facilitate gaseous exchange as need arises.
- Atmospheric air that dissolves in the fluid at the end of tracheoles has more oxygen than the surrounding cells of tracheole epithelium'.
- Oxygen diffuses into these cells along a concentration gradient. '
- Carbon (IV) oxide concentration inside the cells is higher than in the atmospheric .
- Air and diffuses out of the cells along a concentration gradient.
- It is then removed with expired air.

Ventilation in Insects

- Ventilation in insects is brought about by the contraction and relaxation of the abdominal muscles.
- In locusts, air is drawn into the body through the thoracic spiracles and expelled through the abdominal spiracles.
- Air enters and leaves the tracheae as abdominal muscles contract and relax.
- **The muscles contract laterally so the** abdomen becomes wider and when they relax it becomes narrow.
- Relaxation of muscles results in low pressure hence inspiration occurs while contraction of muscles results in higher air pressure and expiration occurs.
- In locusts, air enters through spiracles in the thorax during inspiration and leaves through the abdominal spiracles during expiration.
- This results in efficient ventilation.
- Maximum extraction of oxygen from the air occurs sometimes when all spiracles close and hence contraction of abdominal muscles results in air circulating within the tracheoles.

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- The valves in the spiracles regulate the opening and closing of spiracles.

Observation of Spiracle in Locust

- Some fresh grass is placed in a gas jar.
- A locust is introduced into the jar.
- A wire mesh is placed on top or muslin cloth tied around the mouth of the beaker with rubber band.
- The insect is left to settle.
- Students can approach and observe in silence the spiracles and the abdominal movements during breathing.
- Alternatively the locust is held by the legs and observation of spiracles is made by the aid of hand lens.

Gaseous Exchange in Bony Fish (e.g, Tilapia)

- Gaseous exchange in fish takes place between the gills and the surrounding water.
- The gills are located in an opercular cavity covered by a flap of skin called the operculum.
- Each _gill consists of a number of thin leaf-like lamellae projecting from a skeletal base branchial arch (gill bar) situated in the wall of the pharynx.
- There are four gills within the opercular cavity on each side of the head.
- Each gill is made up of a bony gill arch which has a concave surface facing the mouth cavity (anterior) and a convex posterior surface.
- Gill rakers are bony projections on the concave side that trap food and other solid particles which are swallowed instead of going over and damaging the gill ftlaments.
- Two rows of gill filaments subtend from the convex surface.

Adaptation of Gills for Gaseous Exchange

- Gill filaments are thin walled.
- Gill filaments are very many (about seventy pairs on each gill), to increase surface area.
- Each gill filament has very many gill lamellae that further increase surface area.
- The gill filaments are served by a dense network of blood vessels that ensure efficient transport of gases.
- It also ensures that a favourable diffusion gradient is maintained.
- The direction of flow of blood in the gill lamellae is in the opposite direction to that of the water (counter current flow) to ensure maximum diffusion of gases.

Ventilation

- As the fish opens the mouth, the floor of the mouth is lowered.
- This increases the volume of the buccal cavity.
- Pressure inside the mouth is lowered causing water to be drawn into the buccal cavity.
- Meanwhile, the operculum is closed, preventing water from entering or leaving through the opening.
- As the mouth closes and the floor of the mouth is raised, the volume of buccal cavity decreases while pressure in the opercular cavity increases due to contraction of opercular muscles.
- The operculum is forced to open and water escapes.
- As water passes over the gills, oxygen is absorbed and carbon dioxide from the gills dissolves in the water.

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- As the water flows over the gill filaments oxygen in the water is at a higher concentration than that in the blood flowing, in the gill.
- Oxygen diffuses through the thin walls of gill filaments/lamellae into the blood.
- Carbon (IV) oxide is at a higher concentration in the blood than in the water.
- It diffuses out of blood through walls of gill filaments into the water.

Counter Current Flow

- In the bony fish direction of flow of water over the gills is opposite that of blood flow through the gill filaments .
- This adaptation ensures that maximum amount of oxygen diffuses from the water into the blood in the gill filament.
- This ensures efficient uptake of oxygen from the water.
- Where the flow is along the same direction (parallel flow) less oxygen is extracted from the water.

Observation of Gills of a Bony Fish (Tilapia)

- Gills of a fresh fish are removed and placed in a petri-dish with enough water to cover them.
- A hand lens is used to view the gills.
- Gill bar, gill rakers and two rows of gill filaments are observed.

Gaseous Exchange in an Amphibian - Frog

- An adult frog lives on land but goes back into the water during the breeding season.
- A frog uses three different respiratory surfaces.
- These are the ***skin, buccal cavity and lungs.***

Skin

- The skin is used both in water and on land.
- It is quite efficient and accounts for 60% of the oxygen taken in while on land.

Adaptations of a Frog's Skin for Gaseous Exchange

- The skin is a thin epithelium to allow fast diffusion.
- The skin between the digits in the limbs (i.e. webbed feet) increase the surface area for gaseous exchange.
- It is richly supplied with blood vessels for transport of respiratory gases.
- The skin is kept moist by secretions from mucus glands.
- This allows for respiratory gases to dissolve.
- Oxygen dissolved in the film of moisture diffuses across the thin epithelium and into the blood which has a lower concentration of oxygen.
- Carbon (IV) oxide diffuses from the blood across the skin to the atmosphere along the concentration gradient.

Buccal (Mouth) Cavity

- Gaseous exchange takes place all the time across thin epithelium lining the mouth cavity.
- Adaptations of Buccal Cavity for Gaseous Exchange

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- It has a thin epithelium lining the walls of the mouth cavity allowing fast diffusion of gases.
- It is kept moist by secretions from the epithelium for dissolving respiratory gases.
- It has a rich supply of blood vessels for efficient transport of respiratory gases.
- The concentration of oxygen in the air within the mouth cavity is higher than that of the blood inside the blood vessels.
- Oxygen, therefore dissolves in the moisture lining the mouth cavity and then diffuses into the blood through the thin epithelium.
- On the other hand, carbon (IV) oxide diffuses in the opposite direction along a concentration gradient.

Lungs

- There is a pair of small lungs used for gaseous exchange.

Adaptation of Lungs

- The lungs are thin walled for fast diffusion of gases.
- Have internal foldings to increase surface area for gaseous exchange.
- A rich supply of blood capillaries for efficient transport of gases.
- Moisture lining for gases to dissolve.

Ventilation

Inspiration

- During inspiration, the floor of the mouth is lowered and air is drawn in through the nostrils.
- When the nostrils are closed and the floor of the mouth is raised, air is forced into the lungs.
- Gaseous exchange occurs in the lungs, oxygen dissolves in the moisture lining of the lung and diffuses into the blood through the thin walls.
- Carbon (IV) oxide diffuses from blood into the lung lumen.

Expiration

- When the nostrils are closed and the floor of mouth is lowered by contraction of its muscles, volume of mouth cavity increases.
- Abdominal organs press against the lungs and force air out of the lungs into buccal cavity.
- Nostrils open and floor of the mouth is raised as its muscles relax.
- Air is forced out through the nostrils.

Gaseous Exchange in a Mammal -Human

- The breathing system of a mammal consists of a pair of **lungs** which are thin-walled elastic sacs lying in the thoracic cavity.
- The thoracic cavity consists of vertebrae, sternum, ribs and intercostal muscles.
- The thoracic cavity is separated from the abdominal cavity by the diaphragm.
- The lungs lie within the thoracic cavity.
- They are enclosed and protected by the ribs which are attached to the sternum and the thoracic vertebrae.
- There are twelve pairs of ribs, the last two pairs are called 'floating ribs' because they are only attached to the vertebral column.

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- The ribs are attached to and covered by internal and external intercostals muscles.
- The diaphragm at the floor of thoracic cavity consists of a muscLe sheet at the periphery and a central circular fibrous tissue.
- The muscles of the diaphragm are attached to the thorax wall.
- The lungs communicate with the outside atmosphere through the bronchi, trachea, mouth and nasal cavities.
- The trachea opens into the mouth cavity through the larynx.
- A flap of muscles, the epiglottis, covers the opening into the trachea during swallowing.
- This prevents entry of food into the trachea.
- Nasal cavities are connected to the atmosphere through the external nares(or nostrils)which are lined with hairs and mucus that trap dust particles and bacteria, preventing them from entering into the lungs.
- Nasal cavities are lined with cilia.
- The mucus traps dust particles,
- The cilia move the mucus up and out of the nasal cavities.
- The mucus moistens air as it enters the nostrils.
- Nasal cavities are winding and have many blood capillaries to increase surface area to ensure that the air is warmed as it passes along.
- Each lung is surrounded by a space called the pleural cavity.
- It allows for the changes in lung volume during breathing.
- An internal pleural membrane covers the outside of each lung while an external pleural membrane lines the thoracic wall.
- The pleural membranes secrete pleural fluid into the pleural cavity.
- This fluid prevents friction between the lungs and the thoracic wall during breathing.
- The trachea divides into two bronchi, each of which enters into each lung.
- Trachea and bronchi are lined with rings of cartilage that prevent them from collapsing when air pressure is low.
- Each bronchus divides into smaller tubes, the bronchioles.
- Each bronchiole subdivides repeatedly into smaller tubes ending with fine bronchioles.
- The fine bronchioles end in alveolar sacs, each of which gives rise to many alveoli.
- Epithelium lining the inside of the trachea, bronchi and bronchioles has cilia and secretes mucus.

Adaptations of Alveolus to Gaseous Exchange

- Each alveolus is surrounded by very many blood capillaries for efficient transport of respiratory gases.
- There are very many alveoli that greatly increases the surface area for gaseous exchange.
- The alveolus is thin walled for faster diffusion of respiratory gases.
- The epithelium is moist for gases to dissolve.

Gaseous Exchange Between the Alveoli and the Capillaries

- The walls of the alveoli and the capillaries are very thin and very close to each other.
- Blood from the tissues has a high concentration of carbon (IV) oxide and very little oxygen compared to alveolar air.

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- The concentration gradient favours diffusion of carbon (IV) oxide into the alveolus and oxygen into the capillaries .
- No gaseous exchange takes place in the trachea and bronchi.
- These are referred to as dead space.

Ventilation

- Exchange of air between the lungs and the outside is made possible by changes in the volumes of the thoracic cavity.
- This volume is altered by the movement of the intercostal muscles and the diaphragm.

Inspiration

- The ribs are raised upwards and outwards by the contraction of the external intercostal muscles, accompanied by the relaxation of internal intercostal muscles.
- The diaphragm muscles contract and diaphragm moves downwards.
- The volume of thoracic cavity increases, thus reducing the pressure.
- Air rushes into the lungs from outside through the nostrils.

Expiration

- The internal intercostal muscles contract while external ones relax and the ribs move downwards and inwards.
- The diaphragm muscles relaxes and it is pushed upwards by the abdominal organs. It thus assumes a dome shape.
- The volume of the thoracic cavity decreases, thus increasing the pressure.
- Air is forced out of the lungs.
- As a result of gaseous exchange in the alveolus, expired air has different volumes of atmospheric gases as compared to inspired air.

Table 7.1: Comparison of Inspired and Expired Air (% by volume)

Component	*Inspired*	*Expired %*
Oxygen	21	16
Carbon	0.03	4
Nitrogen	79	79
Moisture	Variable	Saturated

Lung Capacity

- The amount of air that human lungs can hold is known as lung capacity.
- The lungs of an adult human are capable of holding 5,000 cm^3 of air when fully inflated.
- However, during normal breathing only about 500 cm^3 of air is exchanged.
- This is known as the tidal volume.

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- A small amount of air always remains in the lungs even after a forced expiration.
- This is known as the residual volume.
- The volume of air inspired or expired during forced breathing is called vital capacity.

Control of Rate Of Breathing

- The rate of breathing is controlled by the respiratory centre in the medulla of the brain.
- This centre sends impulses to the diaphragm through the phrenic nerve.
- Impulses are also sent to the intercostal muscles.
- The respiratory centre responds to the amount of carbon (IV) oxide in the blood.
- If the amount of carbon (IV) oxide rises, the respiratory centre sends impulses to the diaphragm and the intercostal muscles which respond by contracting in order to increase the ventilation rate.
- Carbon (IV) oxide is therefore removed at a faster rate.

Factors Affecting Rate of Breathing in Humans

- Factors that cause a decrease or increase in energy demand directly affect rate of breathing.
- Exercise, any muscular activity like digging.
- Sickness
- Emotions like anger, flight
- Sleep.

Effects of Exercise on Rate of Breathing

- Students to work in pairs.
- One student stands still while the other counts (his/her) the number of breaths per minute.
- The student whose breath has been taken runs on the sport vigorously for 10 minutes.
- At the end of 10 minutes the number of breaths per minute is immediately counted and recorded.
- It is noticed that the rate of breathing is much higher after exercise than at rest.

Dissection of a Small Mammal (Rabbit) to Show Respiratory Organs

- The rabbit is placed in a bucket containing cotton wool which has been soaked in chloroform.
- The bucket is covered tightly with a lid.
- The dead rabbit is placed on the dissecting board ventral side upwards.
- Pin the rabbit to the dissecting board by the legs.
- Dissect the rabbit to expose the respiratory organs.
- Ensure that you note the following features.
- Ribs, intercostal muscles, diaphragm, lungs, bronchi, trachea, pleural membranes, thoracic cavity.

Diseases of the Respiratory System

Asthma

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- Asthma is a chronic disease characterised by narrowing of air passages.

Causes:

Allergy

- Due to pollen, dust, fur, animal hair, spores among others.
- If these substances are inhaled, they trigger release of chemical substances and they may cause swelling of the bronchioles and bring about an asthma attack.

Heredity

- Asthma is usually associated with certain disorders which tend to occur in more than one member of a given family, thus suggesting' a hereditary tendency.

Emotional or mental stress

- Strains the body immune system hence predisposes to asthma attack.

Symptoms

- Asthma is characterized by wheezing and difficulty in breathing accompanied by feeling of tightness in the chest as a result of contraction of the smooth muscles lining the air passages.

Treatment and Control

- There is no definite cure for asthma.
- The best way where applicable is to avoid whatever triggers an attack (allergen).
- Treatment is usually by administering drugs called bronchodilators.
- The drugs are inhaled, taken orally or injected intravenously depending on severity of attack to relief bronchial spasms.

Bronchitis

- This is an inflammation of bronchial tubes.

Causes

- This is due to an infection of bronchi and bronchioles by bacteria and viruses.
- ***Symptoms***
  - Difficulty in breathing.
  - Cough that produces mucus.
- ***Treatment***
  - Antibiotics are administered.

Pulmonary Tuberculosis

- Tuberculosis is a contagious disease that results in destruction of the lung tissue.

Causes

- Tuberculosis is caused by the bacterium ***Mycobacterium tuberculosis.***
- Human tuberculosis is spread through droplet infection i.e., in saliva and sputum.
- Tuberculosis can also spread from cattle to man through contaminated milk.
- From a mother suffering from the disease to a baby through breast feeding.
- The disease is currently on the rise due to the lowered immunity in persons with HIV and AIDS (Human Immuno Deficiency Syndrome).

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- Tuberculosis is common in areas where there is dirt, overcrowding and malnourishment.

Symptoms

- It is characterised by a dry cough, lack of breath and body wasting.

Prevention

- Proper nutrition with a diet rich in proteins and vitamins to boost immunity.
- Isolation of sick persons reduces its spread.
- Utensils used by the sick should be sterilised by boiling.
- Avoidance of crowded places and living in well ventilated houses.
- Immunisation with B.C.G. vaccine gives protection against tuberculosis.
- This is done a few days after birth with subsequent boosters.

Treatment

- Treatment is by use of antibiotics.

Pneumonia

- Pneumonia is infection resulting in inflammation of lungs.
- The alveoli get filled with fluid and bacterial cells decreasing surface are for gaseous exchange.
- Pneumonia is caused by bacteria and virus.
- More infections occur during cold weather.
- The old and the weak in health are most vulnerable.

Symptoms

- Pain in the chest accompanied by a fever, high body temperatures (39-40°C) and general body weakness.

Prevention

- Maintain good health through proper feeding.
- Avoid extreme cold.

Treatment

- If the condition is caused by pneumococcus bacteria, antibiotics are administered.
- If breathing is difficult, oxygen may be given using an oxygen mask.

Whooping Cough

- Whooping cough is an acute infection of respiratory tract.
- The disease is more common in children under the age of five but adults may also be affected.

Causes

- It is caused by ***Bordetella pertusis*** bacteria and is usually spread by droplets produced when a sick person coughs.

Symptoms:

- Severe coughing and frequent vomiting.
- Thick sticky mucus is produced.
- Severe broncho-pneumonia.
- Convulsions in some cases.

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Prevention

- Children may be immunised against whooping cough by means of a vaccine which is usually combined with those against diphtheria and tetanus.
- It is called "Triple Vaccine" or ***Diptheria, Pertusis*** and ***Tetanus*** (DPT).

Treatment

- Antibiotics are administered.
- To reduce the coughing, the patient should be given drugs.

END OF CHAPTER NOTES

Practical Activities

Observation of permanent slides of terrestrial and aquatic leaves and stems

Leaves

- Observation of T.S. of bean and water lily are made under low and 'medium power objectives. Stomata and air space are seen.
- Labelled drawings of each are made.
- The number and distribution of stomata on the lower and upper leaf surface is noted.
- Also the size of air spaces and their distribution.

Stem

- Prepared slides (TS) of stems of terrestrial and aquatic plants such as croton and reeds are obtained.
- Observations under low power and medium power of a microscope are made.
- Labelled drawings are made and the following are noted:
  - Lenticels on terrestrial stems.
  - Large air spaces (aerenchyma) in aquatic stems.

END OF CHAPTER NOTES

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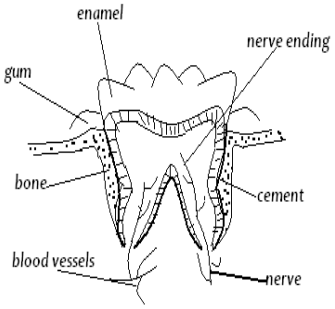
HUMAN BODY SYSTEM

Internal Structure of tooth



Internal structure of a tooth

The tooth consists of two main parts:

Crown: The portion above the gum; it is covered by the enamel.

Root: The portion below the gum; it is covered by the cement.

- The tooth has two roots.

Neck: Is the region at the same level with the gum.

- It forms the junction between the crown and the root.
- It is covered by enamel. Incisors and canines have one root only.
- Premolars have one or two roots while molars have two to three roots each.
- Internally, the bulk of the tooth is made up of dentine which consists of living cells and extends to the root.
- It is composed of calcium salts, collagen and water.
- It is harder than bone but wears out with use.
- This is why it is covered by enamel which is the hardest substance in a mammal's body.

Pulp Cavity: Contains blood vessels which provide nutrients to the dentine and remove waste products.

- It also contains nerve endings which detect heat, cold and pain.

Cement: Fixes the tooth firmly to the jaw bone.

Common Dental Diseases

Dental Carries

- *Dental carries* are the holes or cavities that are formed as acid corrodes enamel and eventually the dentine.
- ***Causes***
  - This is caused by bacteria acting on the food left between teeth and on the cusp.
  - Acids are formed that eventually corrode the enamel.
  - The pulp cavity is eventually reached.
  - A lot of pain is experienced then.
  - The bacteria then infect the pulp cavity and the whole tooth decays.
- ***Treatment***
  - Treatment depends on the extent of the dental caries:

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- Extraction of Tooth.
- Filling - this involves replacing the dentine with amalgam, a mixture of hard elements e.g. silver and tin.
- Root Canal Treatment - This involves surgery and reconstruction.
- It saves severely damaged teeth.
- The nerves in the root canal are surgically severed.
- The tooth is cleaned and filled up with amalgam.

Periodontal Diseases

- These are diseases of the gum.
- The gum becomes inflamed, and starts bleeding.
- Progression of the disease leads to infection of the fibres in the periodontal membranes and the tooth becomes loose.
- This condition is known as pyorrhoea.
- The diseases are caused by poor cleaning of the teeth.
- The accumulation of food particles leading to formation of plaque, lack of adequate vitamin A and C in the diet.

Treatment

- Nutrition - by taking adequate balanced diet rich in vitamins A and C.
- Antibiotics are used to kill bacteria.
- Anti-inflamatory drugs are given.
- Antiseptic is prescribed to use in cleaning the mouth daily to prevent further proliferation of bacteria.
- The plaque is removed-drilled away - a procedure known as scaling.

Care of Teeth

In order to maintain healthy teeth the following points should be observed:

- A proper diet that includes calcium and vitamins, particularly vitamin D is essential.
- The diet should also contain very small quantities of fluorine to strengthen the enamel.
- Large quantities of fluorine are harmful.
- The enamel becomes brown, a condition known as dental flourosis.
- Chewing of hard fibrous foods like carrots and sugar cane to strengthen and cleanse the teeth.
- Proper use of teeth e.g. not using teeth to open bottles and cut thread.
- Regular and thorough brushing of teeth after meals.
- Dental floss can be used to clean between the teeth.
- Not eating sweets and sugary foods between meals.
- Regular visits to the dentist for checkup.
- Washing the mouth with strong salt solution or with any other mouth wash with antiseptic properties.

Digestive System and Digestion in Humans

- Organs that are involved with feeding in humans constitute the digestive system.

Digestive System and Associated Glands

- Human digestive system starts at the mouth and ends at the anus.
- This is the alimentary canal.
- Digestion takes place inside the lumen of the alimentary canal.

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- The epithelial wall that faces the lumen has mucus glands (goblet cells).
- These secrete mucus that lubricate food and prevent the wall from being digested by digestive enzymes.
- Present at specific regions are glands that secrete digestive enzymes.
- The liver and pancreas are organs that are closely associated with the alimentary canal.
- Their secretions get into the lumen and assist in digestions.

Digestive system consists of:

- Mouth.
- Oesophagus.
- Stomach.
- Small intestines - consist of duodenum, the first part next to the stomach, ileum - the last part that ends up in a vestigial caecum and appendix which are nonfunctional.
- Large intestines consist of: colon and rectum that ends in the anus.

Ingestion, Digestion and Absorption

- Feeding in humans involves the following processes:
- *Ingestion*: This is the introduction of the food into the mouth.
- *Digestion:* This is the mechanical and chemical breakdown of the food into simpler, soluble and absorbable units.
- *Absorption:* Taking into blood the digested products.
- *Assimilation:* Use of food in body cells.
- Mechanical breakdown of the food takes place with the help of the teeth.
- Chemical digestion involves enzymes.

Digestion in the Mouth

- In the mouth, both mechanical and chemical digestion takes place.
- Food is mixed with saliva and is broken into smaller particles by the action of teeth.
- Saliva contains the enzyme amylase.
- It also contains water and mucus which lubricate and soften food in order to make swallowing easy.
- Saliva is slightly alkaline and thus provides a suitable pH for amylase to act on cooked starch, changing it to maltose.
- The food is then swallowed in the form of semisolid balls known as boluses.
- Each bolus moves down the oesophagus by a process known as peristalsis.
- Circular and longitudinal muscles along the wall of the alimentary canal contract and relax pushing the food along.

Digestion in the Stomach

- In the stomach, the food is mixed with gastric juice secreted by gastric glands in the stomach wall.
- Gastric juice contains pepsin, rennin and hydrochloric acid.
- The acid provides a low pH of 1.5-2.0 suitable for the action of pepsin.
- Pepsin breaks down protein into peptides.
- Rennin coagulates the milk protein casein.

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- The stomach wall has strong circular and longitudinal muscles whose contraction mixes the food with digestive juices in the stomach.

Digestion in the Duodenum

- In the duodenum the food is mixed with bile and pancreatic juice.
- Bile contains bile salts and bile pigments.
- The salts emulsify fats, thus providing a large surface area for action of lipase.
- Pancreatic juice contains three enzymes:
  - Trypsin which breaks down proteins into peptides and amino acids,
  - Amylase which breaks down starch into maltose, and
  - Lipase which breaks down lipids into fatty acids and glycerol.
- These enzymes act best in an alkaline medium which is provided for by the bile.

Digestion in ileum

- Epithelial cells in ileum secrete intestinal juice, also known as succus entericus.
- This contains enzymes which complete the digestion of protein into amino acids, carbohydrates into monosaccharides and lipids into fatty acids and glycerol.

Absorption

- This is the diffusion of the products of digestion into the blood of the animal.
- It takes place mainly in the small intestines though alcohol and some glucose are absorbed in the stomach.

The ileum is adapted for absorption in the following ways:

- It is highly coiled.
- The coiling ensures that food moves along slowly to allow time for its digestion and absorption.
- It is long to provide a large surface area for absorption.
- The epithelium has many finger-like projections called villi (singular villus).

- They greatly increase the surface area for absorption.
- Villi have microvilli that further increase the surface area for absorption.
- The wall of villi has thin epithelial lining to facilitate fast diffusion of products of digestion.
- Has numerous blood vessels for transport of the end products of digestion.
- Has lacteal vessels; for absorption of fatty acids and glycerol and transport of lipids.

Respiration

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Meaning and Significance of Respiration

- Respiration is the process by which energy is liberated from organic compounds such as glucose.
- It is one of the most important characteristics of living organisms.
- Energy is expended (used) whenever an organism exhibits characteristics of life, such as feeding, excretion and movement.
- Respiration occurs all the time and if it stops, cellular activities are disrupted due to lack of energy.
- This may result in death e.g., if cells in brain lack oxygen that is needed for respiration for a short time, death may occur.
- This is because living cells need energy in order to perform the numerous activities necessary to maintain life.
- The energy is used in the cells and much of it is also lost as heat.
- In humans it is used to maintain a constant body temperature.

Tissue Respiration

- Respiration takes place inside cells in all tissues.
- Every living cell requires energy to stay alive.
- Most organisms require oxygen of the air for respiration and this takes place in the mitochondria.

Mitochondrion Structure and Function

Structure

- Mitochondria are rod-shaped organelles found in the cytoplasm of cells.
- A mitochondrion has a smooth outer membrane and a folded inner membrane.
- The folding of the inner membrane is called cristae and the inner compartment is called the matrix.

Adaptations of Mitochondrion to its Function

- The matrix contains DNA ribosomes for making proteins and has enzymes for the breakdown of pyruvate to carbon (IV) oxide, hydrogen ions and electrons.
- Cristae increase surface area of mitochondrial inner membranes where attachment of enzymes needed for the transport of hydrogen ions and electrons are found.
- There are two types of respiration:
  - Aerobic Respiration
  - Anaerobic. Respiration

Aerobic Respiration

- This involves breakdown of organic substances in tissue cells in the presence of oxygen .

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- All multicellular organisms and most unicellular organisms e.g. some bactena respire aerobically.
- In the process, glucose is fully broken down to carbon (IV) oxide and hydrogen which forms water when it combines with the oxygen.
- Energy produced is used to make an energy rich compound known as adenosine triphosphate (ATP).
- It consists of adenine, an organic base, five carbon ribose-sugar and three phosphate groups.
- ATP is synthesised from adenosine diphosphate (ADP) and inorganic phosphate.
- The last bond connecting the phosphate group is a high-energy bond.
- Cellular activities depend directly on ATP as an energy source.
- When an ATP molecule is broken down, it yields energy.

Process of Respiration

- The breakdown of glucose takes place in many steps.
- Each step is catalysed by a specific enzyme.
- Energy is released in some of these steps and as a result molecules of ATP are synthesised.
- All the steps can be grouped into three main stages:

Glycolysis.

- The initial steps in the breakdown of glucose are referred to as glycolysis and they take place in the cytoplasm.
- Glycolysis consists of reactions in which glucose is gradually broken down into molecules of a carbon compound called pyruvic acid or pyruvate.
- Before glucose can be broken, it is first activated through addition of energy from ATP and phosphate groups.
- This is referred to as phosphorylation.
- The phosphorylated sugar is broken down into two molecules of a 3-carbon sugar (triose sugar) each of which is then converted into pyruvic acid.
- If oxygen is present, pyruvic acid is converted into a 2-carbon compound called acetyl coenzyme A (acetyl Co A).
- Glycolysis results in the net production of two molecules of ATP.
- The next series of reactions involve decarboxylation i.e. removal of carbon as carbon (IV) oxide and dehydrogenation, removal of hydrogen as hydrogen ions and electrons.
- These reactions occur in the mitochondria and constitute the Tri-carboxylic Acid Cycle (T.C.A.) or Kreb's citric acid cycle.
- The acetyl Co A combines with 4-carbon compound with oxalo-acetic acid to form citric acid - a 6 carbon compound.
- The citric acid is incorporated into a cyclical series of reactions that result in removal of carbon (IV) oxide molecules, four pairs of hydrogen, ions and electrons.
- Hydrogen ions and electrons are taken to the inner mitochondria membrane where enzymes and electron carriers effect release of a lot of energy.

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- Hydrogen finally combines with oxygen to form water, and 36 molecules of ATP are synthesised.

Anaerobic Respiration

- Anaerobic respiration involves breakdown of organic substances in the absence of oxygen.
- It takes place in some bacteria and some fungi.
- Organisms which obtain energy by anaerobic respiration are referred to as anaerobes.
- Obligate anaerobes are those organisms which do not require oxygen at all and may even die if oxygen is present.
- Facultative anaerobes are those organisms which survive either in the absence or in the presence of oxygen.
- Such organisms tend to thrive better when oxygen is present e.g. yeast.

Products of Anaerobic Respiration

- The products of anaerobic respiration differ according to whether the process is occurring in plants or animals.

Anaerobic Respiration in Plants

- Glucose is broken down to an alcohol, (ethanol) and carbon (IV) oxide.
- The breakdown is incomplete.
- Ethanol is an organic compound, which can be broken down further in the presence of oxygen to provide energy, carbon (IV) oxide and water.

$$C_6H_{12}O_6 _ 2C_2H_5OH + 2CO_2 + \text{Energy}$$
(Glucose) (Ethanol) (Carbon (IV) oxide)

Fermentation-

- Is the term used to describe formation of ethanol and carbon (IV) oxide from grains.
- Yeast cells have enzymes that bring about anaerobic respiration.

Lactate Fermentation

- Is the term given to anaerobic respiration in certain bacteria that results in formation of lactic acid.

Anaerobic Respiration in Animals

- Anaerobic respiration in animals produces lactic acid and energy.

$$C_6H_1P6 _ 2CH_3CHOH.COOH + \text{energy}$$
(Glucose) (Lactic acid) + energy

- When human muscles are involved in very vigorous activity, oxygen cannot be delivered as rapidly as it is required.
- The muscle respire anaerobically and lactic acid accumulates.
- A high level of lactic acid is toxic.
- During the period of exercise, the body builds up an oxygen debt.
- After vigorous activity, one has to breathe faster and deeper to take in more oxygen.

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- Rapid breathing occurs in order to break down lactic acid into carbon (IV) oxide and water and release more energy.
- Oxygen debt therefore refers to the extra oxygen the body takes in after vigorous exercise.

Practical Activities

To Show the Gas Produced When the Food is burned

- A little food substance e.g., maize flour or meat is placed inside a boiling tube.
- The boiling tube is stoppered using a rubber bung connected to a delivery tube inserted into a test-tube with limewater.
- The food is heated strongly to bum.
- Observations are made on the changes in lime water (calcium hydroxide) as gas is produced.
- The clear lime water turns white due to formation of calcium carbonate precipitate proving that carbon (Iv) oxide is produced.

Experiment to Show the Gas Produced During Fermentation

- Glucose solution is boiled and cooled. Boiling expels all air.
- A mixture of glucose and yeast is placed in a boiling tube, and covered with a layer of oil to prevent entry of air.
- A delivery tube is connected and directed into a test-tube containing lime water.
- The observations are made immediately and after three days the contents are tested for the presence of ethanol.
- A control experiment is set in the same way except that yeast which has been boiled and cooled is used.
- Boiling kills yeast cells.
- The limewater becomes cloudy within 20 minutes.
- This proves that carbon (IV) oxide gas is produced.
- The fermentation process is confirmed after three days when alcohol smell is detected in the mixture.

Experiment to Show Germinating Seeds Produce Heat

- Soaked bean seeds are placed in a vacuum flask on wet cotton wool.
- A thermometer is inserted and held in place with cotton wool .
- The initial temperature is taken and recorded.
- A control experiment is set in the same way using boiled and cooled bean seeds which have been washed in formalin to kill microorganisms.
- Observation is made within three days.
- Observations show that temperature in the flask with germinating seeds has risen.
- The one in the control has not risen.

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Comparison Between Aerobic and Anaerobic Respiration

	Aerobic Respiration	*Anaerobic Respiration*
1. Site	In the mitochondria.	In the cytoplasm.
2. Products	Carbon dioxide and water.	Ethanol in plants and lactic acid in animals
3. Energy yield	38 molecules of A TP (2880 KJ) from each molecule of glucose.	2 molecules of ATP 210KJ from each molecule of glucose.
4. Further reaction	No further reactions on carbon dioxide and water.	Ethanol and lactic acid can be broken down further in the presence of oxygen.

Comparison Between Energy Output in Aerobic and Anaerobic Respiration

- Aerobic respiration results in the formation of simple inorganic molecules, water and carbon (Iv) oxide as the byproducts.
- These cannot be broken down further. A lot of energy is produced.
- When a molecule of glucose is broken down in the presence of oxygen, 2880 KJ of energy are produced (38 molecules of ATP).
- In anaerobic respiration the by products are organic compounds.
- These can be broken down further in the presence of oxygen to give more energy.
- Far less energy is thus produced.
- The process is not economical as far as energy production is concerned.
- When a molecule of glucose is broken down in the absence of oxygen in plants, 210 KJ are produced (2 molecule ATP).
- In animals, anaerobic respiration yields 150 kJ of energy.

Substrates for Respiration

- Carbohydrate, mainly glucose is the main substrate inside cells.
- Lipids i.e. fatty acids and glycerol are also used.
- Fatty acids are used when the carbohydrates are exhausted.
- A molecule of lipid yields much more energy than a molecule of glucose.
- Proteins are not normally used for respiration.
- However during starvation they are hydrolysed to amino acids, dearnination follows and the products enter Kreb's cycle as urea is formed.
- Use of body protein in respiration result to body wasting, as observed during prolonged sickness or starvation.

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- The ratio of the amount of carbon (IV) oxide produced to the amount of oxygen used for each substrate is referred to as Respiratory Quotient (RQ) and is calculated as follows:

$$R.Q. = \frac{\text{Amount of carbon (IV) oxide produced}}{\text{Amount of oxygen used}}$$

- Carbohydrates have a respiratory quotient of 1.0 lipids 0.7 and proteins 0.8.
- Respiratory quotient value can thus give an indication of types of substrate used.
- Besides values higher than one indicate that some anaerobic respiration is taking place.

Application of Anaerobic Respiration in Industry and at Home

Industry

- Making of beer and wines.
- Ethanol in beer comes from fermentation of sugar(maltose) in germinating barley seeds.
- Sugar in fruits is broken down anaerobically to produce ethanol in wines.
- In the dairy industry, bacterial fermentation occurs in the production of several dairy products such as cheese, butter and yoghurt.
- In production of organic acids e.g., acetic acid, that are used in industry e.g., in preservation of foods.

Home

- Fermentation of grains is used to produce all kinds of beverages e.g., traditional beer and sour porridge.
- Fermentation of milk.

End of Topic

Reproduction in Animals

- Sexual reproduction involves the fusion of gametes.
- In animals two individuals are involved, a male and a female.
- Special organs known as gonads produce gametes.
- In males testes produce sperms while in females ovaries produce ova.

- The fusion of male gamete and female gamete to form a zygote is called fertilisation.

There are two types of fertilisation. External and internal.

External fertillsation

- Example in amphibians takes place in water.

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- The male mounts the female and shed sperms on the eggs as they are laid.
- Eggs are covered by slippery jelly-like substance which provides protection.
- Many eggs are released to increase the chances of survival.

Internal fertilisation

- This occurs in reptiles, birds and mammals.
- Fertilisation occurs within the body of the female.
- Fewer eggs are produced because there are higher chances of fertilisation since sperms are released into the female body.

Reproduction in Humans

Structure of female reproduction system

The female reproduction system consist of the following:

Ovaries

- Are two oval cream coloured structures found in lower abdomen below the kidneys.

Oviducts.

- They produce the ova.
- Are tubes which conduct the ova produced by the ovaries to the uterus.
- Fertilisation occurs in the upper part of the oviduct.

Uterus

- The uterus is a hollow muscular organ found in the lower abdomen.
- The embryo develops inside the uterus.
- The inner lining endometrium supplies nutrients to embryo.
- The embryo is implanted into the inner uterine wall- the endometrium which nourishes the embryo.
- The thick muscles of the uterus assist in parturition.

Cervix

- Has a ring of muscles that separates the uterus from the vagina.
- It forms the opening to the uterus

Vagina

- Is a tube that opens to the outside and it acts as the copulatory and birth canal through the vulva.

Structure of male reproductive system

The male reproductive system consists of the following:

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Testis:

- Each testis is a mass of numerous coiled tubes called semniferous tubules.
- Each is enclosed within a scrotal sac that suspends them between the thighs.
- This ensures that sperms are maintained at a temperature lower than that of the main body.

Seminiferous tubules

- The lining of seminiferous tubules consists of actively dividing cells which give rise to sperms.
- Between the seminiferous tubules are interstitial cells which produce the male hormones called androgens e.g. testosterone.
- The seminiferous tubules unite to form the epididymis, which is a coiled tube where sperms are stored temporarily .
- Vas deferens (sperm duct) is the tube through which sperms are carried from testis to urethra.
- Seminal vesicle produces an alkaline secretion which nourishes the spermatozoa.

Prostate gland

- Produces an alkaline secretion to neutralise vaginal fluids.

Cowpers' gland

- Secretes an alkaline fluid.
- All these fluids together with spermatozoa form semen.

Urethra

- Is a long tube through which the semen is conducted during copulation.
- It also removes urine from the bladder.

Penis

- Is an intro-mittent organ which is inserted into the vagina during copulation .

Fertilisation in Animals

- Fertilisation is preceded by copulation in which the erect penis is inserted into the vagina.
- This leads to ejaculation of semen.
- The sperms swim through the female's genital tract to the upper part of the oviduct.
- The head of the sperm penetrates the egg after the acrosome_ releases lytic enzymes t dissolve the egg membrane.
- The tail is left behind.
- Sperm nucleus fuses with that of the ovum and a zygote is formed.
- A fertilisation membrane forms around the zygote which prevents other sperms from penetrating the zygote.

Implantation:

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- After fertilisation the zygote begins to divide mitoticaly as it moves towards the uterus.
- It becomes embedded in the wall of the uterus a process called implantation.
- By this time the zygote is a hollow ball of cells called blastocyst or embryo.
- In the uterus the embryo develops villi which project into uterus for nourishment later the villi and endometrium develop into placenta.

Embryonic membranes

- Embryonic membranes develop around the embryo.
- The outermost membrane is the chorion which forms the finger-like projections (chorionic villi) which supply nutrients to the embryo.
- The amnion surrounds the embryo forming a fluid filled cavity within which the embryo lies.
- Amniotic cavity is filled with amniotic fluid.
- This fluid acts as a shock absorber and protects the foetus against mechanical injury.
- It also regutates temperature.
- The chorionic villi, allantois together with the endometrium from the placenta.
- The embryo is attached to the placenta by a tube called umbilical cord which has umbilical vein and artery.
- The maternal blood in the placenta flows in the spaces lacuna and surrounds capillaries from umbilical vein and artery.
- The umbilical cord increase in length as the embryo develops.

Role of placenta

Protection

- Maternal blood and foetal blood do not mix.
- This ensures that the pathogens and toxins from maternal blood do not reach the foetus.
- The placenta allows maternal antibodies to pass into the foetus, providing the foetus with immunity.

Nutrition

- The placenta facilitates the transfer of nutrients from maternal blood to foetus.

Excretion

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- Placenta facilitates the removal of nitrogenous wastes from the foetus' blood to maternal blood.

Gaseous exchange

- Oxygen from the maternal blood diffuses into the foetal blood while carbon (IV) oxide from foetal blood diffuse into maternal blood.

Production of hormones

- Placenta produces progesterone and oestrogen.

Gestation period

- The period between conception and birth is called gestation.
- In humans gestation takes nine months (40 weeks).
- The embryo differentiates into tissues and organs during this period.

Week 1 to 3:

- Zygote divides to form blastocyst.
- Implantation takes place.
- The three germ layers form endoderm, mesoderm and ectoderm.
- Nervous system starts to form.

Week 4 to 7:

- Development of circulating and digestive systems.
- Further development of nervous system, formation of sensory organs,
- All major internal organs are developed.
- At week 5, heartbeat starts .

Week 8 to 24:

- All organs well developed including sex organs.
- Hair, finger and toe nails grow.
- Foetus move and eyelids open.

Week 25- 30:

- The fully developed foetus responds to touch and noises and moves vigorously.
- The head turns and faces downwards ready for birth.

Week 31-40:

- Foetus increases in size.
- Birth occurs.

Reproductive Hormones

Hormone	***Source***	***Functions***
Follicle Stimulating Hormone (FSH)	Pituitary gland	Development of ovarian follicle; stimulates secretion of oestrogen by the ovary
Luteinising Hormone	Pituitary gland	Causes ovulation; causes development of Graafian follicle into the corpus luteurn; causes secretion of progesterone by the ovary
Prolactin	Pituitary gland	Initiates production and secretion of milk by the

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		mammary glands
Oxytocin	Pituitary gland	Causes contraction of the uterus during parturition (birth)
Progesterone	Corpus luteum in the ovary	Causes contraction of wall of the uterus to thicken after ovulation
Oestrogen	Ovary	Causes changes in the uterine wall in preparation for implantation; initiates development of secondary sexual characteristics
Androgens-Testosterone	Interstitial cells of testis	Stimulates the development of secondary sexual characteristics
Interstitial Cell Hormone (ICSH)	Pituitary gland	Stimulates the interstitial cells of testis to release androgens
Human Chorionic Gonadotrophin (HCG)	Chorionic villi	Stops the degeneration of the corpus luteum for production of oestrogen and progesterone

Secondary Sexual Characteristics

Male

- Testerone is the main androgen that stimulates the development of secondary sexual characteristics.
- Broadening of the shoulders.
- Deepening of the voice due to enlargement of larynx.
- Hair at the pubic area, armpit and chin regions.
- Penis and testis enlarge and produce sperms.
- Body becomes more masculine.

Female

- Enlargement of mammary glands.
- Hair grows around pubic and armpit regions.
- Widening of the hips.
- Ovaries mature and start producing ova.
- Menstruation starts.
- Oestrogen triggers the onset of secondary sexual characteristics.

Sexually transmitted infections (STI)

Disease	*Causative*	*Method of*	*Symptoms*	*Prevention/control*

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		transmission		
Gonorrhoea	Bacterium Neiseeria Gonorrhoea	-Sexual contact - during birth for infants -Sharing towels	-Itching of urethra -yellowish discharge pain as males urinate, vaginal discharge. with odour in females	A void indiscriminate sex. Treat both partners infected A void sharing linen
Syphilis	Bacterium Treponema Palladium	-Sexual contact - During birth for infants. - Sharing towels and linen	Solitary painless ulcer-on genital or mucous -Rashes, muscles and papules on hands, feet lips, genital areas	Treat at primary infection stage -Avoid indiscriminate sex. - A void sharing linen
Trichomoniasi	Protozoan Trichomonas Vaginalis	-Sexual contact -contaminated linen, underwear and toilet seats	Itching of urethra or vagina in females, smelly, yellow discharge	A void sharing linen -Avoid indiscriminate sex -personal hygiene
Hepatitis	Virus Hepatitis B	-Sexual contact -blood contaminated needles and syringes	Fever, nausea, jaundice, loss of appetite, yellow urine	-Avoid indiscriminate sex -use disposable needles and syringes - strict personal hygiene
Candidiasis	Fungus Albicans	-spread through sexual contact - sharing linen and towels	Itching and burning sensation and white discharge from genitals	-Avoid indiscriminate sex - Treat both partners
Herpes (Simplex)	Virus Herpes	-sexual contact kissing, contaminated needles	Lesions on skin and mucous membranes of buccal cavity vagina or head of penis	- A void indiscriminate sex and contaminated needles and syringes.
HIV and Aids	Virus Human Deficiency	-sexual contact -blood -contaminated instruments -Through breast milk and body fluids. -Through birth canal for infants	-chronic diarrhoea -weight loss (more than 10% body weight lost in a month) - constant, persistent cough, skin infectious (herpes zoster)	-Avoid indiscriminate sex. -Use screened blood - No sharing of tooth brushes, razors - Use disposable needles

Menstrual Cycle

- This is characterized by discharge of blood and tissue debris (menses) from the uterus every 28 days.
- This is due to the breakdown of the endometrium which occurs when the level of progesterone falls and the girl starts to menstruate.
- The follicle stimulating hormone (FSH) causes the Graafian follicle to develop and also stimulate the ovary to release oestrogen.

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- Oestrogen hormone triggers the onset of secondary sexual characteristics.
- Luteinising hormone (L.H) causes the mature ovum to be released from the Graafian follicle - a process called ovulation.
- After ovulation progesterone hormone is produced.
- After menstruation, the anterior lobe of the pituitary gland starts secreting the follicle stimulating hormone (FS.H) which causes the Graafian follicle to develop in the ovary.
- It also stimulates the ovary tissues to secrete oestrogen.
- Oestrogen brings about the repair and healing of the inner lining of the uterus (endometrium) which had been destroyed during menstruation.
- Oestrogen level stimulates the pituitary gland to produce (Luteinising Hormone (L.H).
- This hormone makes the mature Graafian follicle to release the ovum into the funnel of oviduct, a process called ovulation.
- After releasing the ovum, the Graafian follicle changes into a yellow body called corpus luteum.
- The luteinising hormone stimulates the corpus luteum to secrete a hormone called progesterone which stimulates the thickening and vascularisation of endometrium.
- This prepares the uterine wall for implantation of the blastocyst.
- If fertilisation takes place, the level of progesterone increases and thus inhibits FSH from stimulating the maturation of another Graafian follicle.
- If fertilisation does not occur, the corpus luteum disintegrates and the level of progesterone goes down.
- The endometrium, sloughs off and menstruation occurs.

Advantages of Reproduction Asexual

- Good qualities from parents are retained in the offspring without variation.
- New individuals produced asexually mature faster.
- Process does not depend on external factors which may fail such as pollination.
- New individuals obtain nourishment from parent and so are able to survive temporarily under unsuitable conditions.
- No indiscriminate spreading of individuals which can result in wastage of offspring.
- Takes a shorter time and leads to rapid colonization.

Disadvantages of asexual reproduction

- New offspring may carry undesirable qualities from parents.
- Offspring may be unable to withstand changing environmental conditions.
- Faster maturity can cause overcrowding and stiff competition.
- Reduced strength and vigour of successive generations.

Advantages of sexual reproduction

- Leads to variations.
- Variations which are desirable often show hybrid vigour.

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- High adaptability of individuals to changing environmental conditions.
- Variations provide a basis for evolutionary changes.

Disadvantages of sexual reproduction

- Fusion is difficult if two individuals are isolated.
- Some variations may have undesirable qualities.
- Population growth is slow.

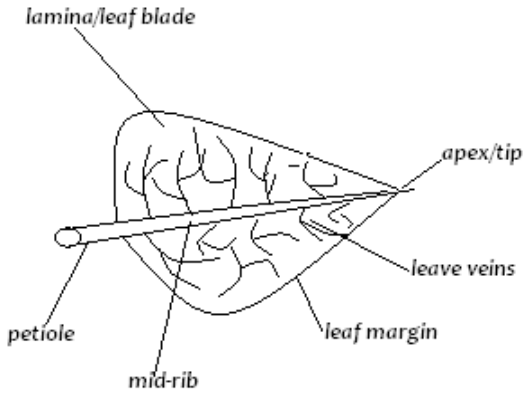
NUTRITION IN PLANTS AND ANIMALS

Nutrition in Plants and Animals

Structure of the Leaf

External Structure

- The external structure of the leaf consists of a leaf stalk or petiole and a broad leaf blade or lamina.
- The lamina has a main vein midrib from which smaller veins originate.
- The outline of the leaf is the margin and the tip forms the apex.



External structure of a leaf

Internal Structure of the Leaf

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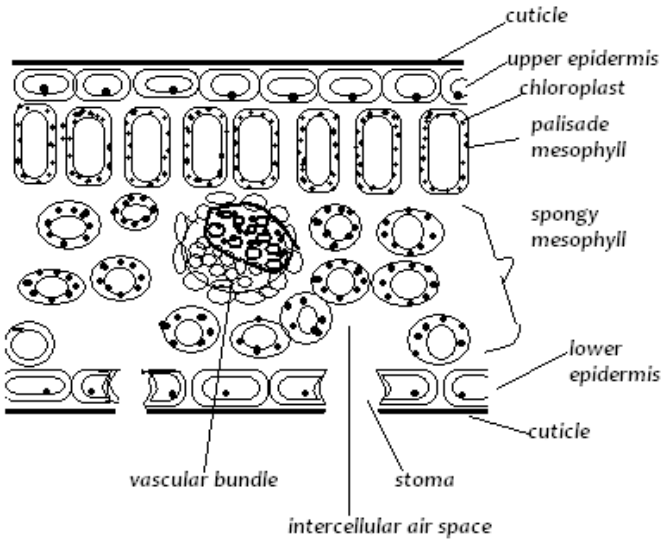

Transvers section of a leaf showing internal structure

Epidermis

- This is the outer layer of cells, normally one cell thick.
- It is found in both the upper and lower leaf surfaces.
- The cells are arranged end to end.
- The epidermis offers protection and maintains the shape of the leaf.
- It is covered by a layer of cuticle which reduces evaporation.

Leaf Mesophyll

Consists of the palisade layer, next to upper epidermis, and the spongy layer next to the lower epidermis.

Palisade Mesophyll Layer

The cells are elongated and arranged close to each other leaving narrow air spaces.
These contain numerous chloroplasts and are the main photosynthetic cells.
In most plants, the chloroplast are distributed fairly uniformly throughout the cytoplasm.
In certain plants growing in shaded habitats in dim light, most chloroplasts migrate to the upper region of the palisade cells in order to maximise absorption of the limited light available.

Spongy Mesophyll Layer

- The cells are spherical in shape.
- They are loosely arranged, with large intercellular spaces between them.
- The spaces are airfilled and are linked to the stomatal pores.
- The spongy mesophyll cells have fewer chloroplasts than the palisade mesophyll cells.

Vascular Bundles

- These are made up of the xylem and the phloem tissues.
- The xylem transports water and mineral salts to the leaves.
- The phloem transports food manufactured in the leaf to the other parts of the plant and from storage organs to other parts.

Adaptations of Leaf for Photosynthesis

- Presence of veins with vascular bundles.

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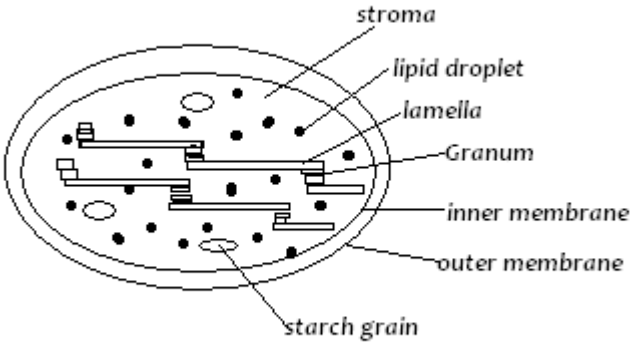
Xylem vessels transport water for photosynthesis.

- Phloem transports manufactured food from leaves to other parts of the plant.
- Leaf lamina is thin to allow for penetration of light over short distance to reach photosynthetic cells.
- Broad lamina provides a large surface area for absorption of light and carbon (IV) oxide.
- Transparent cuticle and epidermal layer allow light to penetrate to mesophyll cells.
- Palisade cells are close to the upper epidermis for maximum light absorption.
- Presence of numerous chloroplasts in palisade mesophyll traps maximum light.
- Chloroplast contain chlorophyll that traps light energy.
- Spongy mesophyll layer has large intercellular air spaces allowing for gaseous exchange.
- Presence of stomata for efficient gaseous exchange (entry of carbon (IV) oxide into leaf and exit of oxygen).
- Mosaic arrangement of leaves to ensure no overlapping of leaves hence every leaf is exposed to light.

Structure and Function of Chloroplasts

- Chloroplasts are large organelles (5 um in diameter) found in the cytoplasm of green plant cells.
- They are visible under the light microscope.
- They contain chlorophyll, a green pigment and other carotenoids which are yellow, orange and red in colour.
- Certain plants have red or purple leaves due to abundance of these other pigments.
- Chlorophyll absorbs light energy and transforms it into chemical energy.
- The other pigments absorb light but only to pass it onto chlorophyll.



Structure of Chloroplast

- The wall of chloroplast consists of an outer and an inner membrane.
- The two make up the chloroplast envelop.
- Inner membrane encloses a system of membranes called lamellae.
- At intervals, the membranes form stacks of fluid filed sacs known as grana (singular granum).
- Chloroplast and other pigments are attached to the grana.

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- In between the lamellae is a gel-like stroma, that contains starch grains and lipid droplets.
- Enzymes for the dark stage reaction (light independent stage) are embedded in the stroma.
- Enzymes for the light dependent stage occur in the grana.

Functions

- .Absorption of light by chlorophyll and other pigments.
- Light stage of photosynthesis occurs on the grana. (transformation of light energy to chemical energy.)
- Carbon fixation to form carbohydrate takes place in the stroma which has enzymes for dark stage of photosynthesis.

Process of Photosynthesis

- Photosynthesis involves a series of chemical reactions, all of which take place inside chloroplasts.
- A general equation for photosynthesis is:

$$\text{Carbon (IV)Oxide+Water} \xrightarrow[\text{chlorophyll}]{\text{light energy}} \text{Glucose+Oxygen}$$

$$6CO_2 + 6H_2O \xrightarrow[\text{chlorophyll}]{\text{light}} C_6H_{12}O + 6O_2$$

- The reaction occurs in two main phases or stages.
- The initial state requires light and it is called the light dependent stage or simply light stage.
- It takes place on the lamellae surfaces.
- Its products are used in the dark stage.
- The dark stage does not require light although it occurs in the light and is called light independent stage.

Light-Stage

- Two reactions take place that produce raw materials for the dark stage:
- Light energy splits the water molecules into hydrogen and oxygen.
- This process is called photolysis.
- The hydrogen is taken up by a hydrogen acceptor called Nicotinamide adenine dinucleotide phosphate (NADP) while oxygen is released as a by-product.

$$2H_2O_{(l)} \xrightarrow[\text{photolysis}]{\text{light energy}} 4H + O_2$$

- Light energy strikes the chlorophyll molecules and sets in motion a series of reactions resulting in the production of a high energy molecule called adenosine triphophate (ATP).

Dark Stage

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- This stage involves the fixation of carbon i.e. the reduction of carbon (IV) oxide by addition of hydrogen to form carbohydrate.
- It uses the products formed during the light stage.
- *ATP*
  Carbon + Hydrogen --- Carbohydrates
- (IV) oxide
- The synthesis of carbohydrates does not take place in a simple straight line reaction as shown in the equation above.
- It involves a series of steps that constitute what is known as the Calvin cycle.
- Carbon (IV) oxide is taken up by a compound described as a carbon (IV) oxide acceptor.
- This is a 5-carbon compound known as ribulose biphosphate and a six carbon compound is formed which is unstable and splits into two three-carbon compounds.
- Hydrogen from the light reaction is added to the three carbon compound using energy (ATP) from the light reaction.
- The result is a three carbon (triose) sugar, (phosphoglycerate or PGA).
- This is the first product of photosynthesis.
- Glucose, other sugars as well as starch are made from condensation of the triose sugar molecules.
- The first product is a 3-carbon sugar which condenses to form glucose (6-C sugar).
- From glucose, sucrose and eventually starch is made.
- Sucrose is the form in which carbohydrate is transported from the leaves to other parts of the plant.
- Starch is the storage product.
- Other substances like oils and proteins are made from sugars.
- This involves incorporation of other elements e.g. nitrogen, phosphorus and sulphur.

Factors Influencing Photosynthesis

- Certain factors must be provided for before photosynthesis can take place.
- The rate or amount of photosynthesis is also influenced by the quantity or quality of these same factors.

Carbon(IV) Oxide Concentration

- Carbon (IV) oxide is one of the raw materials for photosynthesis.
- No starch is formed when leaves are enclosed in an atmosphere without carbon (IV) oxide.
- The concentration of carbon (IV) oxide in the atmosphere remains fairly constant at about 0.03% by volume.
- However, it is possible to vary the carbon (IV) oxide concentration under experimental conditions.
- Increasing the carbon (IV) oxide concentration up to 0.1 % increases the rate of photosynthesis.
- Further increase reduces the rate.

Light Intensity

- Light supplies the energy for photosynthesis.
- Plants kept in the dark do not form starch.
- Generally, increase in light intensity up to a certain optimum, increases the rate of photosynthesis.
- The optimum depends on the habitat of the plant.
- Plants that grow in shady places have a lower optimum than those that grow in sunny places.

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Water

- Water is necessary as a raw material for photosynthesis.
- The amount of water available greatly affects the rate of photosynthesis.
- The more water available, the more the photosynthetic rate, hence amount of food made.
- Effect of water on photosynthesis can only be inferred from the yield of crops.
- It is the main determinant of yield (limiting factor in the tropics).

Temperature

- The reactions involved in photosynthesis are catalysed by a series of enzymes.
- A suitable temperature is therefore necessary.
- The optimum temperature for photosynthesis in most plants is around 30"C.
- This depends on the natural habitat of the plant.
- Some plants in temperate regions have 20°C as their optimum while others in the tropics have 45°C as their optimum temperature.
- The rate of photosynthesis decreases with a decrease in temperature below the optimum.
- In most plants, photosynthesis stops when temperatures approach 0°C although some arctic plant species can photosynthesise at -2°C or even -3°C.
- Likewise, increase in temperature above the optimum decreases the rate and finally the reactions stop at temperatures above 40°c due to enzyme denaturation.
- However, certain algae that live in hot springs e.g. *Oscilatoria* can photosynthesise at 75°C

Chlorophyll

- Chlorophyll traps or harnesses the energy from light.
- Leaves without chlorophyll do not form starch.

Chemical Compounds Which Constitute Living Organisms

- All matter is made up of chemical elements, each of which exists in the form of smaller units called atoms.
- Some of the elements occur in large amounts in living things.
- These include carbon, oxygen, hydrogen, nitrogen, sulphur and phosphorus.
- Elements combine together to form compounds.
- Some of these compounds are organic.
- Organic compounds contain atoms of carbon combined with hydrogen and they are usually complex.
- Other compounds are inorganic.
- Most inorganic compounds do not contain carbon and hydrogen and they are usually less complex.
- Cells contain hundreds of different classes of organic compounds.
- However, there are four classes of organic compounds found in all cells.
- These are: *carbohydrates, lipids, proteins and nucleic acids.*

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Carbohydrates

- Carbohydrates are compounds of carbon, hydrogen and oxygen.
- Hydrogen and oxygen occur in the ratio of 2: 1 as in water.
- Carbohydrates are classified into three main groups: monosaccharides, disaccharides and polysaccharides.

Monosaccharides

- These are simple sugars.
- The carbon atoms in these sugars form a chain to which hydrogen and oxygen atoms are attached.
- Monosaccharides are classified according to the number of carbon atoms they possess.
- The most common monosaccharides are:
  - Glucose - found free in fruits and vegetables.
  - Fructose - found free in fruits and in bee honey.
  - Galactose - found combined in milk sugar.
- The general formula for these monosaccharides is $(CH_2O)n$ where n is 6.
- They have the same number of carbon, hydrogen and oxygen molecules i.e. $C_6H_{12}O_6$.

Properties of Monosaccharides

- They are soluble in water.
- They are crystallisable.
- They are sweet.
- The are all reducing sugars.
- This is because they reduce blue copper (II) sulphate solution when heated to copper oxide which is red in colour and insoluble.

Functions of Monosaccharides

- They are oxidised in the cells to produce energy during respiration.
- Formation of important biological molecules e.g. deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).
- Some monosaccharides are important metabolic intermediates e.g. in photosynthesis and in respiration.
- Monosaccharides are the units from which other more complex sugars are formed through condensation.

Disaccharides

- These contain two monosaccharide units.
- The chemical process through which a large molecule (e.g. a disaccharide) is formed from smaller molecules is called condensation and it involves loss of water.

Common examples of disaccharides include sucrose, maltose and lactose.

Monosaccharide units	*Disaccharides*
Glucose+fructose	Sucrose(cane sugar)
Glucose+glucose	Maltose(malt sugar)
Glucose+galactose	Lactose(milk sugar)

- Disaccharides are broken into their monosaccharide units by heating with dilute hydrochloric acid.
- This is known as hydrolysis and involves addition of water molecules.

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- The same process takes place inside cells through enzymes.

Sucrose+water_--hydrolysis----------glucose+fructose

Properties of Disaccharides

- Sweet tasting.
- Soluble in water.
- Crystallisable.
- Maltose and lactose are reducing sugars while sucrose is non-reducing sugar.
- Sucrose is the form in which carbohydrate is transported in plants:
- This is because it is soluble andjchernically stable.
- Sucrose is a storage carbohydrate in some plants e.g. sugar-cane and sugar-beet.
- Disaccharides are hydrolysed to produce monosaccharide units which are readily metabolised by cell to provide energy.

Polysaccharides

- If many monosaccharides are joined together through condensation, a polysaccharide is formed.
- Polysaccharides may consist of hundreds or even thousands of monosaccharide units.
- Examples of polysaccharides:
  - Starch - storage material in plants.
  - Glycogen is a storage carbohydrate in animals like starch, but has longer chains.
  - Inulin - a storage carbohydrate in some plants e.g. Dahlia.
  - Cellulose - structural carbohydrate in plants.
  - Chitin - forms exoskeleton in arthropods.

Importance and Functions of Polysaccharides

- They are storage carbohydrates - starch in plants glycogen in animals.
- They are hydrolysed to their contituent monosaccharide units and used for respiration. .
- They form structural material e.g. cellulose makes cell walls.
- Cellulose has wide commercial uses e.g.
  - Fibre in cloth industry.
  - Cellulose is used to make paper.
- Carbohydrates combine with other molecules to form important structural compounds in living organisms.
- Examples are:

  Pectins: Combine with calcium ions to form calcium pectate.

  Chitin: Combine with (NH) group. Makes the exoskeleton of arthropods, and walls of fungi.

Lipids

- These are fats and oils.
- Fats are solid at room temperature while oils are liquid.
- They are made up of carbon, oxygen and hydrogen atoms.
- The structural units of lipids are fatty acids and glycerol.

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- Fatty acids are made up of hydrocarbon chain molecules with a carboxyl group (-COOH) at one end.
- In the synthesis of a lipid, three fatty acid molecules combine with one glycerol molecule to form a triglyceride.
- Three molecules of water are lost in the process.
- This is a condensation reaction and water is given off.
- Lipids are hydrolysed e.g. during digestion to fatty acids and glycerol, water is added.

condensation

Glycerol + 3 Fatty *hydrolysis* Lipid + Water acids

Properties

- Fats are insoluble in water but dissolve in organic solvents e.g. in alcohols.
- They are chemically inactive, hence used as food storage compounds.

Functions of Lipids

- ***Structural materials*** - as structural material they make up the cell membrane.
- ***Source of energy*** - they are energy rich molecules.
  One molecule of lipid provides more energy than a carbohydrate molecule.
- ***Storage compound*** - They are stored as food reserves in plants.
- In animals e.g. mammals, all excess food taken is converted to fats which are stored in adipose tissue, and around internal organs such as the heart and kidneys.
- ***Insulation*** - They provide insulation in animals living in cold climates.
  A lot of fat is stored under the skin e.g. blubber in seals.
- ***Protection*** - Complex lipids e.g. wax on leaf surfaces protects the plant against water-loss and overheating.
- **Fats stored around some internal organs acts as shock absorbers**, thus protecting the organs.
- ***Source of Metabolic Water*** **-:-**lipids when oxidised produce metabolic water which supplements water requirements in the body.
  Desert animals e.g. the camel accumulate large quantities of fat in the hump which when oxidised releases metabolic water.

Proteins

- Proteins are the most abundant organic compounds in cells and constitute 50% of total dry weight.
- Proteins are compounds which are made up of carbon, hydrogen, nitrogen, oxygen and sometimes sulphur and phosphorus.
- The structural units of proteins are amino acids.
- The nature of a protein is determined by the types of amino acids it is made of.
- There are about 20 common amino acids that make up proteins.

Essential and Non-Essential Amino Acids

- Essential amino acids are those which cannot be synthesised in the body of an organism and must therefore be provided in the diet.
- There are ten amino acids which are essential for humans.

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- These are valine, leucine, phenylalanine, lysine, tryptophan, isoleucine, methionine, threonine, histidine and arginine.
- Non-essential amino acids are those which the body can synthesise and therefore need not be available in the diet.
- There are ten of them.
- These are glycine, alanine, glutamic acid, aspartic acid, serine, tyrosine, proline, glutamine, arginine and cysteine.
- Proteins are essential in the diet because they are not stored in the body.
- Excess amino acids are deaminated.

Formation of Proteins

- Proteins are made up of many amino acid units joined together through peptide bonds.
- When two amino acids are joined together a dipeptide is formed.
- The chemical process involved is called condensation and a molecule of water is eliminated .
- When many amino acids are joined together a polypeptide chain is formed.
- The nature of a particular protein depends on the types, number and sequence of amino acids from which it is made.

Functions of Proteins

- ***As structural materials proteins-***
  - Are the basic building structures of protoplasms.
  - Proteins in conjunction with lipid form the cell membrane.

  Examples of structural proteins include:

  - Keratin (in hair, nails, hoofs, feathers and wool)
  - Silk in spider's web.
  - Elastin forms ligaments that join bones to each other.
- ***Protective proteins.***
  - Antibodies that protect the body against foreign antigens.
  - Fribrogen and thrombin are involved in clot formation, preventing entry of micro-organisms when blood vessel is cut.

- ***As functional chemical compounds.***
  - Examples are hormones and enzymes that act as regulators in the body.
  - Respiratory pigments.
  - Examples are haemoglobin that transports oxygen in the blood and myoglobin that stores up oxygen in muscles.
  - Contractile proteins - make up muscles, i.e. myosin and actin.
  - Proteins combine with other chemical groups to form important substances e.g. mucin in saliva.
- ***Source of energy.***

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- Proteins are a source of energy in extreme conditions when carbohydrates and fats are not available e.g. in starvation.

Enzymes

- Enzymes are biological catalysts that increase the rate of chemical reaction in the body.
- They are all produced inside cells.
- Some are intracellular and they catalyse reactions within the cells .
- Others are extracellular and are secreted out of the cells where they work. e.g. digestive enzymes.

Properties of Enzymes

- Enzymes are protein in nature.
- Enzymes are specific to the type of reaction they catalyse.
- This is referred to as substrate specificity.
- Enzymes work in very small amounts.
- They remain unchanged after the reaction.
- They catalyse reversible reactions.
- They work very fast (high turnover numbers) e.g. the enzyme catalase works on 600 thousand molecules of hydrogen peroxide in one second.

Naming of enzymes

Enzymes are named by adding the suffix -ase to:

- *Name of substrate that they work on e.g.*
  - carbohydrates - carbohydrases e.g.sucrase.
  - Starch (amylose) - amylase
  - Protein - proteinase (protease)
  - Lipids -lipases
- *Type of chemical reaction catalised e.g.*
  - Oxidation - oxidase
  - Reduction - reductase
  - Hydrolysis - hydrolase

Factors Affecting Enzyme Action

Temperature

- Enzymes are sensitive to temperature changes.
- Generally, the rate of an enzymecontrolled reaction doubles with every $10^{O}C$ increase in temperature.
- However, temperatures above 40°C do not favour enzyme reaction.
- This is because enzymes are denatured by high temperatures.

pH

- Every enzyme has a particular pH range over which it works best.
- Some enzymes work best in acidic media while others function better in alkaline media.
- Many enzymes function well under neutral conditions.

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Enzyme Concentration

- Under conditions where the substrate is in excess, the rate of an enzyme-controlled reaction increases as the enzyme concentration is increased.

Substrate Concentration

- If the concentration of the substrate is increased while that of the enzyme remains constant, the rate of the reaction will increase for sometime and then become constant.
- Any further increase in substrate concentration will not result in corresponding increase in the rate of the reaction.

Enzyme Inhibitors

- These are substances that either compete with substrates for enzyme active sites or combine with enzymes and hence they inhibit the enzyme reaction.
- e.g. certain drugs, cyanide and nerve gas.

Co-factors

- Most enzymes require the presence of other compounds known as co-factors which are non-proteins.
- There are three groups of co-factors.
- Inorganic ions - e.g. iron, magnesium, copper and zinc.
- Complex organic molecules known as prosthetic groups are attached to the enzyme
- e.g. flavin adenine dinucleotide (FAD) derived from vitamin B2 (riboflavin).
- Co-enzymes e.g. coenzyme A is involved in respiration.
- All co-enzymes are derived from vitamins.

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Nutrition in Animals=Heterotrophism

Meaning and Types of Heterotrophism

- This is a mode of nutrition whereby organisms feed on complex organic matter from other plants or animals.
- All animals are heterotrophs.
- Their mode of feeding is also said to be holozoic to distinguish it from other special types of heterotrophic nutrition namely:
    - saprophytism
    - parasitism.
- Saprophytism/saprotrophysim- occurs in most fungi and some forms of bacteria.
- Saprophytes feed on dead organic matter and cause its decomposition or decay.
- Parasitism is a mode of feeding whereby one organism called the parasite feeds on or lives in another organism called the host and harms it.

Modes of Feeding in Animals

- Animals have developed various structures to capture and ingest food.
- The type of structures present depend on the method of feeding and the type of food.
- **Carnivorous** animals feed on whole animals or portions of their flesh.
- **Herbiverous** animals feed on plant material.
- ***Omnivorous*** animals feed on both plants and animal materials.

Feeding in Mammals

- The jaws and teeth of mammals are modified according to the type of food eaten.
- Mammals have different kinds of teeth.
- Each type of teeth has a particular role to play in the feeding process.

Feeding in Mammals

- The jaws and teeth of mammals are modified according to the type of food eaten.
- Mammals have different kinds of teeth.
- Each type of teeth has a particular role to play in the feeding process.
- This condition is described as heterodont.
- The teeth of reptiles and amphibians are all similar in shape and carry out the same function.
- They are said to be homodont.

Types of Mammalian Teeth

- Mammals have four kinds of teeth.
- The incisors are found at the front of the jaw.

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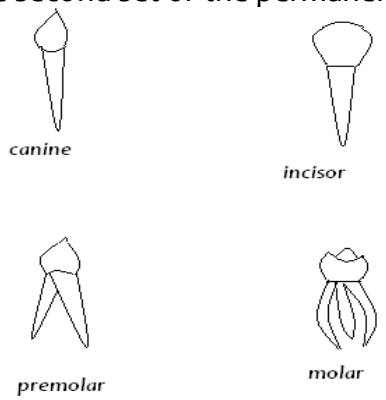
- They are sharp-edged and are used for biting.
- The canines are located at the sides of the jaw.
- They are pointed and are used for tearing and piercing.
- The premolars are next to the canines and the molars are at the back of the jaw.
- Both premolars and molars are used for crushing and grinding.
- Teeth are replaced only once in a lifetime.
- The first set is the milk or deciduous teeth.
- These are replaced by the second set or the permanent teeth.



Different types of teeth in humans

- Dentition refers to the type of teeth, the number and their arrangement in the jaw.
- A dental formula shows the type and number of teeth in each half of the jaw.
- The number of teeth in half of the upper jaw is represented above a line and those on the lower jaw below the line.
- The first letter of each type of teeth is used in the formula i.e. i =incisors, c = canines, pm = premolars and m = molars.
- The total number is obtained by multiplying by two (for the two halves of each jaw).

Adaptation of Teeth to Feeding

- In general, incisors are for cutting, canines for tearing while premolars and molars are for grinding.
- However, specific modifications are observed in different mammals as an adaptation to the type of food they eat.
- Teeth of Herbivores
- Incisors are long and flat with a sharp chisellike edge for cutting.
- The enamel coating is thicker in front than at the back so that as the tooth wears out, a sharp edge is maintained.
- Canines are reduced or absent.
- If absent, the space left is called the diastema.
- The diastema allows the tongue to hold food and push it to the grinding teeth at the back of the mouth.

Premolars and molars:

- These are transversely ridged.
- The ridges on the upper teeth fit into grooves on the lower ones.
- This gives a sideways grinding surface.
- The teeth of herbivores have open roots i.e., wide opening into the pulp cavity.

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- This ensures a continued adequate supply of food and oxygen to the tooth.
- In some herbivores, such as rabbits and elephants, the incisors continue to grow throughout life.

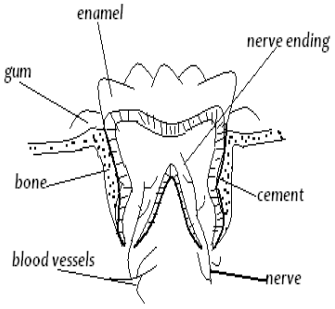
Teeth of Carnivores

- Incisors are reduced in size and pointed.
- They are well suited for grasping food and holding prey.
- Canines are long, pointed and curved.
- They are used for piercing and tearing flesh as well as for attack and defence.
- ***Premolars and molars:*** In general, they are long and longitudinally ridged to increase surface area for crushing .
- ***Carnassial Teeth:*** These are the last premolars on the upper jaw and the first molars on the lower one.
- They are enlarged for cutting flesh.
- They act as a pair of shears.
- They also crush bones.
- The teeth of carnivores have closed roots i.e., only a very small opening of the pulp cavity to allow food and oxygen to keep teeth alive.
- Once broken, no re-growth can take place.

Teeth of Omnivores

- Incisors have a wide surface for cutting.
- Canines are bluntly pointed for tearing.
- Premolars and molars have cusps for crushing and grinding.
- The premolars have two blunt cusps while the molars have three to four.

Internal Structure of tooth



Internal structure of a tooth

The tooth consists of two main parts:

Crown: The portion above the gum; it is covered by the enamel.

Root: The portion below the gum; it is covered by the cement.

- The tooth has two roots.

Neck: Is the region at the same level with the gum.

- It forms the junction between the crown and the root.
- It is covered by enamel. Incisors and canines have one root only.

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- Premolars have one or two roots while molars have two to three roots each.
- Internally, the bulk of the tooth is made up of dentine which consists of living cells and extends to the root.
- It is composed of calcium salts, collagen and water.
- It is harder than bone but wears out with use.
- This is why it is covered by enamel which is the hardest substance in a mammal's body.

Pulp Cavity: Contains blood vessels which provide nutrients to the dentine and remove waste products.

- It also contains nerve endings which detect heat, cold and pain.

Cement: Fixes the tooth firmly to the jaw bone.

Common Dental Diseases

Dental Carries

- *Dental carries* are the holes or cavities that are formed as acid corrodes enamel and eventually the dentine.
- ***Causes***
  - This is caused by bacteria acting on the food left between teeth and on the cusp.
  - Acids are formed that eventually corrode the enamel.
  - The pulp cavity is eventually reached.
  - A lot of pain is experienced then.
  - The bacteria then infect the pulp cavity and the whole tooth decays.
- ***Treatment***
  - Treatment depends on the extent of the dental caries:
  - Extraction of Tooth.
  - Filling - this involves replacing the dentine with amalgam, a mixture of hard elements e.g. silver and tin.
  - Root Canal Treatment - This involves surgery and reconstruction.
  - It saves severely damaged teeth.
  - The nerves in the root canal are surgically severed.
  - The tooth is cleaned and filled up with amalgam.

Periodontal Diseases

- These are diseases of the gum.
- The gum becomes inflamed, and starts bleeding.
- Progression of the disease leads to infection of the fibres in the periodontal membranes and the tooth becomes loose.
- This condition is known as pyorrhoea.
- The diseases are caused by poor cleaning of the teeth.
- The accumulation of food particles leading to formation of plaque, lack of adequate vitamin A and C in the diet.

Treatment

- Nutrition - by taking adequate balanced diet rich in vitamins A and C.
- Antibiotics are used to kill bacteria.
- Anti-inflamatory drugs are given.
- Antiseptic is prescribed to use in cleaning the mouth daily to prevent further proliferation of bacteria.
- The plaque is removed-drilled away - a procedure known as scaling.

Care of Teeth

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In order to maintain healthy teeth the following points should be observed:

- A proper diet that includes calcium and vitamins, particularly vitamin D is essential.
- The diet should also contain very small quantities of fluorine to strengthen the enamel.
- Large quantities of fluorine are harmful.
- The enamel becomes brown, a condition known as dental flourosis.
- Chewing of hard fibrous foods like carrots and sugar cane to strengthen and cleanse the teeth.
- Proper use of teeth e.g. not using teeth to open bottles and cut thread.
- Regular and thorough brushing of teeth after meals.
- Dental floss can be used to clean between the teeth.
- Not eating sweets and sugary foods between meals.
- Regular visits to the dentist for checkup.
- Washing the mouth with strong salt solution or with any other mouth wash with antiseptic properties.

Digestive System and Digestion in Humans

- Organs that are involved with feeding in humans constitute the digestive system.

Digestive System and Associated Glands

- Human digestive system starts at the mouth and ends at the anus.
- This is the alimentary canal.
- Digestion takes place inside the lumen of the alimentary canal.
- The epithelial wall that faces the lumen has mucus glands (goblet cells).
- These secrete mucus that lubricate food and prevent the wall from being digested by digestive enzymes.
- Present at specific regions are glands that secrete digestive enzymes.
- The liver and pancreas are organs that are closely associated with the alimentary canal.
- Their secretions get into the lumen and assist in digestions.

Digestive system consists of:

- Mouth.
- Oesophagus.
- Stomach.
- Small intestines - consist of duodenum, the first part next to the stomach, ileum - the last part that ends up in a vestigial caecum and appendix which are nonfunctional.
- Large intestines consist of: colon and rectum that ends in the anus.

Ingestion, Digestion and Absorption

- Feeding in humans involves the following processes:
- *Ingestion*: This is the introduction of the food into the mouth.
- *Digestion:* This is the mechanical and chemical breakdown of the food into simpler, soluble and absorbable units.
- *Absorption:* Taking into blood the digested products.

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- *Assimilation:* Use of food in body cells.
- Mechanical breakdown of the food takes place with the help of the teeth.
- Chemical digestion involves enzymes.

Digestion in the Mouth

- In the mouth, both mechanical and chemical digestion takes place.
- Food is mixed with saliva and is broken into smaller particles by the action of teeth.
- Saliva contains the enzyme amylase.
- It also contains water and mucus which lubricate and soften food in order to make swallowing easy.
- Saliva is slightly alkaline and thus provides a suitable pH for amylase to act on cooked starch, changing it to maltose.
- The food is then swallowed in the form of semisolid balls known as boluses.
- Each bolus moves down the oesophagus by a process known as peristalsis.
- Circular and longitudinal muscles along the wall of the alimentary canal contract and relax pushing the food along.

Digestion in the Stomach

- In the stomach, the food is mixed with gastric juice secreted by gastric glands in the stomach wall.
- Gastric juice contains pepsin, rennin and hydrochloric acid.
- The acid provides a low pH of 1.5-2.0 suitable for the action of pepsin.
- Pepsin breaks down protein into peptides.
- Rennin coagulates the milk protein casein.
- The stomach wall has strong circular and longitudinal muscles whose contraction mixes the food with digestive juices in the stomach.

Digestion in the Duodenum

- In the duodenum the food is mixed with bile and pancreatic juice.
- Bile contains bile salts and bile pigments.
- The salts emulsify fats, thus providing a large surface area for action of lipase.
- Pancreatic juice contains three enzymes:
  - Trypsin which breaks down proteins into peptides and amino acids,
  - Amylase which breaks down starch into maltose, and
  - Lipase which breaks down lipids into fatty acids and glycerol.
- These enzymes act best in an alkaline medium which is provided for by the bile.

Digestion in ileum

- Epithelial cells in ileum secrete intestinal juice, also known as succus entericus.
- This contains enzymes which complete the digestion of protein into amino acids, carbohydrates into monosaccharides and lipids into fatty acids and glycerol.

Absorption

- This is the diffusion of the products of digestion into the blood of the animal.
- It takes place mainly in the small intestines though alcohol and some glucose are absorbed in the stomach.

The ileum is adapted for absorption in the following ways:

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- It is highly coiled.
- The coiling ensures that food moves along slowly to allow time for its digestion and absorption.
- It is long to provide a large surface area for absorption.
- The epithelium has many finger-like projections called villi (singular villus).

- They greatly increase the surface area for absorption.
- Villi have microvilli that further increase the surface area for absorption.
- The wall of villi has thin epithelial lining to facilitate fast diffusion of products of digestion.
- Has numerous blood vessels for transport of the end products of digestion.
- Has lacteal vessels; for absorption of fatty acids and glycerol and transport of lipids.

Absorption of Glucose and Amino Acids

- Glucose and other monosaccharides as well as amino acids are absorbed through the villi epithelium and directly into the blood capillaries.
- First they are carried to the liver through the hepatic portal vein, then taken to all organs via circulatory system.

Absorption of Fatty Acids and Glycerol

- Fatty acids and glycerol diffuse through the epithelial cells of villi and into the lacteal.
- When inside the villi epithelial cells, the fatty acids combine with glycerol to make tiny fat droplets which give the lacteal a milky appearance.
- The lacteals join the main lymph vessel that empties its contents into the bloodstream in the thoracic region.
- Once inside the blood, the lipid droplets are hydrolysed to fatty acids and glycerol.

Absorption of Vitamins and Mineral Salts

- Vitamins and mineral salts are absorbed into the blood capillaries in' the villi. Water is mainly absorbed in the colon.
- As a result the undigested food is in a semi-solid form (faeces) when it reaches the rectum.
- ***Egestion*:** This is removal of undigested or indigestible material from the body. Faeces are temporarily stored in the rectum then voided through the anus. Opening of the anus is controlled by sphincter muscles
- ***Assimilation:*** This is the incorporation of the food into the cells where it is used for various chemical processes.

Carbohydrates

- used to provide energy for the body.
- Excess glucose is converted to glycogen and stored in the liver and muscles.
- Some of the excess carbohydrates are also converted into fat in the liver and stored in the adipose tissue' (fat storage tissue), in the mesenteries and in the connective tissue under the skin, around the heart and other internal organs.

Proteins

- Amino acids are used to build new cells and repair worn out ones.
- They are also used for the synthesis of protein compounds.

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- Excess amino acids are de-aminated in the liver.
- Urea is formed from the nitrogen part.
- The remaining carbohydrate portion is used for energy or it is converted to glycogen or fat and stored.

Lipids

- Fats are primarily stored in the fat storage tissues.
- When carbohydrates intake is low in the body, fats are oxidised to provide energy.
- They are also used as structural materials e.g. phospholipids in cell membrane. They act as cushion, protecting delicate organs like the heart.
- Stored fats under the skin act as heat insulators.

Summary of digestion in humans

Digestive gland and juice produced	*pH*	*Contents*	*Food substance*	*Products*	*Notes*
Salivary glands (Saliva) p	7.4	Water, and salts			Soften and lubricate food, provide neutral pH.
		Amylase	Starch	Maltose	Glucose if food stays longer mouth.
Stomach (Gastric Juice)	1'.8	Hydrochloric acid	Nucleo-proteins	Nucleic + protein	Not an enzyme but the nuclear proteins. 1. Kills micro-organisms. 2. Provides acidic medium. 3. Activates enzyme pepsinogen and protennin.
		Rennin	Milk protein	Curd coagulated milk (casein)	abundant in infants secreted prorennin.
		Pepsin	Protein	Peptones	Secreted as pepsinogen
Pancrease (Pancreati juice)	8.8	Trypsin	Protein	Peptones	Secreted as trypsinogen activated by enterokinase to trypsin
		Chymotrypsin	Peptones, casein	Amino acids	Secreted as chymotrypsin activated to trypsin.
		Amylase	Starch glycogen	Maltose Maltose	
		Lipase	Lipids	Fatty acids and	PH in duodenum lowered to by acid from stomach
		Sodium bicarbonate			Provides alkaline conditions
		Peptidases (erepsin)	Peptides	Amino acids	Erepsin contains a mixture peptidases
		Invertase		Fructose +	

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Ileum (succus entericus)	8.3	made of sucrase	Sucrose	glucose	
		Lactase	Lactose	Galactose + glucose	
		Maltase	Maltose	Glucose	
		Lipase	Lipids	Fatty acids and	
		Enterokinase			Activates trypsinogen to trypsin

Importance of Vitamins, Mineral Salts, Roughage and Water in Human Nutrition

Vitamins

- These are organic compounds that are essential for proper growth, development and functioning of the body.
- Vitamins are required in very small quantities.
- They are not stored and must be included in the diet.
- Vitamins Band C are soluble in water, the rest are soluble in fat.
- Various vitamins are used in different ways.

Mineral Salts

- Mineral ions are needed in the human body.
- Some are needed in small amounts while others are needed in very small amounts (trace).
- All are vital to human health.
- Nevertheless, their absence results in noticeable mulfunction of the body processes.

Water

- Water is a constituent of blood and intercellular fluid.
- It is also a constituent of cytoplasm.
- Water makes up to 60-70% of total fresh weight in humans.
- No life can exist without water.

Functions of Water

- Acts as a medium in which chemical reactions in the body takes place.
- Acts as a solvent and it is used to transport materials within the body.
- Acts as a coolant due to its high latent heat of vaporisation.
- Hence, evaporation of sweat lowers body temperature.
- Takes part in chemical reactions i.e. hydrolysis.

Vitamins, sources, uses and the deficiency disease resulting from their absence in diet

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Name of Vitamin	*Sources*	*Uses in body*	*Deficiency disease(s)/Disorder*
A (retinol) Soluble	Liver, egg-yolk, carrots, milk, spinach	Synthesis of rhodopsin, Control of growth of epithelium	Hardening of cornea of the eye (xerophthalmia), poor night resistance to diseases of skin and is reduced.
B, (Thiamine)	Yeast, whole grain, liver, kidney, beans, meat, spinach	Formation f the enzyme carboxylase important in conversion of pyruvic respiration.	Beriberi - swelling of the feet; slowing of heartbeat and intestinal disorder.
B2 (Riboflavin)	Whole grain, eggs, milk, groundnuts, cheese, yeast	Formation of f1avoproteins that form enzymes and for	Sores on tongue surface and comers of the mouth.
B3 (Nicotinic	Liver, kidneys, milk, yeast whole grain.	Makes co-enzyme 1 and 2 (NAD & NAD P) It is also co-enzyme A needed in respiration.	Pellagra, inflammation of nervous disorders leading to
B, (Pantothenic acid)	In most natural foods	Forms parts of co-enzyme A.	Poor co-ordination of muscles nervous muscle cramp.
B6 (Pryidoxine) water soluble	Eggs, kidneys, whole grain, vegetables.	Makes a co-enzyme for amino acids metabolism.	Irritability, depression, dermatitis

4

Potassium	Milk, eggs, liver, vegetables, bananas.	In intracellular body fluids buffer and for nerve transmission.	Nervour transmission interfered with.
Chloride p	Table salt, sea foods.	Present in tissue fluid. water balance essential for digestion. Constituent of hydrochloric acid.	
Magnesium	Green vegetables.	Also needed as a co-factor respirator enzymes. Muscle contraction.	
Iodine	Iodised table salt and food.	Constituent of the hormone thyroxine that controls metabolism.	In young animals leads cretinism. Simple goitre adults.
Manganese	Eggs, milk, fish.	Activates certain enzymes.	
Iron	Liver, greet leaves, lean meat, grains, milk.	A constituent of haemoglobin and myoblobin.	Anaemia.

Potassium	Milk, eggs, liver, vegetables, bananas.	In intracellular body fluids buffer and for nerve transmission.	Nervour transmission interfered with.
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Sulphur	Protein foods	A constituent of some needed in synthesis of certain enzymes and phospholipids in cell membranes.	
Copper		Catalyses use of iron, a constituent of cytochrome oxidase (an enzyme)	Needed in very small amounts.
Cobalt		Influences the use of iron (found in Vitamin ~2)'	Needed in very small amounts.
Zi	Fruits and vegetables Seeds of cucubitaceae	Needed for proper growth [illegible] insulin.	Needed in very small amounts.
Fluorine	Water, fruits and vegetables.	Strengthening of enamel	Needed in small amounts.
Molybdenum	Plant seeds	Activates enzyme system in mucleic acid metabolism.	Very small amounts needed, excess is dangerous.
Chromium		Involved in use of glucose.	Needed in small

Roughage

- Roughage is dietary fibre and it consists mainly of cellulose.
- It adds bulk to the food and provides grip for the gut muscles to enhance peristalsis.
- Roughage does not provide any nutritional value because humans and all animals not produce cellulase enzyme to digest cellulose.
- In herbivores symbiotic bacteria in the gut produce cellulase that digests cellulose.

Factors Determining Energy Requirements in Humans

- ***Age:*** Infants, for instance, need a greater proportion of protein than adults.
- ***Sex:*** males generally require more carbohydrates than females.
- The requirements of specific nutrients for females depends on the stage of development in the life cycle.
- **Adolescent girls** require more iron in their diet; expectant and nursing mothers require a lot of proteins and mineral salts.
- ***State of Health*:** A sick individual requires more of certain nutrients e.g. proteins, than a healthy one.
- ***Occupation:*** An office worker needs less nutrients than a manual worker.

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Balanced Diet

- A diet is balanced when it contains all the body's nutrient requirements and in the right amounts or proportions.

A balanced diet should contain the following:

- Carbohydrates
- Proteins
- Lipids
- Vitamins
- Mineral Salts
- Water
- Dietary fibre or roughage

Malnutrition

- This is faulty or bad feeding where the intake of either less or more than the required amount of food or total lack of some food components.

Deficiency Diseases

- Deficiency diseases result from prolonged absence of certain components in the diet.

Examples are:

Marasmus:

- Lack of enough food reuslts in thin arms and legs,
  - severe loss of fluid,
  - general body wasting
  - sunken eyes.

- ***Kwashiorkor –***
  - Lack of protein in the diet of children.
  - The symptoms of kwashiorkor include wasting of the body, red thin hair, swollen abdomen and scaly skin.
- Other deficiency diseases are due to lack of accessory food factors (vitamins and mineral salts.). Such diseases include *rickets, goitre and anaemia.*
- Treatment of these deficiency diseases is by supplying the patient with the component missing in the diet.

THE END

Practical Activities

- Experiments to show that Carbon (IV) Oxide is necessary for Photosynthesis
- Experiment to Show Effect of Light on Photosynthesis
- Experiment to Show the Effect of Chlorophyll on Photosynthesis
- Experiment To Observe Stomata Distribution in Different Leaves

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- Test for Reducing Sugar
- Test for non-reducing sugar
- Test for Lipids;
  (a) Grease Spot Test

  (b) Emulsion Test
- Test for Proteins -Biuret Test
- Experiment To Investigate Presence of Enzyme in Living Tissue
- Dissection of a Rabbit to show the Digestive System

Transport in plants and animal

TRANSPORT IN PLANTS AND ANIMALS.

Introduction

- Transport is the movement of substances within an organism.
- All living cells require oxygen and food for various metabolic processes.
- These substances must be transported to the cells.
- Metabolic processes in the cells produce excretory products which should be eliminated before they accumulate.
- The excretory products should be transported to sites of excretion.
- Organisms like amoeba are unicellular.
- They have a large surface area to volume ratio.
- The body is in contact with the environment.
- Diffusion is adequate to transport substances across the cell membrane and within the organism.
- Large multi-cellular organisms have complex structure where cells are far from each other hence diffusion alone cannot meet the demand for supply and removal of substances.
- Therefore an elaborate transport system is necessary.

Transport in plants

- Simple plants such as mosses and liverworts lack specialized transport system.
- Higher plants have specialized transport systems known as the vascular bundle.
- Xylem transports water and mineral salts .

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- Phloem transports dissolved food substances like sugars.

Internal structure of roots and root hairs

- The main functions of roots are ;
  - Anchorage
  - absorption.
  - storage
  - gaseous exchange.
- The outermost layer in a root is the piliferous layer.
- This is a special epidermis of young roots whose cells give rise to root hairs.
- Root hairs are microscopic outgrowths of epidermal cells.
- They are found just behind the root tip,
- They are one cell thick for efficient absorption of substances.
- They are numerous and elongated providing a large surface area for absorption of water and mineral salts.
- Root hairs penetrate the soil and make close contact with it.
- Below the piliferous layer is the cortex.
- This is made up of loosely packed, thin walled parenchyma cells.
- Water molecules pass through this tissue to reach the vascuiar bundles.
- In some young plant stems, cortex cells contain chloroplasts.
- The endodermis (starch sheath) is a single layer of cells with starch grains.
- The endodermis has a casparian strip which has an impervious deposit controlling the entry of water and mineral salts into xylem vessels.
- Pericyc1e forms a layer next to the endodermis.
- Next to the pericycle is the vascular tissue.
- In the Dicotyledonous root, xylem forms a star shape in the centre, with phloem in between the arms.
- It has no pith. In monocotyledonous root, xylem alternates with phloem and there is a pith in the centre.

Internal structure of a root hair cell

The Stem

- The main functions of the stem are;
  - support and exposure of leaves and flowers to the environment,
  - conducting water and mineral salts
  - conducting manufactured food from leaves to other parts of the plant.
- In monocotyledonous stems, vascular bundles are scattered all over the stem, while in dicotyledonous stems vascular bundles are arranged in a ring.
- Vascular bundles are continuous from root to stems and leaves.
- The epidermis forms a single layer of cells enclosing other tissues.
- The outer walls of the cells have waxy cuticle to prevent excessive loss of water.
- The cortex is a layer next to the epidermis.
- It has collenchyma, parenchyma and schlerenchyma cells.

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Collenchyma

- Is next to the epidermis and has thickened walls at the corners which strengthen the stem.

Parenchyma

- Cells are irregular in shape, thin walled and loosely arranged hence creating intercellular spaces filled with air.
- They are packing tissues and food storage areas.

Sclerenchyma

- Cells are closely connected to vascular bundles.
- These cells are thickened by deposition of lignin and they provide support to plants**.**

Pith

- Is the central region having parenchyma cells.

Absorption of Water and Mineral Salts Absorption of Water

- Root hair cell has solutes in the vacuole and hence a higher osmotic pressure than the surrounding soil water solution.
- Water moves into the root hair cells by osmosis along a concentration gradient.
- This makes the sap in the root hair cell to have a lower osmotic pressure than the surrounding cells.
- Therefore water moves from root hair cells into the surrounding cortex cells by osmosis.
- The process continues until the water gets into the xylem vessels .

Uptake of Mineral Salts

- If the concentration of mineral salts in solution is greater than its concentration in root hair cell, the mineral salts enter the root hair cell by diffusion.
- If the concentration of mineral salts in the root hair cells is greater than in the soil water, the mineral salts enter the root hairs by active transport.
- Most minerals are absorbed in this way.
- Mineral salts move from cell to cell by active transport until they reach the xylem vessel.
- Once inside the xylem vessels, mineral salts are transported in solution as the water moves up due to root pressure, capillary attraction and cohesion and adhesion forces.

Transpiration

- Transpiration is the process by which plants lose water in the form of water vapour into the atmosphere.
- Water is lost through ***stomata, cuticle and lenticels.***
- ***Stomatal transpiration*:**
  - This accounts for 80-90% of the total transpiration in plants.
  - Stomata are found on the leaves.

- ***Cuticular transpiration:***

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- The cuticle is found on the leaves, and a little water is lost through it.
- Plants with thick cuticles do not lose water through the cuticle.

- ***Lenticular transpiration***
  - Is loss' of water through lenticels.
  - These are found on stems of woody plants.
  - Water lost through the stomata and cuticle by evaporation leads to evaporation of water from surfaces of mesophyll cells .
  - The mesophyll cells draw water from the xylem vessels by osmosis.
  - The xylem in the leaf is continuous with xy lem in the stem and root.

Structure and function of Xylem

- Movement of water is through the xylem.
- Xylem tissue is made up of vessels and tracheids.

Xylem Vessels

- Xylem vessels are formed from cells that are elongated along the vertical axis and arranged end to end.
- During development, the cross walls and organelles disappear and a continuous tube is formed.
- The cells are dead and their walls are strengthened by deposition of lignin.
- The lignin has been deposited in various ways.
- This results in different types of thickening
  - Annular.
  - Simple spiral.
  - Double spiral.
  - Reticulate.
- The bordered pits are areas without lignin on xylem vessels and allow passage of water in and out of the lumen to neighbouring cells.

Tracheids

- Tracheids have cross-walls that are perforated.
- Their walls are deposited with lignin.
- Unlike the xylem vessels, their end walls are tapering or chisel-shaped.
- Their lumen is narrower.
- Besides transport of water, xylem has another function of strengthening the plant which is provided by xylem fibres and xylem parenchyma.

Xylem fibres ;

- Are cells that are strengthened with lignin.
- They form wood.

Xylem parenchyma:

- These are cells found between vessels.
- They form the packing tissue.

Forces involved in Transportation of Water and Mineral Salts

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Transpiration pull

- As water vaporises from spongy mesophyll cells into sub-stomatal air spaces, the cell sap of mesophyll cells develop a higher osmotic pressure than adjacent cells.
- Water is then drawn into mesophyll cells by osmosis from adjacent cells and finally from xylem vessels.
- A force is created in the leaves which pulls water from xylem vessels in the stem and root.
- This force is called ***transpiration pull .***

Cohesion and Adhesion:

- The attraction between water molecules is called cohesion.
- The attraction between water molecules and the walls of xylem vessels is called adhesion.
- The forces of cohesion and adhesion maintain a continuous flow of water in the xylem from the root to the leaves.

Capillarity:

- Is the ability of water to rise in fine capillary tubes due to surface tension.
- Xylem vessels are narrow, so water moves through them by capillarity.

Root Pressure:

- If the stem of a plant is cut above the ground level, it is observed that cell sap continues to come out of the cut surface.
- This shows that there is a force in the roots that pushes water up to the stem.
- This force is known as root pressure.

Importance of Transpiration

- Transpiration leads to excessive loss of water if unchecked.

Some beneficial effects are:

- Replacement of water lost during the process.
- Movement of water up the plant is by continuous absorption of water from the soil.
- Mineral salts are transported up the plant.
- Transpiration ensures cooling of the plant in hot weather.
- Excessive loss of water leads to wilting' and eventually death if water is not available in the soil.

Factors Affecting Transpiration

The factors that affect transpiration are grouped into two.

- i.e. environmental and structural.

Environmental factors

Temperature

- High temperature increases the internal temperature of the leaf .

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- which in turn increases kinetic energy of water molecules which increases evaporation.
- High temperatures dry the air around the leaf surface maintaining a high concentration gradient.
- More water vapour is therefore lost from the leaf to the air.

Humidity

- The higher the humidity of the air around the leaf, the lower the rate of transpiration.
- The humidity difference between the inside of the leaf and the outside is called the saturation deficit.
- In dry atmosphere, the saturation deficit is high.
- At such times, transpiration rate is high.

Wind

- Wind carries away water vapour as fast as it diffuses out of the leaves.
- This prevents the air around the leaves from becoming saturated with vapour.
- On a windy day, the rate of transpiration is high.

Light Intensity

- When light intensity is high; more stomata open hence high rate of transpiration.

Atmospheric Pressure

- The lower the atmospheric pressure the higher the kinetic energy of water molecules hence more evaporation.
- Most of the plants at higher altitudes where atmospheric pressure is very low have adaptations to prevent excessive water-loss.

Availability of Water

- The more water there is in the soil, the more is absorbed by the plant and hence a lot of water is lost by transpiration.

Structural Factors

Cuticle

- Plants growing in arid or semi-arid areas have leaves covered with a thick waxy cuticle.

Stomata

- The more the stomata, the higher the rate of transpiration.
- Xerophytes have few stomata which reduce water-loss.
- Some have sunken stomata which reduces the rate of transpiration as the water vapour accumulates in the pits.
- Others have stomata on the lower leaf surface hence reducing the rate of water-loss.
- Some plants have reversed stomatal rhythm whereby stomata close during the day and open at night.
- This helps to reduce water-loss.

Leaf size and shape

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- Plants in wet areas have large surface area for transpiration.
- Xerophytes have small narrow leaves to reduce water-loss.
- The photometer can be used to determine transpiration in different environmental conditions.

Translocation of organic compounds

- Translocation of soluble organic products of photosynthesis within a plant is called translocation.
- It occurs in phloem in sieve tubes.
- Substances translocated include glucose, amino acids, vitamins.
- These are translocated to the growing regions like stem, root apex, storage organs e.g. corms, bulbs and secretory organs such as nectar glands.

Phloem

phloem is made up of;

- sieve tubes,
- companion cells
- parenchyma, a packing tissue
- schlerenchyma, a strengthening tissue

Sieve Tubes

- These are elongated cells arranged end to end along the vertical axis.
- The cross walls are perforated by many pores to make a sieve plate.
- Most organelles disappear and those that remain are pushed to the sides of the sieve tube.
- Cytoplasmic strands pass through the pores in the plate into adjacent cells.
- Food substances are translocated through cytoplasmic strands.

Companion Cells

- Companion cells are small cells with large nuclei and many mitochondria.
- They are found alongside each sieve element.
- The companion cell is connected to the tube through plasmodesmata.
- The mitochondria generate energy required for translocation.

Phloem Parenchyma

- These are parenchyma cells between sieve elements.
- They act as packing tissue.

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Transport in Animals

The Circulatory System

- Large and complex animals have circulatory systems that consist of tubes, a transport fluid and a means of pumping the fluid.
- ***Blood*** is the transport fluid which contains dissolved substances and cells.
- The tubes are blood vessels through which dissolved substances are circulated around the body.
- The heart is the pumping organ which keeps the blood in circulation.

The types of circulatory system exist in animals: open and closed.

- ***In an open circulatory system***;
  - The heart pumps blood into vessels which open into body spaces known as haemocoel.
  - Blood comes into contact with tissues.
- ***A closed circulatory system***;
  - Found in vertebrates and annelids where the blood is confined within blood vessels and does not come into direct contact with tissues.

Transport in Insects

- In an insect, there is a tubular heart just above the alimentary canal.
- This heart is suspended in a pericardial cavity by ligaments.
- The heart has five chambers and extends along the thorax and abdomen .
- Blood is pumped forwards into the aorta by waves of contractions in the heart.
- It enters the haemocoel and flows towards the posterior.
- The blood flows back into the heart through openings in each chamber called ostia.
- The ostia have valves which prevent the backflow of blood.
- Blood is not used as a medium for transport of oxygen in insects.
- This is because oxygen is supplied directly to the tissues by the tracheal system.
- The main functions of blood in an insect are to transport nutrients, excretory products and hormones.

Mammalian Circulatory System

- Mammals have a closed circulatory system where a powerful heart pumps blood into arteries.
- The arteries divide into smaller vessels called arterioles.
- Each arteriole divides to form a network of capillaries inside the tissues.
- The capillaries eventually re-unite to form venules, which form larger vessels called veins.
- The veins take the blood back to the heart.
- Blood from the heart goes through the pulmonary artery to the lungs and then back to the heart through pulmonary vein.
- This circulation is called pulmonary circulation.

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- Oxygenated blood leaves the heart through the aorta and goes to all the tissues of the body.
- From the tissues, deoxygenated blood flows back to the heart through the vena cava.
- This circulation is called systemic circulation.
- In each complete circulation, the blood flows into the heart twice.
- This is called double circulation.
- Some other animals like fish have a single circulation.
- Blood flows only once through the heart for every complete circuit.

Structure and Function of the Heart

- The heart has four chambers:
- Two artria (auricles) and two ventricles.
- The left and right side of the heart are separated by a muscle wall (septum) so that oxygenated and deoxygenated blood does not mix.
- Deoxygenated blood from the rest of the body enters the heart through the vena cava .
- Blood enters the right atrium, then through tricuspid valve into right ventricle.
- Then via semi-lunar valve to the pulmonary artery to the lungs.
- Oxygenated blood from the lungs enters the heart through pulmonary vein.
- It enters the left atrium of the heart, then through bicuspid valve into left ventricle.
- Then via semi-lunar valves to aorta which takes oxygenated blood round the body.
- A branch of the aorta called coronary artery supplies blood to the heart muscle.
- The coronary vein carries blood from the heart muscle to the pulmonary artery which then takes it to the lungs for oxygenation.

Pumping Mechanism of the heart

- The heart undergoes contraction (systole) and relaxation (diastole).

Systole

- When the ventricular muscles contract, the cuspid valves (tricuspid and bicuspid) close preventing backflow of blood into auricles.
- The volume of the ventricles decreases while pressure increases.
- This forces blood out of the heart to the lungs through semi-lunar valves and pulmonary artery, and to the body tissues via semi-lunar valve and aorta respectively.
- At the same time the atria are filled with blood.
- The left ventricle has thicker muscles than the right ventricle, and pumps blood for a longer distance to the tissues.

Diastole

- When ventricular muscles relax, the volume of each ventricle increases while pressure decreases.
- Contractions of atria force the bicuspid and tricuspid valves to open allowing deoxygenated blood from right atrium into right ventricle which oxygenated blood flows from left atrium into the left ventricle.
- Semi-lunar valves close preventing the backflow of blood into ventricles.
- The slight contractions of atria force the , blood flow into ventricles.

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The Heartbeat

- The heart is capable of contracting and relaxing rhythmically without fatigue due to its special muscles called cardiac muscles.
- The rhythmic contraction of the heart arise from within the heart muscles without nervous stimulation.
- The contraction is said to be myogenic.
- The heartbeat is initiated by the pacemaker or sino-artrio-node (SAN) which is located in the right atrium.
- The wave of excitation spreads over the walls of atria.
- It is picked by the artrio-ventricular node which is located at the junction:
- Of the atria and ventricles, from where the purkinje tissue spreads the wave to the walls of the ventricles.
- The heart contracts and relaxes rhythmically at an average rate of 72 times per minute.
- The rate of the heartbeat is increased by the sympathetic nerve, while it is slowed down by the vagus nerve.
- Heartbeat is also affected by hormones e.g. adrenaline raises the heartbeat.

Structure and Function of Arteries,Capillaries and Veins

Arteries

- Arteries carry blood away from the heart.
- They carry oxygenated blood except pulmonary artery which carries deoxygenated blood to the lungs.
- Arteries have a thick, muscular wall, which has elastic and collagen fibres that resist the pressure of the blood flowing in them.
- The high pressure is due to the pumping action of the heart.
- The pressure in the arteries originate from the pumping action of the heart.
- The pulse or number of times the heart beats per minute can be detected by applying pressure on an artery next to the bone.
- e.g. by placing the finger/thumb on the wrist.
- The innermost layer of the artery is called endothelium which is smooth.
- It offers least possible resistance to blood flow.
- Have a narrow lumen .
- The aorta forms branches which supply blood to all parts of the body.
- These arteries divide into arterioles which further divide to form capillaries.

Capillaries

- Capillaries are small vessels whose walls are made of endothelium which is one cell thick.
- This provides a short distance for exchange of substances.
- Capillaries penetrate tissues,
- The lumen is narrow therefore blood flowing in capillaries is under high pressure.
- Pressure forces water and dissolved substances out of the blood to form tissue fluid.

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- Exchange of substances occurs between cells and tissue fluid.

- Part of the tissue fluid pass back into capillaries at the venule end.
- Excess fluid drains into small channels called lymph capillaries which empty their contents into lymphatic vessels.
- Capillaries join to form larger vessels called venules which in turn join to form veins which transport blood back to the heart.

Veins

- Veins carry deoxygenated blood from the tissues to the heart (except pulmonary vein which carries oxygenated blood from the lungs to the heart).
- Veins have a wider lumen than arteries.
- Their walls are thinner than those of arteries.
- **Blood pressure in the veins is low.**
- Forward flow of blood in veins is assisted by contraction of skeletal muscles, hence the need for exercise.
- Veins have valves along their length to prevent backflow of blood.
- This ensures that blood flows towards the heart.
- The way the valves work can be demonstrated on the arm.
- By pressing on one vein with two fingers, leaving one and pushing blood toward the heart then releasing the latter finger, it can be observed that the part in between is left with the vein not being visible.
- This is because bleed does not flow back towards the first finger.

Diseases and Defects of Circulatory System

Thrombosis

- Formation of a clot in the blood vessels is called thrombosis.
- Coronary thrombosis is the most common.
- It is caused by blockage of coronary artery which supplies blood to the heart.
- Blockage may be due to artery becoming fibrous or accumulation of fatty material on the artery walls.
- Narrow coronary artery results in less blood reaching the heart muscles.
- A serious blockage can result in heart attack which can be fatal.
- Heavy intake of fat, alcohol, being overweight and emotional stress can cause coronary thrombosis.
- A blockage in the brain can lead to a stroke causing paralysis of part of the body, coma or even death.
- A healthy lifestyle, avoiding a lot of fat in meals and avoiding alcohol can control the ***disease.***

Arteriosclerosis

- This condition results from the inner walls having materials being deposited there or growth of fibrous connective tissue.
- This leads to thickening of the wall of the artery and loss of elasticity.
- Normal blood flow is hindered.
- Arteriosclerosis can lead to thrombosis or hypertension.

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- A person with hypertension which is also called high blood pressure has his/her blood being pumped more forcefully through the narrow vessels.
- This puts stress on the walls of the heart and arteries.
- Regular exercise, healthy diet and avoiding smoking can help maintain normal blood pressure.

Varicose Veins

- Superficial veins especially at the back of the legs become swollen and flabby due to some valves failing to function properly.
- This results to retention of tissue fluid.
- Regular physical exercise will prevent this condition.
- Repair of valves through surgery can also be done.
- Wearing surgical stockings may ease a mild occurence.

Structure and Function of Blood

Composition of Blood

- The mammalian blood is made up of a fluid medium called plasma with substances dissolved in it.
  - Cellular components suspended in plasma include;
  - erythrocytes (red blood cells),
  - leucocytes (white blood cells)
  - thrombocytes (platelets)
  - blood proteins.

Plasma

- This is a pale yellow fluid consisting of 90% water.
- There are dissolved substances which include;
  - glucose, amino acids, lipids, salts,
  - hormones, urea, fibrinogen, albumen,
  - antibodies, some enzymes suspended cells.
- Serum is blood from which fibrinogen and cells have been removed.

The functions of plasma include:

- Transport of red blood cells which carry oxygen.
- Transport dissolved food substances round the body.
- Transport metabolic wastes like nitrogenous wastes and carbon (IV) oxide in solution about 85% of the carbon (IV) oxide is carried in form of hydrogen carbonates.
- Transport hormones from sites of production to target organs.
- Regulation of pH of body fluids.
- Distributes heat round the body hence regulate body temperature.

Erythrocytes (Red Blood Cells)

- In humans these cells are circular biconcave discs without nuclei.

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- Absence of nucleus leaves room for more haemoglobin to be packed in the cell to enable it to carry more oxygen.
- Haemoglobin contained in red blood cells is responsible for the transport of oxygen.
- Haemoglobin + Oxygen =oxyhaemoglobin
- (Hb) + ($4O_2$) __ (HbO_g)
- Oxygen is carried in form of oxyhaemoglobin.
- Haemoglobin readily picks up oxygen in the lungs where concentration of oxygen is high.
- In the tissues, the oxyhaemoglobin breaks down (dissociates) easily into haemoglobin and oxygen.
- Oxygen diffuses out of the red blood cells into the tissues.
- Haemoglobin is then free to pick up more oxygen molecules.
- The biconcave shape increases their surface area over which gaseous exchange takes place.
- Due to their ability, they are able to change their shape to enable themselves squeeze inside the narrow capillaries.
- There are about five million red blood cells per cu bic millimetre of blood.
- They are made in the bone marrow of the short bones like sternum, ribs and vertebrae.
- In the embryo they are made in the liver and spleen.
- Erythrocytes have a life span of about three to four months after which they are destroyed in the liver and spleen.
- Also in the red blood cells is carbonic anhydrase which assists in the transport of carbon (IV) oxide.

Leucocytes (White Blood Cells)

- These white blood cells have a nucleus.
- They are divided into two:
  - Granulocytes (also phagocytes or polymorphs)
  - Agranulocytes .
- White blood cells defend the body against disease.
- Neutrophils form 70% of the granulocytes.
- Others are eosinophils and basophils.
- About 24% agronulocytes are called lymphocytes, while 4% agranulocytes are monocytes.
- The leucocytes are capable of amoebic movement.
- They squeeze between the cells of the capillary wall to enter the intercellular spaces.
- They engulf and digest disease causing organisms (pathogens) by phagocytosis.
- Some white blood cells may die in the process of phagocytosis.
- The dead phagocytes, dead organisms and damaged tissues form pus.
- Lymphocytes produce antibodies which inactivate antigens.

Antibodies include:

- Antitoxins which neutralise toxins.
- Agglutinins cause bacteria to clump together and they die.
- Lysins digest cell membranes of microorganisms.
- Opsonins adhere to outer walls of microorganisms making it easier for phagocytes to ingest them.
- Lymphocytes' are made in the thymus gland and lymph nodes.

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- There are about 7,000 leucocytes per cubic millimetre of blood.

Platelets (Thrombocytes)

- Platelets are small irregularly shaped cells formed from large bone marrow cells called megakaryocytes.
- There are about 250,000 platelets per cubic millimetre of blood.
- They initiate the process of blood clotting.
- The process of clotting involves a series of complex reactions whereby fibrinogen is converted into a fibrin clot.
- When blood vessels are injured platelets are exposed to air and they release **thromboplastin** which initiates the blood clotting process.
- Thromboplastin neutralises heparin the anti-clotting factor in blood and activates prothrombin to thrombin.
- The process requires calcium ions and vitamin K.
- Thrombin activates the conversion of fibrinogen to fibrin which forms a meshwork of fibres on the cut surface to trap red blood cells to form a clot.
- The clot forms a scab that stops bleeding and protects the damaged tissues from entry of micro-organisms.
- Blood clotting reduces loss of blood when blood vessels are injured.
- Excessive loss of blood leads to anaemia and dehydration.
- Mineral salts lost in blood leads to osmotic imbalance in the body.
- This can be corrected through blood transfusion and intravenous fluid.

ABO Blood Groups

- There are four types of blood groups in human beings: A, B, AB and O.
- These are based on types of proteins on the cell membrane of red blood cells.
- There are two types of proteins denoted by the letters A and B which are antigens.
- In the plasma are antibodies specific to these antigens denoted as *a* and *b.*
- A person of blood group A has A antigens on the red blood cells and *b* antibodies in plasma.
- A person of blood group B has B antigens on red blood cells and *a* antibodies in plasma.
- A person of blood group AB has A and B antigens on red blood cells and no antibodies in plasma .
- A person of blood group a has no antigens on red blood cells and *a* and *b* antibodies in plasma.

Blood groups

Blood	Antigens	Antibodi
A	A	b
B	B	a
AB	AandB	None
o	None	a and b

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Blood Transfusion

Blood transfusion is the transfer of blood from a donor to the circulatory system of the recipient.

A recipient will receive blood from a donor if the recipient has no corresponding antibodies to the donor's antigens.

If the donor's blood and the recipient's blood are not compatible, agglutination occurs whereby red blood cells clump together.

Blood typing

- A person of blood group o can donate blood to a person of any other blood group.
- A person of blood group o is called a universal donor.
- A person of blood group AB can receive blood from any other group.
- A person with blood group AB is called a universal recipient.
- A person of blood group A can only donate blood to another person with blood group A or a person with blood group AB.
- A person of blood group B can only donate blood to somebody with blood group B or a person with blood group AB.
- A person with blood group AB can only donate blood to a person with blood groupAB.
- Blood screening has become a very important step in controlling HIV/AIDS.
- It is therefore important to properly screen blood before any transfusion is done.

Rhesus Factor

- The Rhesus factor is present in individuals with the Rhesus antigen in their red blood cells.
- Such individuals are said to be Rhesus positive (Rh+), while those without the antigen are Rhesus negative (Rh-).
- If blood from an Rh+ individual is introduced into a person who is Rh- , the latter develops antibodies against the Rhesus factor.
- There may not be any reaction after this transfusion.
- However a subsequent transfusion with Rh+ blood causes a severe reaction, and agglutination occurs i.e. clumping of red blood cells.
- The clump can block the flow of blood, and cause death.
- Erythroblastosis foetalis (haemolytic disease of the newborn) results when an Rh- mother carries an Rh+ foetus.
- This arises when the father is Rh+.
- During the latter stage of pregnancy, fragments of Rhesus positive red blood cells of the foetus may enter mother's circulation.
- These cause the mother to produce Rhesus antibodies which can pass across the. placenta to the foetus and destroy foetal red blood cells.
- During the first pregnancy, enough antibodies are not formed to affect the foetus.
- Subsequent pregnancies result in rapid production of Rhesus antibodies by the mother.

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- These destroy the red blood cells of the foetus, the condition called haemolytic disease of the newborn.
- The baby is born anaemic and with yellow eyes (jaundiced).
- The condition can be corrected by a complete replacement of baby's blood with safe healthy blood.

Lymphatic System

- The lymphatic system consists of lymph vessels.
- Lymph vessels have valves to ensure unidirectional movement of lymph.
- Lymph is excess tissue fluid i.e. blood minus blood cells and plasma proteins.
- Flow of lymph is assisted by breathing and muscular contractions.
- Swellings called lymph glands occur at certain points along the lymph vessels.
- Lymph glands are oval bodies consisting of connective tissues and lymph spaces.
- The lymph spaces contain lymphocytes which are phagocytic.
- Lymph has the same composition as blood except that it does not contain red blood cells and plasma proteins.
- Lymph is excess tissue fluid.
- Excess tissue fluid is drained into lymph vessels by hydrostatic pressure.
- The lymph vessels unite to form major lymphatic system.
- The main lymph vessels empty the contents into sub-clavian veins which take it to the heart.

Immune Responses

- Immune response is the production of antibodies in response to antigens.
- An antigen is any foreign material or organism that is introduced into the body and causes the production of antibodies.
- Antigens are protein in nature.
- An antibody is a protein whose structure is complementary to the antigen.
- This means that a specific antibody deals with a specific antigen to make it harmless.
- When harmful organisms or proteins invade the body, lymphocytes produce complementary antibodies, while bone marrow and thymus gland produce more phagocytes and lymphocytes respectively.

Types of Immunity

- There are two types of **immunity; natural and artificial.**

Natural Immunity is also called innate immunity.

- **It** is inherited from parent to offspring.

Artificial Immunity can be natural or induced.

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- When attacked by diseases like chicken pox, measles and mumps, those who recover from these diseases develop resistance to any subsequent infections of the same diseases.
- This is natural acquired immunity.

Artificial Acquired Immunity:

- When attenuated (weakened) or dead microorganisms are introduced into a healthy person.
- The lymphocytes synthesis the antibodies which are released into the lymph and eventually reach the blood.
- The antibodies destroy the invading organisms.
- The body retains 'memory' of the structure of antigen.
- Rapid response is ensured in subsequent infections.
- Vaccines generally contain attenuated disease causing organisms.

Artificial Passive Acquired Immunity:

- Serum containing antibodies is obtained from another organism, and confers immunity for a short duration.
- Such immunity is said to be passive because the body is not activated to produce the antibodies.

Importance of Vaccination

- A vaccine is made of attenuated, dead or nonvirulent micro-organism that stimulate cells in the immune system to recognise and attack disease causing agent through production of antibodies.
- Vaccination protects individuals from infections of many diseases like smallpox, tuberculosis and poliomyelitis.
- Diseases like smallpox, tuberculosis and tetanus were killer diseases but this is no longer the case.
- Diphtheria Pertussis Tetanus (DPT) vaccine protects children against diphtheria, whooping cough and tetanus.
- Bacille Calmette Guerin (BCG) vaccine is injected at birth to children to protect them against tuberculosis.
- Measles used to be a killer disease but today, a vaccine injected into children at the age of rune months prevents it.
- At birth children are given an inoculation through the mouth of the poliomyelitis vaccine.

Allergic Reactions

- An allergy is a hypersensitive reaction to an antigen by the body.
- The antibody reacts with the antigen violently.
- People with allergies are oversensitive to foreign materials like dust, pollen grains, some foods, some drugs and some air pollutants.
- Allergic reactions lead to production of **histamine** by the body.
- Histamine causes swelling and pain.
- Allergic reactions can be controlled by avoiding the allergen and administration of anti-histamine drugs.

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END OF NOTES

Reproduction in plants and animal

REPRODUCTION IN PLANTS AND ANIMALS

Introduction

- The process by which mature individuals produce offspring is called reproduction.
- Reproduction is a characteristic of all living organisms and prevents extinction of a species.
- There are two types of reproduction: sexual and asexual reproduction.

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- Sexual reproduction involves the fusion of male and female gametes to form a zygote.
- Asexual reproduction does not involve gametes.

Cell Division

- Cell division starts with division of nucleus.
- In the nucleus are a number of thread-like structures called chromosomes, which occur in pairs known as homologous chromosomes.
- Each chromosome contains-genes that determine the characteristics of an organism.
- The cells in each organism contains a specific number of chromosomes.

There are two types of cell division:

Mitosis –

- This takes place in all body cells of an organism to bring about increase in number of cells, resulting in growth and repair.
- The number of chromosomes in daughter cells remain the same as that in the mother cell.

Meiosis –

- This type of cell division takes place in reproductive organs (gonads) to produce gametes.
- The number of chromosomes in the gamete is half that in the mother cell.

Mitosis

- Mitosis is divided into four main stages.
- Prophase, Metaphase, Anaphase and Telophase.
- These stages of cell division occur in a smooth and continuous pattern.

Interphase

- The term interphase is used to describe the state of the nucleus when the cell is just about to divide.
- During this time the following take place:
- Replication of genetic material so that daughter cells will have the same number of chromosomes as the parent cell.
- Division of cell organelles such as mitochondria, ribosomes and centrioles.
- Energy for cell division is synthesised and stored in form of Adenosine Triphosphate (ATP) to drive the cell through the entire process.
- During. interphase, the following observations can be made:
- Chromosomes are seen as long, thin, coiled thread-like structures.
- Nuclear membrane and nucleolus are intact.

Prophase

- The chromosomes shorten and thicken.
- Each chromosome is seen to consist of a pair of chromatids joined at a point called centromere.
- Centrioles (in animal cells) separate and move to opposite poles of the cell.

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- The centre of the nucleus is referred to as the equator.
- Spindle fibres begin to form, and connect the centriole pairs to the opposite poles.
- The nucleolus and nuclear membrane disintegrate and disappear.

Metaphase

- Spindle fibres lengthen.
- In animal cells they attach to the centrioles at both poles.
- Each chromosome moves to the equatorial plane and is attached to the spindle fibres by the centromeres.
- Chromatids begin to separate at the centromere.

Anaphase

- Chromatids separate and migrate to the opposite poles due to the shortening of spindle fibres .
- Chromatids becomes a chromosome.
- In animal cell, the cell membrane starts to constrict.

Telophase

- The cell divides into two.
- In animal cells it occurs through cleavage of cell membrane.
- In plants cells, it is due to deposition of cellulose along the equator of the cell.(Cell plate formation).
- A nuclear membrane forms around each set of chromosome.
- Chromosomes later become less distinct.

Significance of Mitosis

- It brings about the growth of an organism:
- It brings about asexual reproduction.
- Ensures that the chromosome number is retained.
- Ensures that the chromosomal constitution of the offspring is the same as the parents.

Meiosis

- Meiosis involves two divisions of the parental cell resulting into four daughter cells.
- The mother cell has the diploid number of chromosomes.
- The four cells (gametes) have half the number of chromosomes (haploid) that the mother cell had.
- In the first meiotic division there is a reduction in the chromosome number because homologous chromosomes and not chromatids separate.
- Each division has four stages ***Prophase, Metaphase, Anaphase and Telophase.***

Interphase

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- As in mitosis the cell prepares for division.
- This involves replication of chromosomes, organelles and build up of energy to be used during the meiotic division.

First Meiotic division

Prophase I

- Homologous chromosomes lie side by side in the process of synapsis forming pairs called bivalents.
- Chromosomes shorten and thicken hence become more visible.
- Chromosomes may become coiled around each other and the chromatids may remain in contact at points called chiasmata (singular chiasma).
- Chromatids cross-over at the chiasmata exchanging chromatid portions. Important genetic changes usually result.

Metaphase I

- Spindle fibres are fully formed and attached to the centromeres.
- The bivalents move to the equator of the spindles.

Anaphase I

- Homologous chromosomes separate and migrate to opposite poles.
- This is brought about by shortening of spindle fibres hence pulling the chromosomes.
- The number of chromosomes at each pole is half the number in the mother cell.

Telophase I

- Cytoplasm divides to separate the two daughter cells.

Second Meiotic Division

- Usually the two daughter cells go into a short resting stage (interphase)
- but sometimes the chromosomes remain condensed and the daughter cells go straight into metaphase of second meiotic division.
- The second meiotic division takes place just like mitosis.

Prophase II

- Each chromosome is seen as a pair of chromatids.

Metaphase II

- Spindle forms and are attached to the chromatids at the centromeres.
- Chromatids move to the equator.

Anaphase II

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- Sister chromatids separate from each other
- Then move to opposite poles, pulled by the shortening of the spindle fibres.

Telophase II

- The spindle apparatus disappears.
- The nucleolus reappears and nuclear membrane is formed around each set of chromatids.
- The chromatids become chromosomes.
- Cytoplasm divides and four daughter cells are formed.
- Each has a haploid number of chromosomes.

Significance of Meiosis

- Meiosis brings about formation of gametes that contain half the number of chromosomes as the parent cells.
- It helps to restore the diploid chromosomal constitution in a species at fertilisation.
- It brings about new gene combinations that lead to genetic variation in the offsprings.

Asexual Reproduction

- Asexual reproduction is the formation of offspring from a single parent.
- The offspring are identical to the parent.

Types of asexual reproduction.

- Binary fission in amoeba.
- Spore formation in *Rhizopus.*
- Budding in yeast.

Binary fission

- This involves the division of the parent organism into two daughter cells.
- The nucleus first divides into two and then the cytoplasm separates into two portions
- Binary fission also occurs in bacteria, *Paramecium, Trypanosoma* and *Euglena.*

Spore formation in Rhizopus

- *Rhizopus* is a saprophytic fungus which grows on various substrate such as bread, rotting fruits or other decaying organic matter.
- The vegetative body is called mycelium which has many branched threads called hyphae.
- Horizontal hyphae are called stolons.
- Vertical hyphae are called sporangiophore.
- The tips of sporangiophore become swollen to form sporangia, the spore bearing structure.
- Each sporangium contains many spores.
- As it matures and ripens, it turns black in colour.
- When fully mature the sporangium wall burst and release spores which are dispersed by wind or insects.

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- When spores land on moist substratum, they germinate and grow into a new *Rhizopus* and start another generation.

Spore formation in ferns

- The fern plant is called a sporophyte.
- On the lower side of the mature leaves are sari (Singular: sorus) which bear spores.

Budding in Yeast

- Budding involves the formation of a protrusion called a bud from the body of the organism.
- The bud separates from the parent cell, in yeast budding goes on so fast and the first bud starts to form another bud before the separation.
- A short chain or mass of cells is formed.

Sexual Reproduction in Plants

- In flowering plants, the flower is the reproductive organ which is a specialised shoot consisting of a modified stem and leaves.
- The stem-like part is the pedicel and receptacle, while modified leaves form corolla and calyx.

Structure of a flower

- A typical flower consists of the following parts:

Calyx –

- made up of sepals.
- They enclose and protect the flower when it is in a bud. Some flowers have an outer whorl made of sepal-like structures called epicalyx.

Corolla –

- consists of petals. The petals are brightly coloured in insect - pollinated flowers.

Androecium –

- Is the male part of the flower. It consists of stamens.
- Each stamen consists of a filament whose end has an anther.
- Inside the anther are pollen sacs which contain pollen grains.

Gynoecium (pistil) –

- Is the female part of the flower.
- It consists of one or more carpels.

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- Each carpel consists of an ovary, a sty le and a stigma.
- The ovary contains ovules which become seeds after fertilisation.
- A ***monocarpous pistil*** has one carpel e.g. beans.
- ***A polycarpous pistil*** has many carpels.
- If the carpes are free, it is called ***apocarpous*** as in rose and *Bryophyllum,*
- In carpels that are fused it is called ***syncarpous*** as in *Hibiscus.*
- ***A complete flower*** has all the four floral parts.
- A regular flower can be divided into two halves by any vertical section passing through the centre. e.g. morning glory.
- Irregular flower can be divided into two halves in only one plane e.g. crotalaria.

Pollination

- This is the transfer of pollen grains from the anther to the stigma.

Types of pollination

- ***Self pollination*** is the transfer of pollen grains from the anther of one flower to the stigma of the same flower.
- ***Cross-pollination*** is the transfer of pollen grains from the anther of one flower to the stigma of a different flower, of the same species.

Agents of pollination

- Agents of pollination include wind, insects, birds and mammals.
- Insect pollinators include bees, butterflies and mosquitoes.

Mechanisms that hinder self-pollination

- Stamens ripen early and release their pollen grains before the stigma, mature. This is called protandry e.g. in sunflower.
- The stigma matures earlier and dries before the anthers release the pollen grains.
- This is called protogyny and is common in grasses.
- Self sterility or incompatibility
- Pollen grains are sterile to the stigma of the same flower, e.g. in maize flower.
- Shorter stamens than pistils.

Fertilisation in Plants

- The pollen grain contains the generative nucleus and a tube nucleus.
- When the pollen grain lands on the stigma, it absorbs nutrient and germinates forming a pollen tube.
- This pollen tube grows through the style pushing its way between the cells.
- It gets nourishment from these cells.
- The tube nucleus occupies the position at the tip of the growing pollen tube.

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- The generative nucleus follows behind the tube nucleus, and divides to form two male gamete nuclei.
- The pollen tube enters the ovule through the micropyle.
- When the pollen tube penetrates the ovule disintegrates and the pollen tube bursts open leaving a clear way for the male nuclei.
- One male nucleus fuses with the egg cell nucleus to form a diploid zygote which develops into an embryo.
- The other male gamete nucleus fuses with the polar nucleus to form a triploid nucleus which forms the primary endosperm.
- This is called double fertilisation.

After fertilisation the following changes take place in a flower:

- The integuments develops into seed coat (testa).
- The zygote develops into an embryo.
- The triploid nucleus develops into an endosperm.
- The ovules become seeds.
- The ovary develops into a fruit.
- The ovary wall develops into pericarp.
- The style, dries up and falls off leaving a scar.
- The corolla, calyx and stamens dry up and fall off.
- In some the calyx persists.

Fruit formation

- Fruit development without fertilisation is called parthenocarpy
- e.g. as in pineapples and bananas.
- Such fruits do not have seeds.

Classification of fruits

- False fruits develops from other parts such as calyx, corolla and receptacle,
- e.g. apple and pineapple which develops from an inflorescence.
- True fruits develop from the ovary, e.g. bean fruit (pod).
- True fruits can be divided into fleshy or succulent fruits e.g. berries and drupes and dry fruits.
- The dry ones can be divided into Dehiscent which split open to release seeds and indehiscent which do not open.

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Types of fruits

Type of fruit	*Structure*	*Example*
Berry Fleshy-	Ovary fleshy, thin skinned juicy with many	Tomato, orange, Sodom
Drupe fleshy-	Outer layer fleshy, inner layer hard, endosing more seeds	Mango, plum
Pod Dehiscent (dry)	Ovary wall thin, contains many seeds. Splits	Bean, pea
Schizocarp (dry)	The ripe fruit breaks up into small one seeded	Castor oil
Caryopsil Dry	Pericarp and seed coat are fused to form thin covering	Maize grain
Cypsela Dry indehiscent	One seeded fruit. The calyx persists	Bidens, Tridax
Pome	Outer fleshy layer develops from calyx and	Pear, apple
Multiple fruit	Formed from several flowers in a cluster	Pineapple
Achene	Ovary wall separated from seed	Sunflower

Placentation

- This is the arrangement of the ovules in an ovary.

Marginal placentation:

- The placenta appears as one ridge on the ovary wall e.g. bean.

Parietal placentation:

- The placenta is on the ridges on ovary wall.
- Ovules are in them e.g. pawpaw.

Axile placentation:

- The placenta is in the centre.
- Ovary is divided into a number of loculi. e.g. orange.

Basal placentation.

- The placenta is formed at the base of the ovary e.g. sunflower.

Free Central placentation.

- Placenta is in the centre of the ovary.
- There are no loculi e.g. in primrose.

Methods of fruit and seed dispersal

Animal dispersal

- Fleshy fruits are eaten by animals.

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- Animals are attracted to the fruits by the bright colour, scent or the fact that it is edible.
- The seeds pass through the digestive tract undamaged and are passed out with faeces. E.g. tomatoes and guavas.
- Such seeds have hard, resistant seed coats.
- Others have fruits with hooks or spines that stick on animal fur or on clothes.
- Later the seeds are brushed of or fall off on their own e.g. *Bidens pilosa* (Black jack).

Wind dispersal

- Fruits and seeds are small and light in order to be carried by air currents.
- A fruit that is a capsule e.g. tobacco split or has pores at the top e.g. Mexican poppy.
- The capsule is attached to along stalk when swayed by wind the seeds are released and scattered.
- Some seeds have hairy or feather-like structures which increase their surface area so that they can be blown off by the wind e.g. *Sonchus.*
- Others have wing-like structures e.g. Jacaranda and Nandi Flame.
- These extensions increase the surface area of fruits and seeds such that they are carried by the wind.

Water dispersal

- Fruits like coconut have fibrous mescocarp which is spongy to trap air, the trapped air make the fruit light and buoyant to float on water.
- Plants like water lily produce seeds whose seed coats trap air bubbles.
- The air bubbles make the seeds float on water and are carried away.
- The pericarp and seed coat are waterproof.

Self dispersal (explosive) Mechanism

- This is seen in pods like bean and pea.
- Pressure inside the pod forces it to open along lines of weakness throwing seeds away from parent plant.

Reproduction in Animals

- Sexual reproduction involves the fusion of gametes.
- In animals two individuals are involved, a male and a female.
- Special organs known as gonads produce gametes.
- In males testes produce sperms while in females ovaries produce ova.

- The fusion of male gamete and female gamete to form a zygote is called fertilisation.

There are two types of fertilisation. External and internal.

External fertillsation

- Example in amphibians takes place in water.
- The male mounts the female and shed sperms on the eggs as they are laid.

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- Eggs are covered by slippery jelly-like substance which provides protection.
- Many eggs are released to increase the chances of survival.

Internal fertilisation

- This occurs in reptiles, birds and mammals.
- Fertilisation occurs within the body of the female.
- Fewer eggs are produced because there are higher chances of fertilisation since sperms are released into the female body.

Reproduction in Humans

Structure of female reproduction system

The female reproduction system consist of the following:

Ovaries

- Are two oval cream coloured structures found in lower abdomen below the kidneys.

Oviducts.

- They produce the ova.
- Are tubes which conduct the ova produced by the ovaries to the uterus.
- Fertilisation occurs in the upper part of the oviduct.

Uterus

- The uterus is a hollow muscular organ found in the lower abdomen.
- The embryo develops inside the uterus.
- The inner lining endometrium supplies nutrients to embryo.
- The embryo is implanted into the inner uterine wall- the endometrium which nourishes the embryo.
- The thick muscles of the uterus assist in parturition.

Cervix

- Has a ring of muscles that separates the uterus from the vagina.
- It forms the opening to the uterus

Vagina

- Is a tube that opens to the outside and it acts as the copulatory and birth canal through the vulva.

Structure of male reproductive system

The male reproductive system consists of the following:

Testis:

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- Each testis is a mass of numerous coiled tubes called semniferous tubules.
- Each is enclosed within a scrotal sac that suspends them between the thighs.
- This ensures that sperms are maintained at a temperature lower than that of the main body.

Seminiferous tubules

- The lining of seminiferous tubules consists of actively dividing cells which give rise to sperms.
- Between the seminiferous tubules are interstitial cells which produce the male hormones called androgens e.g. testosterone.
- The seminiferous tubules unite to form the epididymis, which is a coiled tube where sperms are stored temporarily .
- Vas deferens (sperm duct) is the tube through which sperms are carried from testis to urethra.
- Seminal vesicle produces an alkaline secretion which nourishes the spermatozoa.

Prostate gland

- Produces an alkaline secretion to neutralise vaginal fluids.

Cowpers' gland

- Secretes an alkaline fluid.
- All these fluids together with spermatozoa form semen.

Urethra

- Is a long tube through which the semen is conducted during copulation.
- It also removes urine from the bladder.

Penis

- Is an intro-mittent organ which is inserted into the vagina during copulation .

Fertilisation in Animals

- Fertilisation is preceded by copulation in which the erect penis is inserted into the vagina.
- This leads to ejaculation of semen.
- The sperms swim through the female's genital tract to the upper part of the oviduct.
- The head of the sperm penetrates the egg after the acrosome_ releases lytic enzymes t dissolve the egg membrane.
- The tail is left behind.
- Sperm nucleus fuses with that of the ovum and a zygote is formed.
- A fertilisation membrane forms around the zygote which prevents other sperms from penetrating the zygote.

Implantation:

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- After fertilisation the zygote begins to divide mitoticaly as it moves towards the uterus.
- It becomes embedded in the wall of the uterus a process called implantation.
- By this time the zygote is a hollow ball of cells called blastocyst or embryo.
- In the uterus the embryo develops villi which project into uterus for nourishment later the villi and endometrium develop into placenta.

Embryonic membranes

- Embryonic membranes develop around the embryo.
- The outermost membrane is the chorion which forms the finger-like projections (chorionic villi) which supply nutrients to the embryo.
- The amnion surrounds the embryo forming a fluid filled cavity within which the embryo lies.
- Amniotic cavity is filled with amniotic fluid.
- This fluid acts as a shock absorber and protects the foetus against mechanical injury.
- It also regutates temperature.
- The chorionic villi, allantois together with the endometrium from the placenta.
- The embryo is attached to the placenta by a tube called umbilical cord which has umbilical vein and artery.
- The maternal blood in the placenta flows in the spaces lacuna and surrounds capillaries from umbilical vein and artery.
- The umbilical cord increase in length as the embryo develops.

Role of placenta

Protection

- Maternal blood and foetal blood do not mix.
- This ensures that the pathogens and toxins from maternal blood do not reach the foetus.
- The placenta allows maternal antibodies to pass into the foetus, providing the foetus with immunity.

Nutrition

- The placenta facilitates the transfer of nutrients from maternal blood to foetus.

Excretion

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- Placenta facilitates the removal of nitrogenous wastes from the foetus' blood to maternal blood.

Gaseous exchange

- Oxygen from the maternal blood diffuses into the foetal blood while carbon (IV) oxide from foetal blood diffuse into maternal blood.

Production of hormones

- Placenta produces progesterone and oestrogen.

Gestation period

- The period between conception and birth is called gestation.
- In humans gestation takes nine months (40 weeks).
- The embryo differentiates into tissues and organs during this period.

Week 1 to 3:

- Zygote divides to form blastocyst.
- Implantation takes place.
- The three germ layers form endoderm, mesoderm and ectoderm.
- Nervous system starts to form.

Week 4 to 7:

- Development of circulating and digestive systems.
- Further development of nervous system, formation of sensory organs,
- All major internal organs are developed.
- At week 5, heartbeat starts .

Week 8 to 24:

- All organs well developed including sex organs.
- Hair, finger and toe nails grow.
- Foetus move and eyelids open.

Week 25- 30:

- The fully developed foetus responds to touch and noises and moves vigorously.
- The head turns and faces downwards ready for birth.

Week 31-40:

- Foetus increases in size.
- Birth occurs.

Reproductive Hormones

Hormone	***Source***	***Functions***
Follicle Stimulating Hormone (FSH)	Pituitary gland	Development of ovarian follicle; stimulates secretion of oestrogen by the ovary
Luteinising Hormone	Pituitary gland	Causes ovulation; causes development of Graafian follicle into the corpus luteurn; causes secretion of progesterone by the ovary
Prolactin	Pituitary gland	Initiates production and secretion of milk by the

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		mammary glands
Oxytocin	Pituitary gland	Causes contraction of the uterus during parturition (birth)
Progesterone	Corpus luteum in the ovary	Causes contraction of wall of the uterus to thicken after ovulation
Oestrogen	Ovary	Causes changes in the uterine wall in preparation for implantation; initiates development of secondary sexual characteristics
Androgens-Testosterone	Interstitial cells of testis	Stimulates the development of secondary sexual characteristics
Interstitial Cell Hormone (ICSH)	Pituitary gland	Stimulates the interstitial cells of testis to release androgens
Human Chorionic Gonadotrophin (HCG)	Chorionic villi	Stops the degeneration of the corpus luteum for production of oestrogen and progesterone

Secondary Sexual Characteristics

Male

- Testerone is the main androgen that stimulates the development of secondary sexual characteristics.
- Broadening of the shoulders.
- Deepening of the voice due to enlargement of larynx.
- Hair at the pubic area, armpit and chin regions.
- Penis and testis enlarge and produce sperms.
- Body becomes more masculine.

Female

- Enlargement of mammary glands.
- Hair grows around pubic and armpit regions.
- Widening of the hips.
- Ovaries mature and start producing ova.
- Menstruation starts.
- Oestrogen triggers the onset of secondary sexual characteristics.

Sexually transmitted infections (STI)

Disease	*Causative*	*Method of*	*Symptoms*	*Prevention/control*

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		transmission		
Gonorrhoea	Bacterium Neiseeria Gonorrhoea	-Sexual contact - during birth for infants -Sharing towels	-Itching of urethra -yellowish discharge pain as males urinate, vaginal discharge. with odour in females	A void indiscriminate sex. Treat both partners infected A void sharing linen
Syphilis	Bacterium Treponema Palladium	-Sexual contact - During birth for infants. - Sharing towels and linen	Solitary painless ulcer-on genital or mucous -Rashes, muscles and papules on hands, feet lips, genital areas	Treat at primary infection stage -Avoid indiscriminate sex. - A void sharing linen
Trichomoniasi	Protozoan Trichomonas Vaginalis	-Sexual contact -contaminated linen, underwear and toilet seats	Itching of urethra or vagina in females, smelly, yellow discharge	A void sharing linen -Avoid indiscriminate sex -personal hygiene
Hepatitis	Virus Hepatitis B	-Sexual contact -blood contaminated needles and syringes	Fever, nausea, jaundice, loss of appetite, yellow urine	-Avoid indiscriminate sex -use disposable needles and syringes - strict personal hygiene
Candidiasis	Fungus Albicans	-spread through sexual contact - sharing linen and towels	Itching and burning sensation and white discharge from genitals	-Avoid indiscriminate sex - Treat both partners
Herpes (Simplex)	Virus Herpes	-sexual contact kissing, contaminated needles	Lesions on skin and mucous membranes of buccal cavity vagina or head of penis	- A void indiscriminate sex and contaminated needles and syringes.
HIV and Aids	Virus Human Deficiency	-sexual contact -blood -contaminated instruments -Through breast milk and body fluids. -Through birth canal for infants	-chronic diarrhoea -weight loss (more than 10% body weight lost in a month) - constant, persistent cough, skin infectious (herpes zoster)	-Avoid indiscriminate sex. -Use screened blood - No sharing of tooth brushes, razors - Use disposable needles

Menstrual Cycle

- This is characterized by discharge of blood and tissue debris (menses) from the uterus every 28 days.
- This is due to the breakdown of the endometrium which occurs when the level of progesterone falls and the girl starts to menstruate.
- The follicle stimulating hormone (FSH) causes the Graafian follicle to develop and also stimulate the ovary to release oestrogen.

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- Oestrogen hormone triggers the onset of secondary sexual characteristics.
- Luteinising hormone (L.H) causes the mature ovum to be released from the Graafian follicle - a process called ovulation.
- After ovulation progesterone hormone is produced.
- After menstruation, the anterior lobe of the pituitary gland starts secreting the follicle stimulating hormone (FS.H) which causes the Graafian follicle to develop in the ovary.
- It also stimulates the ovary tissues to secrete oestrogen.
- Oestrogen brings about the repair and healing of the inner lining of the uterus (endometrium) which had been destroyed during menstruation.
- Oestrogen level stimulates the pituitary gland to produce (Luteinising Hormone (L.H).
- This hormone makes the mature Graafian follicle to release the ovum into the funnel of oviduct, a process called ovulation.
- After releasing the ovum, the Graafian follicle changes into a yellow body called corpus luteum.
- The luteinising hormone stimulates the corpus luteum to secrete a hormone called progesterone which stimulates the thickening and vascularisation of endometrium.
- This prepares the uterine wall for implantation of the blastocyst.
- If fertilisation takes place, the level of progesterone increases and thus inhibits FSH from stimulating the maturation of another Graafian follicle.
- If fertilisation does not occur, the corpus luteum disintegrates and the level of progesterone goes down.
- The endometrium, sloughs off and menstruation occurs.

Advantages of Reproduction Asexual

- Good qualities from parents are retained in the offspring without variation.
- New individuals produced asexually mature faster.
- Process does not depend on external factors which may fail such as pollination.
- New individuals obtain nourishment from parent and so are able to survive temporarily under unsuitable conditions.
- No indiscriminate spreading of individuals which can result in wastage of offspring.
- Takes a shorter time and leads to rapid colonization.

Disadvantages of asexual reproduction

- New offspring may carry undesirable qualities from parents.
- Offspring may be unable to withstand changing environmental conditions.
- Faster maturity can cause overcrowding and stiff competition.
- Reduced strength and vigour of successive generations.

Advantages of sexual reproduction

- Leads to variations.
- Variations which are desirable often show hybrid vigour.

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- High adaptability of individuals to changing environmental conditions.
- Variations provide a basis for evolutionary changes.

Disadvantages of sexual reproduction

- Fusion is difficult if two individuals are isolated.
- Some variations may have undesirable qualities.
- Population growth is slow.

Practical Activities

Examining the stages of mitosis

- About 2 mm of a root tip of onion bulb is cut off and placed on a microscope slide.
- A stain e.g. aceto-orcein is added and the root tip macerated using a scapel.
- A cover slip is added and observations made.
- Different stages of mitosis can be observed.

Examining the stages of meiosis

- An unopened bud of *Tradescantia* is obtained
- The anther is removed and placed on a microscope slide.
- A few drops of hydrochloric acid and acetic-orcein stain are added.
- A cover slip is placed on the anther.
- Pressing the cover slip gives a thin squash, which is observed under the microscope.
- Different stages of meiosis are observed.

To observe the structure of Rhizopus

- *Rhizopus* grow on moist bread left under suitable temperature
- A piece of moist bread is placed on a petri-dish or enclosed in a plastic bag and observe daily for four days.
- Under a low power microscope the sporangia and stolons can be observed.

To examine spores on sori of ferns

- Obtain the fern plant.
- Detach a frond from the plant and observe the under-side using a hand lens to see the raised brown patches - the sori.
- Open up the sorus to observe the sporangia.

Examine insect and wind pollinated flowers

- Obtain insect pollinated flowers e.g. crotalaria, hibiscus/Ipomea, Solanum, incunum.
- Note the scent, colour and nectar guides.
- A description of the calyx, corolla, androecium and gynoecium is made.
- Obtain a wmd pollinated flower e.g,' maize, star-grass, sugar-cane, Kikuyu grass.
- Observe the glumes, spikes and spikelet.

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- Examine a single floret, and identify the androecium and gynoecium.

Classifying fruits

- Obtain different fruits - oranges, mangoes, maize, castor oil, bean pod, black jack .
- Observe the fruits, classify them into succulent, dry-dehiscent or indehiscent.

Dissection of Fruits

- Obtain an orange and a mango fruit.
- Make a transverse section.
- Observe the cut surface and draw and label the parts.
- Note that the fruit is differentiated into epicarp, mesocarp and endocarp.
- Obtain a pod of a legume.
- Open up the pod and observe the exposed surface.
- Draw and label the parts.
- Note that the fruit wall is not differentiated.

Dispersal of fruits and seeds

- Obtain animal dispersal fruits, like oranges, tomatoes, black jack, sodom apple.
- Identify the way by which each is adapted to dispersal by animals.
- Obtain wind dispersed fruit/seed

e.g. Nandi flame, Jacaranda *Sonchus,* cotton seed, *Tecoma.*

END

The cell

THE CELL

Introduction

- The cell is the basic unit of an organism.

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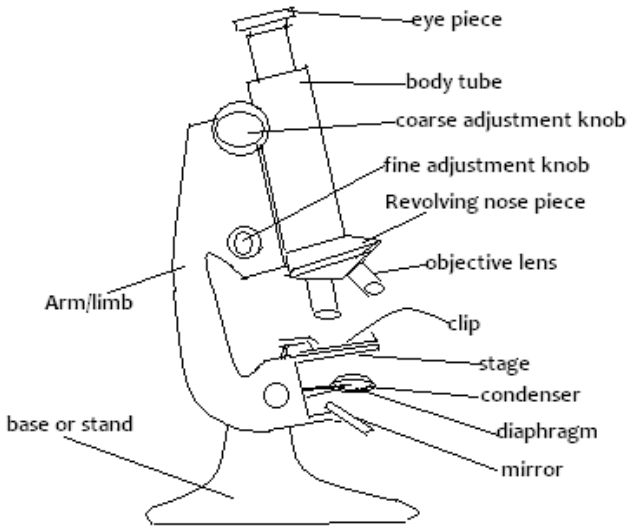
- All living organisms are made up of cells.
- Some organisms are made up of one cell and others are said to be multicellular.
- Other organisms are made of many cells and are said to be multicellular.
- Cells are too little to see with the naked eye.
- They can only be seen with the aid of a microscope.

The microscope

The microscope is used to magnify objects.

Magnification

- The magnifying power is usually inscribed on the lens.
- To find out how many times a specimen is magnified, the magnifying power of the objective lens is multiplied by that of the eye piece lens.
- If the eye piece magnification lens is x10 and the objective lens is x4, the total magnification is x40.
- Magnification has no units.
- It should always have the multiplication sign.e.g.x40



The light microscope

Microscope parts and their functions

Parts	*Function(s)*
Eye piece	Has a lens which contributes to the magnification of the object under
Coarse adjustment	Moves the body tube up and down for long distances and it brings the into focus.
Fine adjustment knob	Moves the body tube and brings the image into fine focus.
Body tube Ji	Holds the eye piece and the revolving nose piece. It directs light from objective lenses to the eye piece lens.
Revolving nose piece	Holds and brings objective lenses into position.

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Objective lens	Contributes to the magnification of the object.
Arm/limb	It is for handling the microscope and also tilting it.
Stage	Is the flat platform onto which the slide with the object is placed.
Clips	They hold the slide firmly onto the stage.
Condenser	Concentrates light onto the object.
Diaphragm	~egulates the amount of light passing through the object.
Mirror	Reflects light into the condenser.
Hinge screw	Fixes the arm to the base and allows for tilting of the arm.
Base/stand	Provides support to the microscope.

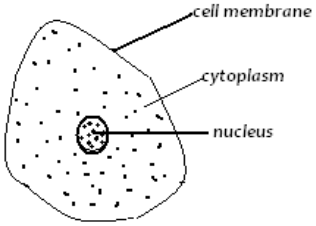
To View the Object

- Turn the low power objective lens until it clicks into position.
- Looking through the eye piece, ensure that enough light is passing through by adjusting the mirror.
- This is indicated by a bright circular area known as the field of view.
- Place the slide containing the specimen on stage and clip it into position.
- Make sure that the specimen is in the centre of the field of view.
- Using the coarse adjustment knob, bring the low power objective lens to the lowest point.
- Turn the knob gently until the specimen comes into focus.
- If finer details are required, use the fine adjustment knob.
- When using high power objective always move the fine adjustment knob upwards.

Care of a Microscope

- Great care should be taken when handling it.
- Keep it away from the edge of the bench when using it.
- Always hold it with both hands when moving it in the laboratory.
- Clean the lenses with special lens cleaning paper.
- Make sure that the low power objective clicks in position in line with eye piece lens before and after use.
- Store the microscope in a dust-proof place free of moisture.

Cell Structure as Seen Through the Light Microscope



A typical animal cell

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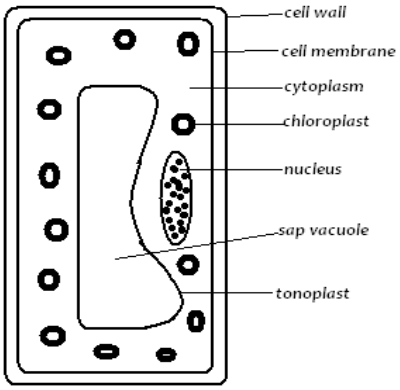

Generalised plant cell

The cell as seen above has the following:

Cell membrane (Plasma membrane):

- This is a thin membrane enclosing cell contents.
- It controls the movement of substances into and out of the cell.

Cytoplasm:

- This is a jelly-like substance in which chemical processes are carried out.
- Scattered all over the cytoplasm are small structures called organelles.
- Like an animal cell, the plant cell has a cell membrane, cytoplasm and a nucleus.

vacuole.

- Plant cells have permanent, central **vacuole.** It contains cell sap where sugars and salts are stored.

Cell wall:

- This is the outermost boundary of a plant cell.
- It is made of cellulose.
- Between the cells is a middle lamella made of calcium pectate.

Chloroplasts;

- With special staining techniques it is possible to observe **chloroplasts.**
- These are structures which contain chlorophyll, the green pigment responsible for trapping light for photosynthesis.

The Electron Microscope (EM)

- Capable of magnifying up to 500,000 times.
- The specimen is mounted in vacuum chamber through which an electron beam is directed.
- The image is projected on to a photographic plate.
- The major **disadvantage** of the electron microscope is that it **cannot be used to observe living objects**.
- However, it provides a **higher magnification** and **resolution** (ability to see close points as separate) than the light microscope so that specimen can be observed in more detail.

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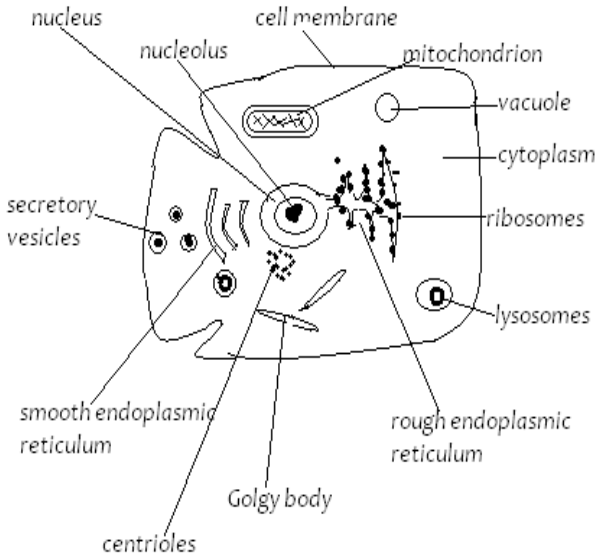
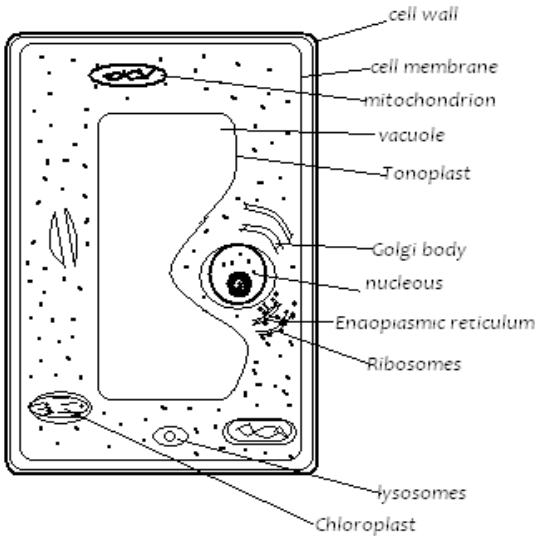
Cell Structure as Seen Through Electron Microscope

A generalised anumal cell as seen under Electron microscope





A generalised plant cell as seen under the electron microscope

The Plasma Membrane

- Under the electron microscope, the plasma membrane is seen as a double layer.
- This consists of a lipid layer sandwiched between two protein layers.
- This arrangement is known as the unit membrane and the shows two lipid layers with proteins within.
- Substances are transported across the membrane by active transport and diffusion.

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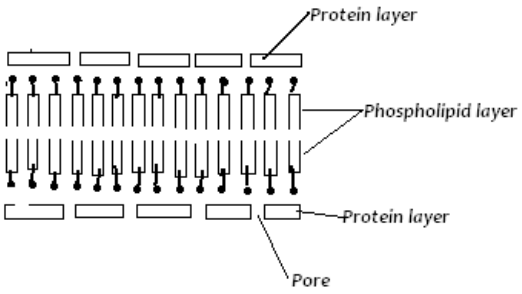

Cell membrane Structure

The Endoplasmic Reticulum (ER)

- This is a network of tubular structures extending throughout the cytoplasm of the cell.
- It serves as a network of pathways through which materials are transported from one part of the cell to the other.
- An ER encrusted with ribosomes it is referred to as **rough endoplasmic reticulum**.
- An ER that lacks ribosomes is referred to as **smooth endoplasmic reticulum.**
- The rough endoplasmic reticulum transports **proteins** while the smooth endoplasmic reticulum transports **lipids.**

The Ribosomes

- These are small spherical structures attached to the ER.
- They consist of protein and ribonucleic acid (RNA).
- They act as sites for the synthesis of proteins.

Goigi Bodies

- Golgi bodies are thin, plate-like sacs arranged in stacks and distributed randomly in the cytoplasm.
- Their function is packaging and transportation of glycol-proteins.
- They also produce lysosomes.

Mitochondria

- Each mitochondrion is a rod-shaped organelle.
- Made up of a smooth outer membrane and a folded inner membrane.
- The foldings of the inner membrane are called **cristae.**
- They increase the surface area for respiration.
- The inner compartments called the matrix.
- Mitochondria are the sites of cellular respiration, where energy is produced.

Lysosomes

- These are vesicles containing hydrolytic enzymes.
- They are involved in the breakdown of micro-organisms, foreign macromolecules and damaged or worn-out cells and organelles ..

The Nucleus

- The nucle s is surrounded by a nuclear membrane which is a unit membrane.
- The nuclear membrane has pores through which materials can move to the surrounding cytoplasm.
- The nucleus contains proteins and nucleic acid deoxyribonucleic acid (DNA) and RNA.

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- The chromosomes are found in the nucleus.
- They are the carriers of the genetic information of the cell.
- The nucleolus is also located in the nucleus but it is only visible during the non-dividing phase of the cell.

The Chloroplasts

- These are found only in photosynthetic cells.
- Each chloroplast consists of an outer unit. membrane enclosing a series of interconnected membranes called lamellae.
- At various points along their length the lamellae form stacks of disc like structures called grana.
- The lamellae are embedded in a granular material called the stroma.
- The chloroplasts are sites of photosynthesis.
- The light reaction takes place in the lamellae while the dark reactions take place in the stroma.

Comparison between animal cell and plant cell

Plant Cell	***Animal Cell***
• Has a cell wall and a cell membrane.	· Has cell membrane only.
• Nucleus at periphery.	• Nucleus at the center.
• Have	• Have no chloroplasts.
• Has a large central vacuole.	· Has no vacuoles, they are small and scattered.
• Are usually large.	• Are usually small.
• Are regular in shape.	· Irregular in shape.
• Has no centriole.	· Has centrioles.
• Stores starch, oils and protein.	· Store glycogen and fats.

Cell Specialisation

Cells are specialised to perform different functions in both plants and animals.

Example;

- Palisade cells have many chloroplasts for photosynthesis.
- Root hair cells are long and thin to absorb water from the soil.
- Red blood cells have haemoglobin which transports oxygen.
- Sperm cells have a tail to swim to the egg.
- Multicellular organisms cells that perform the same function are grouped together to form a tissue.
- Each tissue is therefore made up of cells that are specialised to carry out a particular function.

Animal Tissues- Examples of animal tissues

Type of Tissue	Functions	Characteristics
I. Epithelial Tissue	Covering. allowing movement of materials	
(a) Squamous	Covering of internal organs. lining for	Thin flat cells.
(b) Columnar	Secretion. absorption e.g. in the	Cells that are longer than they
(c) stratified	Covering surfaces, protection e.g. the	Several layers of epithelial cells
(d) Cuboidal	Absoption e.g. in the kidney tubules.	squamous. cuboidal or cube like cells.
2. Muscular Tissue	Contraction, bringing about	Contists of units called

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(a) Striated voluntary (b) Smooth involuntary (c) Cardiac muscle	Contract and allow movement. cover internal organs; allow peristalsis. Cause contraction of the heart.	Are multicleated; have controlled by voluntary Are spindle-shaped. controlled by involuntary contract rhythmically; are (ability to contract is within)
3. Supporting Tissue (a) Cartilage (b) Bone	Support the body. provide a rigid framework, protect soft tissue.	Cells that produce hard
4. Blood	Transport of materials. protection	A complex tissue consisting of of cells suspended in a fluid
5. Nerve Tissue	Receive stimuli and transmit impulses; co-ordinate body activities	Consists of cells called which are interconnected axons to enable transmission

Plant Tissues
Example of plant tissues

Type of Tissue	***Functions***	***Characteristics***
L *Meristematic*	Undergo division and cause e.g. increase in length and girth	Small thin-walled celis, contain a lot of cytoplasm; found mostly at the tip of shoots and roots.
2. *Parenchyma*	Photosynthesis gaseous support; storage.	Thin walled cells; vary in shape and size; many intercellular spaces.
3. *Collenchyma*	Strengthening.	Thickened walls; no intercellular spaces; found in cortex of stems.
4. *Sclerenchyma*	Strengthening.	Vary in shape; thick cell walls; are usually dead.
5. *Vascular* (a) Xylem (b) Phloem	Transport materials. Transport of water and mineral Transport of organic materials (manufactured food).	Tubular vessels and trancheids joined end to end. Sieve elements joined to each other through sieve pores.

Organs

- An organ is made up of different tissues
- e.g. the heart, lungs, kidneys and the brain in animals and roots, stems and leaves in plants.

Organ systems

- Organs which work together form an organ system.
- Digestive, excretory, nervous and circulatory in animals and transport and support system in plants.

organism

- Different organ systems form an organism.

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Practical Activities

Observation and Identification of parts of a light microscope and their functions

- A light microscope is provided.
- Various parts are identified and observed.
- Drawing and labelling of the microscope is done.
- Functions of the parts of the mircroscope are stated.
- Calculations of total magnification done using the formula.

- **Eye piece lens maginification** x **objective lens rnaginification.**

Preparation and Observation of Temporary Slides of Plant Cells

- A piece of epidermis is made from the fleshy leaf of an onion bulb. It is placed on a microscope slide and a drop of water added.
- A drop of iodine is added and a cover slip placed on top.
- Observations are made, under low and medium power objective.
- The cell wall and nucleus stain darker than other parts.
- A labelled drawing is made.
- The following are noted: **Nucleus, cell wall, cytoplasm and cell membrane.**

Observation of permanent slides of animal cells

- Permanent slides of animal cells are obtained e.g, of cheek cells, nerve cells and muscle cells.
- The slide is mounted on the microscope and observations made under low power and medium power objectives.
- Labelled drawings of the cells are made.
- A comparison between plant and animal cell is made.

Observation and Estimation of Cell Size and Calculation of Magnification of Plant Cells.

- Using the low power objective, a transparent ruler is placed on the stage of the microscope.
- An estimation of the diameter of the field of view is made in millimeters.
- This is converted into micrometres (1mm=1000u)
- A prepared slide of onion epidermal cells is mounted.
- The cells across the centre of the field of view are counted from left and right and top to bottom.
- The diameter of field of view is divided by the number of cells lying lengthwise to give an estimate of the length and width of each cell.

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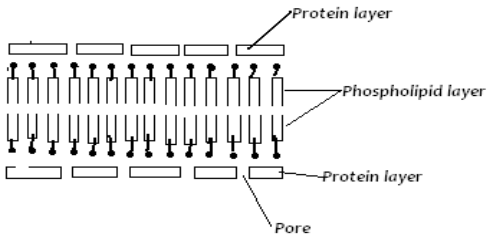
Cell Physiology

Meaning of cell physiology

- The term physiology refers to the functions that occur in living organisms.
- Cell physiology refers to the process through which substances move across the cell membrane.
- Several physiological processes take place inside the cell.e.g. respiration.
- Oxygen and glucose required enter the cell while carbon (IV) oxide and water produced leave the cell through the cell membrane.

Structure and properties of cell membrane

- The cell membrane is the protective barrier that shelter cellular contents.
- Movement of all substances into and out of the cells takes place across the cell membrane.
- It is made up of protein and lipid molecules.
- Lipid molecules have phosphate group attached to it on one end.
- They are then referred to phospholipids.
- The phospholipids are arranged to form a double layer.
- The ends with phosphate group face outwards.
- the proteins are scattered throughout the lipid double layer.
- Some of these proteins act as carrier molecules that channel some material in and outside the cells.
- The cell membrane allows certain molecules to pass through freely while others move through with difficulty and still others do not pass through at all.
- This is selective permeability and the cell membrane is described as semi-permeable.



Cell membrane Structure

Properties of cell membrane

Permeability

- The cell membrane is semi-permeable.
- it allows small molecules that are soluble in lipid to pass through with more ease than water soluble molecules.

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- this is due to the presence of the phospholipids double layer.

Polarlity

- The cell membrane has electrical charges across its surface.it has positive charged ions on the outside and negatively charged ions on the inside.this property contributes to electrical impulses sent along nerve cells.
- Sensitivity to changes in temperature and pH
- Very high temperatures destroy the semi-permeability nature of the cell membrane because the proteins are denatured by extreme pH values have the same effect on the membrane permeability.
- Physiological processes
- Some of the physiological processes include diffusion, osmosis and active transport.

Diffusion

- Diffusion is the movement of molecules or ions from a region of high concentration to a region of low concentration aided by a concentration gradient..
- diffusion continues to occur as long as there is a difference in concentration between two regions (concentration gradient).
- Stops when an equilibrium is reached i.e., when the concentration of molecules is the same in both regions.
- Diffusion is a process that occurs inside living organisms as well as the external environment..
- Does not require energy.

Factors Affecting Diffusion

- **Concentration Gradient**

An increase in the concentration of molecules at one region results in a steeper concentration gradient which in turn increases the rate of diffusion.

- **Temperature**

High temperature increases kinetic energy of molecules. They move faster hence resulting in an increase in rate of diffusion, and vice versa.

- **Size of Molecules or Ions**

The smaller the size of molecules or ions, the faster their movement hence higher rate of diffusion.

- **Density**

The denser the molecules or ions diffusing, the slower the rate of diffusion, and vice versa.

- **Medium**

The medium through which diffusion occurs also affects diffusion of molecules or ions. For example, diffusion of molecules through gas and liquid media is faster than through a solid medium.

- **Distance**

This refers to the thickness or thinness of surface across which diffusion occurs. Rate of diffusion is faster when the distance is small i.e., thin surface.

- **Surface Area to Volume Ratio**

The larger the surface area to volume ratio, the faster the rate of diffusion.

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For example, in small organisms such as Amoeba the surface area to volume ratio, is greater hence faster diffusion than in larger organisms.

Role of Diffusion in Living Organisms

Some processes that depend on diffusion include the following:

- Gaseous exchange: Movement of gases through respiratory surfaces is by diffusion.
- Absorption of materials into cells Cells obtain raw materials and nutrients from the surrounding tissue fluid and blood through diffusion, e.g., glucose needed for respiration diffuses from blood and tissue fluid into cells.
- Excretion: Removal of metabolic waste products like carbon (IV) oxide, and ammonia out of cells is by diffusion.
- Absorption of the end-products of digestion from the intestines is by diffusion.

Osmosis

- Osmosis is the movement of water molecules from a region of high water concentration to a region of low water concentration through a semi-permeable membrane.
- Osmosis is a special type of diffusion that involves the **movement of water molecules only** and **not solute molecules.**

- Osmosis takes place in cells across the cell membrane as well as across non-living membranes
- e.g. cellophane or visking tubing which are also semi-permeable,
- It is purely a physical process.
-

Factors Affecting Osmosis

- **Size of solute molecules-**

Osmosis' occurs onlywhen solute molecules are too large to pass through a semi-permeable membrane.

- **Concentration Gradient .**

Osmosis occurs when two solutions of unequal solute concentration are separated by a semi-permeable membrane.

- **Temperature ,.**

High temperatures increase movement of water molecules hence influence osmosis. However, too high temperatures denature proteins in cell membrane and osmosis stops.

- **Pressure**

Increase in pressure affects movement of water molecules.
As pressure increases inside a plant cell, osmosis decreases.

Roles of Osmosis in Living Organisms

The following processes depend on osmosis in living organisms:

- Movement of water into cells from the surrounding tissue fluid and also from cell to cell.
- Absorption of water from the soil and into the roots of plants.
- Support in plants especially herbaceous ones, is provided by turgor pressure, which results from intake of water by osmosis.
- Absorption of water from the alimentary canal in mammals.
- Re-absorption of water in the kidney tubules.
- Opening and closing stomata.

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Water Relations in Plant and Animal Cells

- The medium (solution) surrounding cells or organisms is described by the terms **hypotonic, hypertonic** and **isotonic.**
- A solution whose solute concentration is more than that of the cell sap is said to be **hypertonic**.
  A cell placed in such a solution loses water to the surroundings by osmosis.
- A solution whose solute concentration is less than that of the cell sap is said to be **hypotonic**.
  A cell placed in such a solution gains water from the surroundings by osmosis.
- A solution which has the same solute concentration as the cell sap is said to be **isotonic**.
  When a cell is placed in such a solution there will be no net movement of water either into or out of the cell.

Osmotic Pressure

- The term osmotic pressure describes the tendency of the solution with a high solute concentration to draw water into itself when it is separated from distilled water or dilute solution by a semi-permeable membrane.
- Osmotic pressure is measured by an osmometer.
- When plant cells are placed in distilled water or in a hypotonic solution, the osmotic pressure in the cells is higher than the osmotic pressure of the medium.
- This causes the water to enter the cells by osmosis.
- The water collects in the vacuole which increases in size.
- As a result the cytoplasm is pushed outwards and it in turn presses the cell membrane next to the cell wall.
- This builds up water pressure (hydrostatic pressure) inside the cell.
- When the cell is stretched to the maximum, the cell wall prevents further entry of water into the cell.
- Then the cell is said to be **fully** turgid.
- The hydrostatic pressure developed is known as **turgor** pressure.

Plasmolysis

- When a plant cell is placed in a hypertonic medium, it loses water by osmosis.
- The osmotic pressure of the cell is lower than that of the medium.
- The vacuole decreases in size and the cytoplasm shrinks as a result of which the cell membrane loses contact with the cell wall.
- The cell becomes flaccid. The whole process is described as plasmolysis.
- **Incipient** plasmolysis is when a cell membrane just begins to lose contact with the cell wall.
- Plasmolysis can be reversed by placing the cell in distilled water or hypotonic solution.
- However, full plasmolysis may not be reversed if cell stays in that state for long.

Wilting

- The term wilting describes the drooping of leaves and stems of herbaceous plants after considerable amounts of water have been lost through transpiration.
- It is observed in hot dry afternoons or in dry weather.
- This is when the amount of water lost through transpiration exceeds the amount absorbed through the roots.

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- Individual cells lose turgor and become plasmolysed and the leaves and stems droop.
- The condition is corrected at night when absorption of water by the roots continue while transpiration is absent.
- Eventually, wilting plants may die if the soil water is not increased through rainfall or watering.

Water Relations in Plants and Animals

Haemolysis

- Haemolysis is the bursting of cell membrane of red blood cells releasing their haemoglobin.
- It occurs when red blood cells are placed in distilled water or hypotonic solution.
- This is because the cell membrane does not resist further entry of water by osmosis after maximum water intake.

Crenation

- Takes place when red blood cells are placed in hypertonic solution.
- They lose water by osmosis, shrink and their shape gets distorted.
    - **Animal cells** have mechanisms that regulate their salt water balance (osmoregulation) to prevent above processes that lead to death of cells.
    - An Amoeba placed in distilled water, i.e. hypotonic solution, removes excess water using a contractile vacuole.
    - The rate of formation of contractile vacuoles increases.

Active Transport

- Active transport is the movement of solutes such as .glucose, amino acids and mineral ions;
- From an area of their low concentration to an area of high concentration.
- It is movement against a concentration gradient and therefore energy is required.
- As such it only takes place in living organisms.
- The energy needed comes from respiration.
- Certain proteins in the cell surface membrane responsible for this movement are referred to as **carrier proteins or channel proteins.**
- The shape of each type of carrier protein is specific to the type of substances conveyed through it.
- It has been shown that the substance fits into a particular slot on the protein molecule,
- As the protein changes from one form of shape to another the substance is moved across and energy is expended.

Factors Affecting Active Transport

Availability of oxygen

- Energy needed for active transport is provided through respiration.
- An increase in the amount of oxygen results in a higher rate of respiration.
- If a cell is deprived of oxygen active transport stops .

Temperature

- Optimum temperature is required for respiration, hence for active transport.
- Very high temperatures denature respiratory enzymes.
- Very low temperatures inactivate enzymes too and active transport stops.

Availability of carbohydrates

- Carbohydrates are the main substrates for respiration.

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- Increase in amount of carbohydrate results in more energy production during respiration and hence more active transport.
- Lack of carbohydrates causes active transport to stop.

Metabolic poisons

- Metabolic poisons e.g. cyanide inhibit respiration and stops active transport due to lack of energy.

Role of Active Transport in Living Organisms

Processes requiring active transport:

- Absorption of mineral salts from the soil into plant roots.
- Absorption of end products of digestion e.g. glucose and amino acids from the digestive tract into blood stream.
- Excretion of metabolic products e.g.urea from the cells.
- Re-absorption of useful substances and mineral salts back into blood capillaries from the kidney tubules.
- Sodium-pump mechanism in nerve cells.
- Re-absorption of useful materials from tissue fluid into the blood stream.

Practical Activities

1.**Experiment to Demonstrate Diffusion**

- Various coloured substances such as: dyes, plant extracts and chemicals like potassium pennanganate are used.
- Potassium manganate (VII) crystals are introduced to the bottom of a beaker filled with water using a glass tubing or drinking straw which is then removed.
- Observations are made and the disappearance of the crystals and subsequent uniform colouring of water noted.

2.**Experiment to Demonstrate Osmosis Using a Visking Thbing**

- A strip of visking tubing 8-10 cm is cut and tied at one end using strong thread.
- About 2 ml of 25% sucrose solution is put inside and the other end tied with thread.
- The tubing is washed under running water and then blotted to dry.
- It is immersed in a beaker containing distilled water and left for at least one hour or overnight.
- It will then be observed that the visking tubing has greatly increased in size and has become firm.
- A control experiment can be set up using distilled water inside the visking tubing in place of sucrose solution.

3.Experiment to Show Osmosis using Living Tissue

- Irish potato tubers are peeled and scooped out to make hollow space at the centre.
- Sucrose solution is placed inside the hollow, and the potato tuber placed in a beaker or petri-dish with distilled water. A conttrol is set using a boiled potato.
- Another one using distilled water inside hollow in place of sugar solution.
- The experiment is left for 3 hours to 24 hours.

4.Experiment to Demonstrate Turgor and Plasmolysis in Onion Epidermal Cells

- Two strips of onion epidermis are obtained.

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- One is placed on a slide with distilled water while the other is placed on a slide with 25% sucrose solution and a coverslip placed on top of each.
- The mounted epidermis is observed under low power microscope and then left for 30 minutes.
- After 30 minutes, observations are made again.

The cells in distilled water have greatly enlarged. Cells in 25% sucrose have shrunk.

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